

Transcendence of a planar reflection–group growth series and its relaxed companion

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Abstract

Let $W = \langle R_0, R_1, R_2 \rangle$ be the group generated by the reflections in the three sides of a right triangle in $\mathbb{Q}^2$ with positive unequal legs, and let $U(x) = \sum_{n \geq 0} u_n x^n$ be its geodesic word-length growth series (the integer sequence OEIS A396406 [42]) with relaxed companion V . We prove:

- (i) each catalytic building-block q -series $\Sigma_0, \Sigma_1, S_0, S_1$ is, by the Pólya–Carlson theorem, either a rational function whose poles are roots of unity or has $|q| = 1$ as a natural boundary and is transcendental over $\mathbb{Q}(q)$. For the two denominators Σ_1, S_1 the first alternative is excluded with no growth hypothesis, by the infinitude of their level sets $\Sigma_1 = 1, S_1 = 1$ in $(0, 1)$ against the finiteness of the zero set of a nonzero rational function (Corollary 3); that route consumes the same single analytic input as everything else here, Lemma 10. For the two numerators Σ_0, S_0 the exclusion is instead under a coefficient-growth hypothesis, verified exactly for the first 34 square indices for Σ_1 and for all four blocks in the uniform form of Hypothesis 5 by exact integer arithmetic out to degree 1156, and not proved (Theorem 2, Remark 6); what the pole route reaches for the numerators is the joint statement that Σ_0 and S_0 are not both rational;
- (ii) the common growth rate is $\beta_2 = 1.4916177871\dots$, and $\frac{3}{2}$ is excluded;
- (iii) the travel resolvent $\Sigma_0/(1 - \Sigma_1)$ is transcendental over $\mathbb{Q}(x)$, unconditionally (Theorem 55), on three inputs all proved here: the pole set, from the rectangle bound of Lemma 10; the non-vanishing of the denominator block at the travel poles, $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$, by an elementary explicit chain with all constants numerical and no cited asymptotic theorem (Appendix A); and the non-vanishing of both numerators, from the exact identity $\Sigma_0(q_m)S_0(q_m) = 2q_m/(1 - q_m)$ (Proposition 21), a consequence of the q -Pythagorean identity at a zero of the q -cosine. The block assembly itself is developed in §5: the travel and the bulk block are one resolvent of the semiseparable kernel $q^{\max(a,b)}$ seen twice, one three-term recursion covers both, and this proves the travel telescoping $G_0^T = \Sigma_0/(1 - \Sigma_1)$, the bulk dressing $b_0 = S_0/(1 - S_1)$, the reciprocity $t_0 = b_1$ (equivalently $\det P = 1$), the identity $t_1 = P_{12}/S_e$, the Möbius factorisation of the gap-marked bulk resolvent, and the *shape* of the local form of V at a travel pole. What is refuted is the product form of its numerator: the marker junction is a max-coupling of rank at least 3. The relaxed growth series V is therefore transcendental over $\mathbb{Q}(x)$ conditional on one statement, (R-J), that a junction pairing does not vanish at infinitely many travel poles (Theorem 56); (R-J) holds at the dominant pole and is open in general. For the true series U we prove the analytic half in full and U 's transcendence conditionally (Theorem 61), on the invariance of the travel block (T) and on the same (R-J); the former hypothesis (L) is now a theorem. All of this rests on the strand-walk transfer model (M). Its local cost law is now a theorem inside the crossing optimisation (Lemma 40, Corollary 41), and its marker clause is corrected: with d_L the last bulk deposit

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and d_R the first travel deposit, the junction cost is $\max(|d_L - 1|, |d_R|)$, which depends on the sign of d_L and not on its magnitude alone; the form $\max(|d_L| - 1, |d_R|)$ carried by earlier versions is false on every cell with $d_L \leq -2$ and $|d_R| \leq |d_L|$ (Remark 44). The shield law is half proved, the inequality $c \geq L - 1$ per gap run (Proposition 45); the reverse inequality, the decomposition (M3), and the crossing optimisation itself remain verified and not proved. The amplitude estimate $(\star)$ ($|P_{12}(q_m)|w_m^3 < 2$ at every travel pole) is proved: the gate block is the symmetric second difference of the q -cosine at its own zero, a two-interval uniform certificate bounds it by $0.619\tau^{3/2} < \tau^{3/2}/\sqrt{2}$ for every pole with $\tau \leq 5 \cdot 10^{-3}$, and the six poles above that threshold are checked directly at 120-digit precision. An exact identity, $S_e = \frac{q^Z}{2}\mathfrak{s}(Z) - P_{12}$, turns the same certificate into the denominator bound $|S_e| \geq 0.63\sqrt{\tau}$, so the gate closes at every travel pole. The turning-point amplitude atom on the U -side, long the special-function obstacle, is an explicit rational inequality *formally verified in Lean 4 / Mathlib [27]* (Lemma 59), as is the whole site cost chain of the model, Lemma 40 with Corollaries 41, 42 and 43 (`SiteCost.lean`, `MarkedSite.lean`, `PairingMatrix.lean`, `Realisation.lean`, `CutCross.lean`);

- (iv) an orthogonal, analysis-free route via Christol’s theorem would settle U without any of the above: the p -kernel of $(u_n \bmod p)$ is computed to be maximally non-automatic at $p = 3, 5$ and machine-checked at $p = 3$, though a proof for dense-support series is open.

Every analytic input consumed by V and by U is proved here, and so now is the algebra of the block assembly; what stands between either series and an unconditional statement is combinatorial, and is stated as a hypothesis in the theorem that uses it. On U every coefficient is derived elementarily and machine-checked, the remainders are at the standard simple-saddle level, and the gate $(\star)$ and the denominator bound) is proved, as is the passage from the gate to $B_U(q_m) \neq 0$; the passage from there to a pole of U is not. The precise rigor standing, including the arithmetic model of the committed certificates, is stated in the text. In sharp contrast with the rational growth of hyperbolic groups and the algebraic growth series of Parry’s wreath products, the geometric word metric places this virtually-lamplighter group outside every algebraic class.

1 Introduction

A right triangle $T \subset \mathbb{Q}^2$ with legs $a \neq b$ has three sides; the reflections across them generate a group $W = \langle R_0, R_1, R_2 \rangle \leq \text{Aff}(\mathbb{R}^2)$. Its *growth series* $U(x) = \sum_n u_n x^n$ counts group elements by minimal word length in $\{R_0, R_1, R_2\}$. Because the legs are unequal the dihedral angles are irrational multiples of π (Niven [32]), so W is a generic triangle group [20] rather than a Coxeter group, and; unlike the rational Coxeter case governed by Steinberg’s formula [43]; its growth is not a priori rational. The sequence $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \dots$ is OEIS A396406; it coincides with the Fibonacci numbers F_{n+3} for $1 \leq n \leq 9$ and first deviates at $n = 10$. This paper settles the arithmetic nature of the travel resolvent $\Sigma_0/(1 - \Sigma_1)$, which is transcendental over $\mathbb{Q}(x)$ with every input proved self-contained (Theorem 55), and reduces both growth series to combinatorial statements about the strand-walk model: the relaxed companion V is transcendental over $\mathbb{Q}(x)$ conditional on the junction non-vanishing (R-J) of Theorem 56, and the true series U conditional on the travel-block invariance (T) and the same (R-J) (Theorem 61); both additionally on the transfer model (M) of Hypothesis 47. The algebraic layer between the blocks and those two series, which earlier versions of this paper carried as the hypotheses (R) and (L), is proved in §5, and so is the local cost law of (M), together with the correction of its marker clause (§5.6). The analytic side is closed on both: the amplitude estimate $(\star)$ and the gate (numerator bound and denominator lower bound) are proved at every travel pole in Appendix B (Theorem 80, Lemma 81, Corollary 82), and the two numerator non-vanishing statements the resolvents need are proved in §4.2.

Context. For word-growth series of groups, rationality is the norm in the classical families: hyperbolic groups have rational growth series with respect to every finite generating set (Cannon [7]), Coxeter systems are governed by Steinberg’s rational formula [18, §5.12], and for wreath products (including lamplighter-type groups) Parry computed the growth series for natural generating sets and found them *algebraic* [35]. Provably transcendental growth series are rare; the classical example is Stoll’s [44]: the higher Heisenberg groups admit both rational and transcendental growth series, the transcendental one arising from a carefully constructed generating set. The group studied here is virtually the lamplighter $\mathbb{Z} \wr \mathbb{Z}$ [4], and its generating set is not constructed but imposed by Euclidean geometry; the three side-reflections of a right triangle. The results below show that for this natural geometric system the growth series is transcendental, by an explicit mechanism (a natural boundary for the building blocks; an accumulation of poles for the series themselves). This places the example in sharp contrast with the algebraicity of Parry’s wreath-product series: the geometric word metric moves a virtually-lamplighter group outside every algebraic class. The higher-dimensional picture is subtler than a clean dimensional dichotomy [5]: the abstract right-angled Coxeter envelope of an n -orthoscheme has rational growth (rate $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$), but the geometric group is an amenable proper quotient that matches this rational envelope only to a finite depth before deviating below it. The plane is the one dimension in which the deviation has been computed past its onset; there it is the transcendental series treated here; and whether every higher dimension is likewise eventually transcendental is an open problem.

The relaxed/true dichotomy. The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller, analytically tamer *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\geq 0}$ a cycle count. The inequality $\ell_T \leq \ell_R + 2c$ is proved in [4]; the reverse inequality is conjectural there, and is verified on the 275 823 elements of the radius-24 ball. The relaxed counting function is the companion V ; one has $v_n \geq u_n$, equality for $n \leq 7$, first strict at $n = 8$. Both are governed by a linear q -difference (dilation $t \mapsto q^2 t$) *catalytic functional equation* (§2) that is q -holonomic and not of polynomial-kernel type.

Method. The catalytic transfer is assembled from four integer q -series $\Sigma_0, \Sigma_1, S_0, S_1$ with j^2 -gapped coefficients. Pólya–Carlson gives a dichotomy for them (rational with roots-of-unity poles, or a natural boundary and transcendence over $\mathbb{Q}(q)$). For the two denominators the rational alternative is excluded outright: their level sets $\Sigma_1 = 1$ and $S_1 = 1$ in $(0, 1)$ are infinite, and a rational function that is not identically zero has finitely many zeros in $\mathbb{C}$ (Corollary 3). For the two numerators it is excluded by super-polynomial coefficient growth along a sparse set of indices, verified for the first 34 square indices for Σ_1 and out to degree 1156 for all four blocks, and not proved (§3). Both V and U acquire poles at the *travel poles* q_m , the roots of $\Sigma_1^T(q) = 1$; transcendence follows from accumulation of these poles at $q = 1$ once a certain auxiliary integral T_2 is shown to be small (§4). What the theorems consume is only $|T_2| \rightarrow 0$, and that is proved self-contained in Lemma 10 by an absolute bound on a rectangle: no saddle, no path through a saddle, no cited asymptotic theorem. Its quantitative part is a verified enclosure of the contour majorant, in interval arithmetic and quantified over parameter boxes, on $\tau \in [2/101^2, 0.02]$, together with an explicit-constant estimate below that; nothing there is a scan over sample values of τ . It is the single analytic input of this kind, shared by V (§6) and U (§7). The sharp size of T_2 , with the constant $\sqrt{2}/36$, is recorded separately in Remark 9 as a numerically certain statement whose proof is incomplete (Remark 11); nothing here depends on it.

The U -side carries one extra ingredient. Writing the cocycle off-diagonal P_{12} in exact closed form as a Hahn–Exton q -Bessel function $J_{3/2}^{(3)}$, the transcendence gate reduces to one amplitude

bound on that series at the travel poles. This bound is *proved* (Appendix B): P_{12} is the symmetric second difference of the q -cosine at its own zero (Proposition 73), so the gate is a second-derivative estimate on a window of width τ , delivered by a two-interval uniform certificate; an exact identity then converts the same certificate into the denominator lower bound, closing the gate (Corollary 82). The gate now also delivers $B_U(q_m) \neq 0$ at every travel pole (Theorem 50). What it does *not* supply is the passage from there to a pole of U , because the assembled residue is a junction pairing and not a multiple of B_U (Proposition 49); that passage, and the invariance of the travel block under the cycle marker, are the two unproved inputs (R-J) and (T) of Theorem 61 (Remark 63). The sharp constants behind the picture are *derived* in Remark 16: the pole–zero offset $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$ and $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the gate threshold. (The turning-point atom feeding that analysis, the rational envelope $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2}) \in [1, \frac{3}{2}]$ of Lemma 59, is formally verified in Lean 4.) U and V therefore rest on the same self-contained *analysis*; no special-function input separates them, and after §5 they share the junction hypothesis (R-J) as well. What separates them is the single combinatorial input (T), which V does not use.

Section 8 records an independent arithmetic route: by Christol’s theorem [9, 10], if $(u_n \bmod p)$ is not p -automatic for one prime p then U is transcendental over $\mathbb{Q}(x)$; the p -kernel is maximally non-automatic at $p = 3, 5$.

Appendix C treats a question separate from all of the above: whether the *number* β_2 is transcendental. That is open. What the appendix contributes is a reduction of it to a value statement for a Hahn–Exton q -Bessel function at an algebraic base, and an exact location of the obstruction, at the arithmetic ratio $\rho = \frac{1}{2}$, one notch beyond the range in which q -irrationality criteria are proved. No statement in the body depends on it.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q -difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q ,

$$\Sigma_1(q) = \sum_{i \geq 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!} \eta_i, \quad S_e(q) = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q; q)_{2j}}, \quad (1)$$

and their companions Σ_0, S_0 , where $\eta_j = e^{-B_j}$ is the Gaussian-decaying form factor of §4 (equivalently $\eta_j = q^{j^2+j} [2(1-q)]^j (2j)! / [(q; q)_{2j} W^{2j}]$; the identification of this display with $1 - \Sigma_1 = \cos(Z; q^2)$ of Proposition 17 is the collapse used in Lemma 12). We write $\Sigma_1^T = \Sigma_1$ when the travel rôle is to be emphasised. The relaxed travel resolvent is $G_0^T = \Sigma_0 / (1 - \Sigma_1)$ (Proposition 27), and U inherits its denominator $1 - \Sigma_1$ (Proposition 1); how the two blocks assemble into V and U , and how far that assembly is proved, is §5. The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \quad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$ (Lemma 12, proved in §4 once the collapse machinery is in place).

Proposition 1 (Invariance of the travel block). *The denominator $1 - \Sigma_1$ is the identical, cycle-marker-free object in the generating functions of U and of V ; hence U and V share the travel poles q_m .*

This proposition is not proved in this paper. The argument is a strand-walk argument (every pure-travel element has cycle count 0, so the two length gradings agree on it) and is written in

the supplement [6]; there it rests in turn on two statements that the supplement does not prove, namely the localisation of cycles (a pure-travel element has $c = 0$), checked exhaustively on the 3869 pure-travel elements of true length ≤ 16 , and the metric identity $\ell_T = \ell_R + 2c$, whose upper bound is proved in [4] and whose lower bound is open. Its standing is therefore that of a verified conjecture, and every statement below that invokes it inherits that standing; this is recorded explicitly in Theorem 61 and in Remark 63.

3 Transcendence of the blocks

The four blocks are the k -recursion sums

$$\Sigma_k = \sum_{j \geq 0} A_{k+2j} \prod_{i < j} C_{k+2i}, \quad S_k = \sum_{j \geq 0} \alpha_{k+2j} \prod_{i < j} \gamma_{k+2i}, \quad (2)$$

$$A_k = \frac{2q}{1 - q^{k+1}}, \quad C_k = \frac{2q^{k+3}}{1 - q^{k+2}} - \frac{2q^{k+2}}{1 - q^{k+1}}, \quad \alpha_k = \frac{2q^{k+1}}{1 - q^{k+1}}, \quad \gamma_k = \alpha_{k+1} - \alpha_k,$$

Σ_0, Σ_1 carrying the travel rôle and S_0, S_1 the bulk rôle; Σ_1 and S_1 are the denominators of (1) and Σ_0, S_0 their numerators. Every factor is a Lambert series with integer coefficients. The factor C_{1+2i} starts at order $2i + 3$, so $\prod_{i < j} C_{1+2i}$ starts at order $\sum_{i < j} (2i + 3) = j^2 + 2j$ and, with A_{1+2j} starting at order 1, the j -th summand of Σ_1 starts at order exactly $(j + 1)^2$; only the finitely many $j \leq \sqrt{n} - 1$ reach $[q^n]$. Each coefficient is therefore a finite integer sum, and $\Sigma_1 = 2q + 2q^3 - 4q^4 + 6q^5 - 4q^6 + \dots$. That integrality, with convergence on $|q| < 1$, is the whole input to Pólya–Carlson.

Theorem 2. *Let Σ be any of $\Sigma_0, \Sigma_1, S_0, S_1$. Then Σ has integer coefficients and converges in $|q| < 1$, and exactly one of the following holds:*

- (a) Σ is a rational function whose poles are roots of unity; or
- (b) $|q| = 1$ is a natural boundary of Σ , and Σ is transcendental over $\mathbb{Q}(q)$.

Alternative (a) is excluded, and hence (b) holds, for any block whose coefficients grow super-polynomially along a subsequence. For Σ_1 that growth is the hypothesis

$$|[q^{(j+1)^2}] \Sigma_1| \geq 2^{j+1} \quad \text{for infinitely many } j, \quad (3)$$

which is not proved here: see Remark 6.

Proof. The coefficients are integers by the order count above: each is a finite sum of integers. Convergence in $|q| < 1$ is read off the closed forms: each block is a series whose j -th term carries the Gaussian factor $q^{j^2 + O(j)}$ against a denominator $(q; q)_{2j}$ or $(q; q)_{2j+1}$ bounded below in modulus by $|(q; q)_\infty| > 0$ on compact subsets of the punctured disc (for Σ_1 use $1 - \Sigma_1 = \cos(Z; q^2) = \sum_{j \geq 0} (-1)^j q^{j^2} (2(1 - q))^j / (q; q)_{2j}$ of Proposition 17, in which the pole of $Z^2 = 2(1 - q)/q$ at the origin is cancelled), so each series converges absolutely and locally uniformly there. Hence Pólya–Carlson [8] applies and gives the dichotomy (a)/(b): a power series with integer coefficients convergent in the unit disc is either a rational function whose poles are roots of unity, or has the unit circle as a natural boundary; a function with a natural boundary is transcendental over $\mathbb{Q}(q)$. In case (a) the coefficient sequence satisfies a linear recurrence all of whose characteristic roots are roots of unity, hence $|[q^n] \Sigma| \leq Cn^d$ for some C, d ; a subsequence growing faster than every polynomial in n is therefore incompatible with (a). Under (3), $|[q^n] \Sigma_1| \geq 2^{\sqrt{n}}$ along $n = (j + 1)^2$, which is such a subsequence. $\square$

For the two denominators the growth hypothesis is not needed at all: alternative (a) is excluded by the pole set instead.

Corollary 3 (the pole route to alternative (a)). *Alternative (a) of Theorem 2 fails for Σ_1 and fails for S_1 ; hence for each of the two denominators $|q| = 1$ is a natural boundary and the block is transcendental over $\mathbb{Q}(q)$. It also fails for at least one of the two numerators Σ_0, S_0 . Neither statement uses (3) or Hypothesis 5.*

Proof. A rational function that is not identically zero has finitely many zeros in $\mathbb{C}$. The constraint is global: unlike the identity theorem it requires no accumulation point inside a domain, so a zero set that is merely infinite already contradicts it, wherever in $\mathbb{C}$ the zeros sit and wherever they accumulate.

Suppose Σ_1 satisfies (a), so that on $|q| < 1$ it agrees with a rational function whose poles are roots of unity. Then $1 - \Sigma_1$ is rational, and it is not identically zero, because $\Sigma_1(0) = 0$: every factor A_k, C_k of (2) vanishes at $q = 0$, equivalently $1 - \Sigma_1 = \cos(Z; q^2)$ has constant term 1 (Proposition 17). A nonzero rational function has finitely many zeros; but $\{q \in (0, 1) : \Sigma_1(q) = 1\}$ is infinite by Lemma 12. Alternative (a) is therefore impossible and Theorem 2 leaves (b). The same argument applies to S_1 , with $1 - S_1 = S_e$ and $S_1(0) = 0$ (Proposition 17) and with Lemma 13 in place of Lemma 12.

For the numerators, suppose Σ_0 and S_0 both satisfy (a). Then $\Sigma_0 S_0$ is rational and, by Proposition 21, agrees with the rational function $2q/(1 - q)$ at every travel pole, hence at infinitely many points of $(0, 1)$ (Lemma 12); two rational functions agreeing at infinitely many points are equal. They are not: $\Sigma_0 S_0 = 4q^2 + 8q^3 + 4q^4 + \dots$ vanishes to order 2 at $q = 0$ and $2q/(1 - q)$ to order 1. This says nothing about either numerator separately, (14) constraining only the product. $\square$

The two lemmas quoted are proved in §4 and the two identities in §4.1; of Theorem 2 they use only its first clause, that the blocks converge on $|q| < 1$, so the dependency graph is acyclic. The series data $\Sigma_1 = 2q + 2q^3 - 4q^4 + \dots$, $S_1 = 2q^2 + 2q^4 - 2q^6 + \dots$, $\Sigma_0 S_0 = 4q^2 + 8q^3 + 4q^4 + \dots$ and the two collapses $1 - \Sigma_1 = \cos(Z; q^2)$, $1 - S_1 = S_e$ are recomputed in exact integer arithmetic at two truncation degrees in `f2_pole_route.py`.

Remark 4 (what the pole route costs). Corollary 3 makes Theorem 2 unconditional for Σ_1 and S_1 , and the two routes to it are of different kinds. (3) is checked in exact integer arithmetic, with no precision to state, at 34 indices, and remains a finite verification of a statement with infinitely many instances. The pole route runs on Lemma 12, hence on Lemma 10, and that lemma is proved on all of $\tau \in (0, 0.02]$: a verified enclosure of the contour majorant, in interval arithmetic and quantified over parameter boxes rather than sampled, on $\tau \geq 2/101^2$, and an explicit-constant estimate below that. The two infinitude lemmas consume it along the infinite sequences $\tau^{(m)} = 2/(m\pi)^2$ and $\tau^{[m]}$, $m \geq 4$, all of which lie in that range. So the pole route carries no unproved arithmetic input, while (3) does; and for Σ_0 and for S_0 individually, (4) remains the only route. What the corollary buys besides the label is a shorter dependency list: the exclusion of (a) for the two denominators consumes the same single analytic input as Theorems 55, 56, 61 and Corollary 58, instead of a second input of its own.

The hypothesis (3) concerns Σ_1 alone. Its counterpart for the other three blocks is the following uniform statement, which is what alternative (a) has to be weighed against for Σ_0 and S_0 , the two blocks Corollary 3 does not reach individually.

Hypothesis 5 (coefficient growth). Let Σ be any of $\Sigma_0, \Sigma_1, S_0, S_1$. Then

$$|[q^n]\Sigma| \geq 2^{\sqrt{n}-1} \quad \text{for infinitely many } n. \quad (4)$$

Statement (3) is the Σ_1 instance of (4) in the sharper form that names the witnessing indices $n = (j+1)^2$. Either form excludes alternative (a) for the block concerned, $2^{\sqrt{n}-1}$ outgrowing every polynomial in n .

Remark 6 (why (3) is expected, and how far it is verified). (3) is verified, not proved, and the reason it is expected is exact rather than heuristic. By the order count above, the j -th summand of the telescoping $\Sigma_1 = \sum_{j \geq 0} (\prod_{i < j} C_{1+2i}) A_{1+2j}$ has lowest degree exactly $(j+1)^2$, with coefficient exactly $(-1)^j 2^{j+1}$ (the leading 2 of A_{1+2j} against the leading -2 of each of the j factors C_{1+2i}), and summands of index $> j$ start above that degree. Hence

$$[q^{(j+1)^2}] \Sigma_1 = (-1)^j 2^{j+1} + \left(\text{the degree-}(j+1)^2 \text{ tails of the } j \text{ summands of index } < j \right),$$

a *finite* correction to an exactly known leading term: (3) says precisely that those j tails do not cancel it. They do not merely fail to cancel; they reinforce. Exact integer evaluation of (2) truncated at degree 1156 gives

$$[q^{(j+1)^2}] \Sigma_1 = 2, -4, 14, -52, 178, -856, 4626, -27524, 150214, -816268, \dots$$

and confirms (3) for every $j \leq 33$, with the sign $(-1)^j$ of the leading term at every one of them and a margin that grows without bound: $|[q^{(j+1)^2}] \Sigma_1| / 2^{j+1}$ is 797 at $j = 9$, $1.2 \cdot 10^8$ at $j = 20$ and $2.7 \cdot 10^{14}$ at $j = 33$, where the value is $-4649405079364022238518260$ against the bound 2^{34} ; the successive ratios $|[q^{(j+2)^2}] \Sigma_1| / |[q^{(j+1)^2}] \Sigma_1|$ rise from 2 to 6.22 over that range. Hypothesis 5 holds for all four blocks with a witness in every window $[(j+1)^2, (j+2)^2)$ for every $j \leq 32$, the range being cut off by the truncation degree and not by a failure. The witnessing diagonals are $n = (j+1)(j+2)$ for S_1 and $n = (j+1)^2$ for S_0 , each carrying $|[q^n] \Sigma| \geq 2^{j+1}$ throughout $4 \leq j \leq 32$ and $3 \leq j \leq 32$ respectively; for Σ_0 the leading diagonal cancels: its values at $n = (j+1)^2$ are 2, 2, 2, 10, 18, 34, 266, 306, 8170, 330, which drop below 2^{j+1} at $j = 1, \dots, 5$ and again at $j = 9$, so its witnesses are off-diagonal and (4) is the only form available for it.

The computation is exact integer arithmetic in $\mathbb{Z}[[q]]/(q^{1161})$ (`blocks_growth.py`); no floating point enters, so no precision is quoted. It is checked against a second and independent derivation of the same coefficients from the q -trigonometric closed forms of Proposition 17 and Lemma 72 (agreement to degree 1160 for each of Σ_1, S_1, S_0), and against the bulk block $S_0/(1-S_1) = 2q+2q^2+6q^3+2q^4+18q^5+6q^6+42q^7+\dots$. For $j \leq 9$ the bound on Σ_1 is also machine-checked in Lean 4 (`PolyaCarlson.lean`, theorem `coeff_bound`); that proof uses `native_decide` and therefore depends, besides `propext`, on the compiler-trust axiom `PolyaCarlson.coeff_bound._native.native_decide.ax_1_1`, which is outside the three standard axioms.

No finite verification can establish a statement quantified over infinitely many indices. (3) is numerically certain and unproved, and the transcendence conclusion (b) of Theorem 2 inherits that standing wherever it is the route used, which after Corollary 3 is Σ_0 and S_0 . What would close it is a lower bound on the degree- $(j+1)^2$ coefficient of the j preceding Lambert products, uniform in j : the statement is finite for each j and the obstruction is uniformity, not any single case.

The dichotomy of Theorem 2 uses no analytic gate. The exclusion of alternative (a) rests on (3) for Σ_0 and S_0 , and on the pole route of Corollary 3, hence on the analytic input of §4, for Σ_1 and S_1 . Neither transfers directly to V or U , which are *ratios* of blocks in which the boundary singularities can cancel; those require the pole-accumulation argument below, which is independent of (3).

4 The steepest-descent estimate

The one analytic input for both V and U is the smallness of an auxiliary integral. Let $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n \geq 1} \varphi_n y^{2n}$ with $\varphi_n = \frac{(-1)^{n+1} \zeta(2n)}{n(2\pi)^{2n}}$. The form factor B_s is *defined* by the closed form

(6) of Lemma 7; the familiar expansion $B_s \sim \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$, $P_n(M) = \sum_{m=1}^M m^{2n}$ the Faulhaber sum, is its Euler–Maclaurin *asymptotic* series (equality at integer $2s$, divergent otherwise; Lemma 7(v)) and is used only truncated. The block S_e decomposes as $S_e = \cos W - T_2$ with $W = w e^{-\tau/2}$ and

$$T_2 = \sum_{n \geq 1} (-1)^n g_n \frac{W^{2n}}{\Gamma(2n+1)}, \quad g_s = 1 - e^{-B_s}. \quad (5)$$

What the theorems consume is only $T_2 \rightarrow 0$; its sharp size is recorded separately in Remark 9 and is not used.

Lemma 7 (amplitude regularity). *For $\operatorname{Re} s > -\frac{1}{2}$ set $M = 2s$ and*

$$B_s = \frac{\tau}{4} M(M+1) - M \log \tau - \log \Gamma(M+1) + \log(q; q)_\infty - \log(q^{M+1}; q)_\infty - s \varphi(\tau), \quad (6)$$

$\varphi(y) = \log \frac{\sinh(y/2)}{y/2}$ as above, each q -Pochhammer logarithm meaning the sum of principal logarithms $\sum_{j \geq 0} \operatorname{Log}(1 - q^{M+1+j})$: every factor has $|q^{M+1+j}| < 1$, hence $\operatorname{Re}(1 - q^{M+1+j}) > 0$, so this is the analytic branch on the half-plane (the logarithm of the product would differ by multiples of $2\pi i$). Then B_s is analytic on $\operatorname{Re} s > -\frac{1}{2}$, and at every integer $n \geq 0$

$$B_n = \sum_{i=1}^{2n} \varphi(i\tau) - n \varphi(\tau), \quad (7)$$

so (6) is the analytic interpolation of the collapse coefficients $\eta_n = e^{-B_n}$ of (1). On the strip

$$\mathcal{S} = \{-\frac{1}{2} < \operatorname{Re} s \leq 2w, |\Im s| \leq 2w\}, \quad \tau \leq 2/\pi^2,$$

one has the explicit bounds

$$|B_s| \leq 30.3 \sqrt{\tau}, \quad |B'_s| \leq 7.6 \tau, \quad (8)$$

and, on the smaller region $|M| \leq 2w$ which contains the saddles, the truncation bound that Remark 9 consumes:

$$|B_s - \varphi_1 \tau^2 (P_1(M) - s)| \leq 0.02 \tau^{3/2}, \quad \varphi_1 = \frac{1}{24}. \quad (9)$$

Consequently $|e^{-B_s}| \leq e^{30.3\sqrt{\tau}}$, the amplitude $A := 1 - e^{-B_s}$ satisfies $|A| \leq 1 + e^{30.3\sqrt{\tau}} \rightarrow 2$, and along any rectifiable $\Gamma \subset \mathcal{S}$ of length $\ell(\Gamma) \leq cw$,

$$V_\Gamma(A) \leq \sup_\Gamma |e^{-B_s}| \int_\Gamma |B'_s| |ds| \leq e^{30.3\sqrt{\tau}} \cdot 7.6 \tau \cdot cw = O(\sqrt{\tau}) \rightarrow 0,$$

using $\tau w = \sqrt{2\tau}$. The strip contains both saddles $s^* = -\frac{1}{4} \pm i(\frac{w}{2} + O(\sqrt{\tau}))$ of Remark 9 and the $O(w)$ -length portion of its steepest-descent contour through them (which is all the $\ell(\Gamma) \leq cw$ clause above requires, the Gaussian width being $O(\sqrt{w})$) and, being built from the larger argument $w = W e^{\tau/2}$, it contains the corresponding W -strip as well.

Proof. (i) Closed form, and (7). Unwinding $\eta_n = q^{n^2+n} [2(1-q)]^n (2n)! / [(q; q)_{2n} W^{2n}]$ with $W^2 = (2/\tau) e^{-\tau}$ gives $\eta_n = e^{-\tau n^2} [(1-q)\tau]^n (2n)! / (q; q)_{2n}$. Now $1 - q^i = i\tau e^{-i\tau/2} \frac{\sinh(i\tau/2)}{i\tau/2}$, i.e. $\log(1 - q^i) = \log(i\tau) - \frac{i\tau}{2} + \varphi(i\tau)$; summing over $1 \leq i \leq M$,

$$\log(q; q)_M = \log \Gamma(M+1) + M \log \tau - \frac{\tau}{4} M(M+1) + \sum_{i=1}^M \varphi(i\tau). \quad (10)$$

Since $\log[(1-q)\tau] = -\frac{\tau}{2} + 2\log\tau + \varphi(\tau)$, substituting (10) into $-\log\eta_n$ collapses every elementary term and leaves exactly (7); and rewriting $\log(q;q)_M = \log(q;q)_\infty - \log(q^{M+1};q)_\infty$ in (10) turns (7) into (6) for all s . Every ingredient of (6) is analytic in s for $\operatorname{Re} s > -\frac{1}{2}$: there $|q^{M+1}| < 1$ keeps $(q^{M+1};q)_\infty$ zero-free and the branch above canonical, and $\operatorname{Re}(M+1) > 0$ keeps $M+1$ off the poles of Γ . (Both singular factors do in fact cancel at $M = -1$, so B_s continues past $\operatorname{Re} s = -\frac{1}{2}$; we never need that.)

(ii) *Splitting into Gamma blocks with exact cancellation.* By the Weierstrass product $\frac{\sinh(y/2)}{y/2} = \prod_{k \geq 1} (1 + \frac{y^2}{4\pi^2 k^2})$ we have $\varphi(y) = \sum_{k \geq 1} \log(1 + y^2/(4\pi^2 k^2))$, so with $R_k := 2\pi k/\tau$ and $\prod_{n=1}^M (1 + n^2 a^2) = a^{2M} \Gamma(M+1-\frac{i}{a})\Gamma(M+1+\frac{i}{a})/[\Gamma(1-\frac{i}{a})\Gamma(1+\frac{i}{a})]$ at $a = 1/R_k$, (7) becomes

$$B_s = \sum_{k \geq 1} G(M; R_k) - s\varphi(\tau), \quad G(M; R) := \int_0^M [\psi(1+u+iR) + \psi(1+u-iR) - 2\log R] du, \quad (11)$$

the integral along the segment $[0, M]$ fixing the branch (legitimate since $|M| < R_1$ puts the segment far from the poles of $\psi(1+u \pm iR)$ at $u = -1 - j \mp iR$). *The $-2\log R$ is not a normalisation: it is the exact cancellation of the two leading Stirling terms*, and it is what makes the k -sum converge; without it each block would grow like $2M \log R$.

Identity (11) is proved at integers, where the i -sum is finite and the interchange of $\sum_i$ and $\sum_k$ is trivial; it extends to all s as follows.

Domain. Both sides are analytic on the half-plane $\operatorname{Re} M > -1$: the left by (i), the right because the segment $[0, M]$ then stays in $\operatorname{Re} u > -1$ while the poles of $\psi(1+u \pm iR)$ lie in $\operatorname{Re} u \leq -1$, and because $\sum_k G$ converges locally uniformly there, all but finitely many k having $R_k > 2|M|$ and hence $|G(M; R_k)| = O(|M|^3/R_k^2)$. (The disc $|M| < R_1$ of step (iii) is where the *quantitative* majorant lives; it is not the domain of analyticity, and the recursion below would walk out of it.)

Recursion. Write $\tilde{B} = B_s + s\varphi(\tau)$ for either side with the elementary term restored. Both obey the *same* unit recursion in M : on the left $\frac{\tau}{4}[(M+1)(M+2) - M(M+1)] - \log\tau - \log(M+1) + \log(1 - q^{M+1}) = \varphi((M+1)\tau)$ term by term, and on the right $G(M+1; R) - G(M; R) = \log(1 + (M+1)^2/R^2)$, which sums over k to the same $\varphi((M+1)\tau)$ by the Weierstrass product. (For B_s itself both increments are $\varphi((M+1)\tau) - \frac{1}{2}\varphi(\tau)$, the extra term being common to the two sides.)

Conclusion. Hence the difference D is 1-periodic on $\operatorname{Re} M > -1$, so it continues to a 1-periodic *entire* function. Its Fourier coefficients satisfy $|c_m| \leq e^{2\pi m y} \sup_{0 \leq x \leq 1} |D(x+iy)|$ for every real y , and D grows at most polynomially: on the left by Stirling, on the right by splitting the k -sum at $R_k = 2|M|$: the $O(|M|\tau)$ low terms are each $O(|M|\log|M|)$ by Stirling and the remainder is $O(|M|^3 \sum_{R_k > 2|M|} R_k^{-2}) = O(|M|^2\tau)$. Letting $y \rightarrow -\infty$ for $m > 0$ and $y \rightarrow +\infty$ for $m < 0$ gives $c_m = 0$ for $m \neq 0$, so D is constant; and $D(0) = 0$ by (7) at $n = 0$, so $D \equiv 0$.

(iii) *The integrand.* For $|z| \geq 1$ with $|\arg z| \leq \frac{3\pi}{4}$ one has $|\psi(1+z) - \log z - \frac{1}{2z}| \leq |z|^{-2}$ (the constant is 0.13 on that sector; the bound fails without limit as $\arg z \rightarrow \pi$, which is why the sector, not a distance to the cut, is the right hypothesis). On $\mathcal{S}$, $|M| \leq 4\sqrt{2}w$ and $|M|/R_1 = \frac{4\sqrt{2}w\tau}{2\pi} = \frac{4}{\pi}\sqrt{\tau} \leq 0.58$, so every $R_k > |M|$; and along the segment $|\arg(u \pm iR_k)| \leq 120^\circ < 135^\circ$ with $|u \pm iR_k| \geq R_1 - |M| > 13$, so the estimate applies throughout. Adding the two conjugate terms, the logarithms combine with $-2\log R$ to $\log(1 + u^2/R^2)$ and the $\frac{1}{2z}$ terms to $u/(u^2 + R^2)$, whence for $|u| \leq |M|$

$$|\psi(1+u+iR) + \psi(1+u-iR) - 2\log R| \leq \frac{|M|^2 + |M|}{R^2 - |M|^2} + \frac{2}{(R - |M|)^2} =: \gamma(R).$$

(iv) *Summation.* Hence $|G(M; R_k)| \leq |M|\gamma(R_k)$ and, from (11),

$$|B_s| \leq |M| \sum_{k \geq 1} \gamma(R_k) + |s|\varphi(\tau), \quad |B'_s| \leq 2 \sum_{k \geq 1} \gamma(R_k) + \varphi(\tau),$$

the second because $\partial_M G = \psi(1+M+iR) + \psi(1+M-iR) - 2\log R$ and $M = 2s$. Both sums are dominated by $\sum_k R_k^{-2} = \tau^2/24$; carrying the finite corrections explicitly at $|M| = 4\sqrt{2}w$, $\tau \leq 2/\pi^2$ gives $|M| \sum_k \gamma(R_k) + |s|\varphi(\tau) \leq 30.3\sqrt{\tau}$ and $2 \sum_k \gamma(R_k) + \varphi(\tau) \leq 7.6\tau$, which is (8). Both ratios are *increasing* in τ on the range (30.20, 25.14, 23.26, 22.21, 21.64 and 7.55, 6.28, 5.81, 5.55, 5.41 at $\tau = 2/\pi^2$, 0.1, 0.05, 0.02, 0.005, decreasing to the limits $\frac{64}{3}$ and $\frac{16}{3}$ as $\tau \rightarrow 0$), so each attains its maximum at the right endpoint $\tau = 2/\pi^2$, and the stated constants hold throughout.

For (9): expand the integrand of G one order further, now *keeping the third Stirling term* $-\frac{1}{12z^2}$, which is what supplies the linear part of P_1 : its integral is $\frac{M}{6(M^2+R^2)} = \frac{M}{6R^2} + O(|M|^3/R^4)$. With $\int_0^M \log(1+u^2/R^2)du = \frac{M^3}{3R^2} + O(|M|^5/R^4)$ and $\int_0^M u/(u^2+R^2)du = \frac{M^2}{2R^2} + O(|M|^4/R^4)$ this gives $R^2 G(M; R) \rightarrow P_1(M) = \frac{M^3}{3} + \frac{M^2}{2} + \frac{M}{6}$, so with $\sum_k R_k^{-2} = \tau^2/24$ exactly,

$$B_s = \varphi_1 \tau^2 (P_1(M) - s) + O(|M|^5 \tau^4) + O(|s| \tau^4).$$

On $|M| \leq 2w$ every remainder is $O(\tau^{3/2})$, and the constant is 0.02 (worst 0.0177 at $\tau = 2/\pi^2$, decreasing).

(v) *What is not used.* The τ -expansion $B_s = \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$, P_n the Faulhaber sum, is the Euler–Maclaurin *asymptotic* expansion of (6): it is *divergent* at non-integer $2s$, since $P_n(2s)$ carries a Bernoulli part of size $(2n)! e^{2\pi i \Im 2s} / (2\pi)^{2n+1}$. Nothing above sums it; it is used only in the truncated form “leading term plus remainder”, as in Remark 9, where the $n = 1$ term at the saddle supplies the constant $\sqrt{2}/36$. $\square$

The algebraic skeleton of this proof is machine-checked in Lean 4 (`BboundedSkeleton.lean`, ten theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`). Five carry the arithmetic content: the Gauss sum, the summation (10) built on it, the telescoping cancellation (7), the scaling identity $(\tau^2 w^3)^2 = 8\tau$ (exact for w , only $e^{-3\tau/2}$ -approximate for W), and the strip condition $|M|/R_1 < 1$. The other five (the per-factor rewriting, the collapse of $\log[(1-q)\tau]$, the cancellation $2M \log a + 2M \log R = 0$ at $aR = 1$, the w -definition, and the bounded-variation arithmetic) are *rearrangements of their hypotheses*, verified as such: they certify that the paper’s algebra is what it claims, not the transcendental facts fed into them. Those facts (the Weierstrass product, the digamma remainder, the convergence of the k -sum, and the continuation argument of step (ii)) are not formalised.

Remark 8 (sharpness). The majorant of step (iv) is lossy by about a factor 6: the true extremes on $\mathcal{S}$ are $\operatorname{Re} B_s \geq -5.03\sqrt{\tau}$ and $|B'_s| \leq 5.46\tau$, and $\sup_{\mathcal{S}} |A| = 9.02$ at $\tau = 2/\pi^2$ (`star_gate_certificate.py`, part (7)). Only the proved bounds (8) are used anywhere.

The sketch exhibits the saddles explicitly and derives the constants by Cauchy’s theorem, Stirling’s formula and elementary Gaussian integration, and it would end by invoking a second-order remainder estimate for a simple non-degenerate saddle, [34, Ch. 4, Thm. 7.1]. Its hypotheses are *not* verified here (the saddle is flat rather than non-degenerate, and the amplitude’s bounded variation is available only on a bounded piece of the path), which is one of the reasons the sketch is not a proof (Remark 11). No theorem of this paper rests on it.

Remark 9 (the sharp size of T_2 ; HEURISTIC). Numerically, $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin X - \frac{161}{1296} \tau \cos X + O(\tau^{3/2})$ uniformly in the phase, for any argument X with $\tau X^2 \in [2e^{-\tau}, 2]$; the constant and the exponent of the remainder are certain to eight digits over $m \leq 1024$. The sketch below indicates why, by steepest descent, but it does *not* constitute a proof, for the reasons set out in Remark 11. Accordingly this statement carries no weight anywhere in this paper: the only property of T_2 that the theorems consume is the crude bound $|T_2| \rightarrow 0$ of Lemma 10, which is proved, and which is what Lemma 12 cites.

Sketch, not a proof; see Remark 11. Since $\cos W = \sum_{n \geq 0} (-1)^n W^{2n} / (2n)!$, (5) gives the exact alternating sum $S_e = \cos W - T_2 = \sum_{n \geq 0} (-1)^n e^{-B_n} W^{2n} / (2n)!$ (with $B_0 = 0$). Let $b(z) = e^{-B_z} W^{2z} / \Gamma(2z + 1)$, analytic for $\operatorname{Re} z > -\frac{1}{2}$ (by (6) and Lemma 7; *not* by termwise polynomiality, the expansion being divergent off z). As $\pi \csc \pi z$ has residue $(-1)^n$ at each integer, Cauchy's theorem gives $S_e = \frac{1}{2\pi i} \oint b(z) \pi \csc(\pi z) dz$ over a contour enclosing $\{0, 1, 2, \dots\}$. Write the integrand as $e^{h(z)} \pi \csc \pi z$ with $h(z) = 2z \log W - \log \Gamma(2z + 1)$; including the $\csc$ phase, the exponent is stationary where $2 \log W - 2\psi(2z + 1) \pm i\pi = 0$, so by $\psi(w) = \log w + O(1/w)$ there are two simple, non-degenerate saddles

$$z_{\pm}^* = \pm \frac{i}{2} W (1 + O(1/W)), \quad h''(z_{\pm}^*) = \pm \frac{4i}{W} + O(W^{-2}) \neq 0.$$

Deform the contour to the steepest-descent paths through $z_{\pm}^*$; this is legitimate because b decays super-exponentially as $\operatorname{Re} z \rightarrow +\infty$ and $\csc$ suppresses $|\Im z| \rightarrow \infty$. A direct Stirling evaluation shows that at the modulation-free level $B \equiv 0$ the two saddles sum to $\cos W$ *exactly*: the divergent factors $e^{\pm \pi W/2}$ from $\csc(\pi z_{\pm}^*)$ and from $1/\Gamma(\pm iW + 1)$ cancel identically, each saddle contributing $\frac{1}{2} e^{\pm iW}$. Restoring the modulation scales each saddle by $e^{-B_{z_{\pm}^*}}$; since $P_1(2z) = \frac{2z(2z+1)(4z+1)}{6}$ and $\varphi_1 = \frac{1}{24}$,

$$B_{z_+^*} = \varphi_1 \tau^2 (P_1(iW) - \frac{iW}{2}) + O(\tau) \quad (\text{the } \tau^{3/2} \text{ part of the remainder is (9); the } O(\tau) \text{ also absorbs } B_{z_+^*} - B_{iW/2} \text{ and})$$

(using $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau} e^{-3\tau/2}$, the factor $e^{-3\tau/2}$ being absorbed in the $O(\tau)$), and $B_{z_-^*} = \overline{B_{z_+^*}}$. Hence

$$S_e = \frac{1}{2} e^{-B_{z_+^*}} e^{iW} + \frac{1}{2} e^{-B_{z_-^*}} e^{-iW} + E = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + E.$$

The remainder E is $O(\tau)$ by the classical second-order estimate for a simple saddle [34, Ch. 4, Thm. 7.1]: parametrising the descent path by arclength s , the integrand equals $e^{h(z^*)} e^{-s^2} (1 + O(sW^{-1/2}) + O(\tau))$, the next saddle coefficient is $\propto h'''(z^*) = O(W^{-2}) = O(\tau)$, and the bounded-variation hypothesis on the amplitude e^{-B_z} along the path is supplied by Lemma 7. This citation is the proof's one non-elementary step. The saddle prediction matches S_e to $O(\tau)$ over $\tau = 0.02, 0.01, 0.005$ numerically (`saddle_verify.py`). Thus $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau)$, with $\sin W = \sin w + O(\sqrt{\tau})$. $\square$

Lemma 10 (absolute bound on T_2). *For $\tau \leq 0.02$, $|T_2| \leq 0.17 \tau^{1/4}$; in particular $|T_2| \leq 0.064 < 1$ there, and $|T_2| \rightarrow 0$ as $\tau \rightarrow 0$. The argument uses no saddle, no path through a saddle, and no cited asymptotic theorem: only the Mellin–Barnes residues, a rectangle contained in the strip $\mathcal{S}$ of Lemma 7, and the amplitude bounds proved there. The same holds at any argument X with $\tau X^2 \in [2e^{-\tau}, 2]$; the constant above is the one for $X = w$, which is the case Lemma 12 uses and the worse of the two.*

Proof. Since $\operatorname{Res}_{s=n} \pi / \sin \pi s = (-1)^n$ and $h(s) = X^{2s} g_s / \Gamma(2s + 1)$ is analytic on $\operatorname{Re} s > -\frac{1}{2}$ (Lemma 7(i)), the terms of (5) are the residues of $h(s) \pi / \sin \pi s$ at $s = 1, 2, \dots$. Split the series at $n = \lfloor 2w \rfloor$.

The tail. At integers, $B_n = \sum_{i \leq 2n} \varphi(i\tau) - n\varphi(\tau) \geq n\varphi(\tau) \geq 0$ by (7) and the monotonicity of φ , so $0 \leq g_n = 1 - e^{-B_n} \leq 1$ and $|\sum_{n > \sigma_*} (-1)^n g_n X^{2n} / (2n)!| \leq \sum_{n > \sigma_*} X^{2n} / (2n)!$. This is a sawtooth in τ , jumping each time $2w$ crosses an integer, with a superexponentially decreasing envelope: $7.5 \cdot 10^{-10}$ at $\tau = 0.02$ and $1.9 \cdot 10^{-11}$ at $\tau = 0.0127$. It is bounded together with the contour integral below and is nowhere the binding term.

The head. The remaining residues are enclosed by the rectangle $R = \{\frac{1}{2} \leq \operatorname{Re} s \leq \sigma_*, |\Im s| \leq X/2\}$, where σ_* is the largest half-odd-integer below $2w$ (so $|\sin \pi s| = \cosh \pi t \geq 1$ on the right edge; taking the edge at $2w$ itself would put it through a pole whenever $2w \in \mathbb{Z}$, e.g. at $\tau = 0.02$), which

lies *wholly inside* $\mathcal{S}$, so the amplitude bounds of Lemma 7 apply at every point of ∂R ; no bound on B_s outside $\mathcal{S}$ is needed anywhere. On $\operatorname{Re} s = \frac{1}{2}$ we have $|\sin \pi s| = \cosh \pi t \geq 1$, so no pole is grazed. On the horizontal edges the two exponentials cancel: $|\pi/\sin \pi s| \leq \pi/\sinh(\pi X/2) \leq 2.01\pi e^{-\pi X/2}$ (valid once $X \geq 1.69$, hence throughout $\tau \leq 0.02$, where $X \geq 9.9$), while $1/|\Gamma(2\sigma + 1 + iX)|$ carries the compensating $e^{+\pi X/2}$; the resulting kernel is $O(\sqrt{2\pi/X})$, attaining 0.7887 at $\sigma = \frac{1}{2}$ against $\sqrt{2\pi/X} = 0.7927$ when $\tau = 0.02$.

For the amplitude: where $|2s| \leq 2w$, (9) gives $|B_s| \leq \frac{\tau^2}{24} \left| \frac{M^3}{3} + \frac{M^2}{2} - \frac{M}{3} \right| + 0.02\tau^{3/2}$ with $M = 2s$, hence $|g_s| = |1 - e^{-B_s}| \leq |B_s|e^{|B_s|}$; elsewhere on ∂R , $|g_s| \leq 1 + e^{30.3\sqrt{\tau}}$ by (8). Write $\mathcal{B}(\tau, X)$ for the resulting majorant: the boundary integral of those bounds against $|X^{2s}/\Gamma(2s+1)| |\pi/\sin \pi s|$, divided by 2π , plus the tail. The prose above fixes the geometry; what remains is to bound $\mathcal{B}$ on the whole region $\tau \in (0, 0.02]$, $\tau X^2 \in [2e^{-\tau}, 2]$, and that is done in two ranges.

(1) $\tau \geq 2/101^2$: *verified enclosure*. On $w = \sqrt{2/\tau} \in [10, 101]$, $X \in [we^{-\tau/2}, w]$, the majorant is enclosed in MPFR interval arithmetic with outward directed rounding at every operation, by a branch and bound on the parameter box: a box is accepted when the enclosed supremum of $\mathcal{B}$ over the whole box does not exceed the infimum of $0.17\tau^{1/4}$ on it, and is bisected otherwise. The contour integrals are enclosed cellwise, each cell contributing its integrand enclosure times its width, so what is returned bounds the integral rather than estimating it; the cells tile each edge exactly, and the initial boxes are cut at the half-integer steps of σ_* , so no box straddles a jump of the contour. The scan terminates with 232 accepted boxes covering the range and no failures, and returns $\mathcal{B} \leq 0.16978\tau^{1/4}$, a box-level figure comparing the largest bound on a box with the smallest $\tau^{1/4}$ on it. Run at 96 and again at 160 bits it produces the same box decomposition and the same figure. Pointwise at $\tau = 0.02$, $X = w = 10$, where the cross-check below locates the maximum of the ratio, the enclosure is $\mathcal{B} \leq 0.061685 = 0.164031\tau^{1/4}$, and the scan repeated against the constant 0.166 in place of 0.17 also terminates without failures. (`tools/t2abs_iv`; the earlier adaptive-quadrature certificate `t2abs_certificate.py` shares no code with it and is kept as a cross-check, its seven tabulated ratios 0.1623, 0.0428, 0.0308, 0.0226, 0.0173, 0.0136, 0.0115 at $\tau = 2 \cdot 10^{-2}, \dots, 10^{-5}$ being reproduced from above.)

(2) $\tau < 2/101^2$: *explicit constants*. No finite scan reaches $\tau \rightarrow 0$. Here $X \geq 101e^{-1/101^2} > 100$, $w \leq Xe^{1/X^2} \leq 1.0001X$ and $\tau \leq 2/X^2$, so $0.17\tau^{1/4} = 0.17 \cdot 2^{1/4}w^{-1/2} \geq 0.2021X^{-1/2}$, and it suffices to bound $\mathcal{B}$ by that.

On the left edge $2s + 1 = 2 + 2it$, and $|\Gamma(1 + iy)|^2 = \pi y/\sinh \pi y$ makes the kernel exact: $|X^{2s}/\Gamma(2s+1)| |\pi/\sin \pi s| = X\pi\sqrt{g(\pi t)}/\sqrt{1+4t^2}$ with $g(u) = \tanh u/u \leq \min(1, 1/u)$, hence at most $X\pi$ for $t \leq 1/\pi$ and at most $\frac{1}{2}X\sqrt{\pi}t^{-3/2}$ beyond. There $|M| = \sqrt{1+4t^2} \leq 1+2t \leq 1+X < 2w$, so (9) applies along the whole edge; with $\frac{(1+2t)^3}{3} + \frac{(1+2t)^2}{2} + \frac{1+2t}{3} = \frac{8}{3}t^3 + 6t^2 + \frac{14}{3}t + \frac{7}{6}$ integrated against $t^{-3/2}$, and collecting the range $t \leq 1/\pi$ and the $0.02\tau^{3/2}$ remainder on the same power of X , the edge contributes $L \leq 0.03033X^{-1/2}$ for $X \geq 100$.

On a horizontal edge put $a = 2\sigma + 1$, $u = a/X$ and $\psi(u) = u - \arctan u - \frac{u}{2}\log(1+u^2)$, so $\psi'(u) = -\frac{1}{2}\log(1+u^2)$ and ψ is negative, decreasing and concave on $u > 0$. Stirling with Binet's remainder, $\log \Gamma(z) = (z - \frac{1}{2})\log z - z + \frac{1}{2}\log 2\pi + \mu(z)$ with $|\mu(z)| \leq \mu(\operatorname{Re} z) \leq 1/(12\operatorname{Re} z)$ (the kernel of Binet's first integral is positive on $(0, \infty)$), cancels every term of size $a \log X$ and leaves

$$\frac{X^{2\sigma}}{|\Gamma(a + iX)|} \cdot \frac{\pi}{\sinh(\pi X/2)} = \frac{\sqrt{2\pi}}{1 - e^{-\pi X}} X^{-1/2}(1+u^2)^{1/4} e^{X\psi(u)} e^{-\operatorname{Re} \mu}, \quad |\operatorname{Re} \mu| \leq \frac{1}{12a} \leq \frac{1}{24}. \quad (12)$$

With $d\sigma = \frac{X}{2}du$ and $|M| \leq X\sqrt{1+u^2}$, (9) applies for $u \leq \sqrt{3}$, where $|M| \leq 2X \leq 2w$, and gives $|g_s| \leq 1.00449[(1+u^2)^{3/2}/(18X) + 0.33502X^{-2}]$; for $u > \sqrt{3}$ only $|g_s| \leq 1 + e^{30.3\sqrt{\tau}} \leq 2.5352$ is used. Two elementary inequalities close the u -integral: $\psi(u) \leq -u^3/12$ on $[0, 1]$, since $\log(1+x) \geq x/2$

there; and $\psi(u) \leq \psi(1) + \psi'(1)(u-1)$ for $u \geq 1$, by concavity. With $\int_0^\infty e^{-Xu^3/12} du = \Gamma(\frac{4}{3})(12/X)^{1/3}$ and $(1+u^2)^{7/4} \leq 2^{7/4}$ on $[0, 1]$, one edge contributes $H \leq 0.50142 X^{-5/6} + 1.06902 X^{-11/6} + 4.5 \cdot 10^{-9} e^{-0.131(X-100)}$.

The right edge and the tail are exponentially small. On $\operatorname{Re} s = \sigma_*$ one has $|\pi/\sin \pi s| \leq \pi$ and $a = 2\sigma_* + 1 > 4X - 1$, while $\log |\Gamma(a + iY)| \geq (a - \frac{1}{2}) \log a - Y^2/a - a + \frac{1}{2} \log 2\pi - \frac{1}{12a}$ for $|Y| \leq X$; since $c \mapsto c(1 - \log c)$ decreases and $a/X \in [3.99, 4.011]$, the kernel there is at most $\pi e^{-1.2805X}$. And $n_0 = \lfloor \sigma_* \rfloor + 1 > 2X - 1$ with $(2n)! \geq (2n/e)^{2n}$ makes the tail at most $1.07 e^{-1.51X}$. Both are below 10^{-50} throughout. Collecting, for $X \geq 100$,

$$\mathcal{B} \leq 0.00966 X^{-1/2} + 0.15961 X^{-5/6} + 0.34028 X^{-11/6},$$

which is at most $0.2021 X^{-1/2}$ as soon as $0.15961 X^{-1/3} + 0.34028 X^{-4/3} \leq 0.2021 - 0.00966 = 0.19244$; the left side decreases and equals 0.03512 at $X = 100$. The constants of this range are regenerated, and the two ψ inequalities and the rearrangement (12) independently checked against `mpmath`'s complex Γ , in `t2abs_smalltau.py` at 40 and again at 60 digits.

At $\tau = \tau_4 = 0.012665$, the largest value Lemma 12 uses, the bound is 0.0144 , and what Lemma 12 consumes there is only $|T_2| < 1$. $\square$

Remark 11 (why Remark 9 is not used). The steepest-descent sketch is recorded because it produces the sharp constant, but it is not a proof, on three counts. (i) It asserts that the two saddles sum to $\cos W$ *exactly*; they do not: the leading saddle term carries a phase $e^{-i/(24W)}$, so the two sum to $\cos W + \frac{1}{24W} \sin W + O(W^{-2})$. (ii) It sizes the first saddle correction by $h'''(z^*) = O(W^{-2})$; but the saddle is *flat* ($h'' = 4i/W \rightarrow 0$), so the expansion parameter is W^{-1} and the correction is $c_1 = i/(24W) + O(W^{-2})$. Errors (i) and (ii) are each about three quarters of the quantity being computed and cancel exactly, which is why the stated constant is nevertheless right; taken literally the chain yields $\sqrt{2}/144$, four times too small. (iii) The deformation is justified by the claim that $\csc$ suppresses $|\Im z| \rightarrow \infty$, which is false for the exact B , as the numbers in the previous proof show; and the uniform remainder is attributed to a steepest-descent error estimate whose hypotheses (a parameter-independent phase and amplitude, an explicit path, an explicit truncation order) are not verified here. A complete proof needs the two-term flat-saddle expansion together with an error estimate on the curved path; that is not written, and nothing in this paper waits on it.

Lemma 12 (infinitude and accumulation of the travel poles). *The set $\{q \in (0, 1) : \Sigma_1^T(q) = 1\}$ is infinite, and together with Lemma 74 it accumulates exactly at $q = 1$. In particular there is a travel pole with $w \in (m\pi, (m+1)\pi)$ for every large m .*

Proof. By Proposition 17, $1 - \Sigma_1^T = \cos(Z; q^2)$ with $Z^2 = z_0^2 e^\tau$. The collapse above defines the radius-free coefficients $e^{-B_n} = q^{n^2+n} [2(1-q)]^n (2n)! / [(q; q)_{2n} W^{2n}]$ through $S_e = \cos(z_0; q^2) = \sum_{n \geq 0} (-1)^n e^{-B_n} W^{2n} / (2n)!$; replacing z_0^2 by $Z^2 = z_0^2 e^\tau$ multiplies the n -th coefficient by $e^{n\tau}$, i.e. replaces W by $We^{\tau/2} = w$ exactly:

$$1 - \Sigma_1^T = \sum_{n \geq 0} (-1)^n e^{-B_n} \frac{w^{2n}}{(2n)!} = \cos w - T_2(w),$$

with $T_2(\cdot)$ the series (5) evaluated at argument w in place of W ; the coefficients e^{-B_n} are the *same* ones, so no re-derivation of the amplitude is involved. Only the argument moves, and it moves inside the region where the two estimates are already proved: Lemma 7 is stated on a strip built from the larger argument w (and containing the W -strip), so it applies verbatim, and Lemma 10 is stated for any argument with $\tau X^2 \in [2e^{-\tau}, 2]$, hence at $X = w$, where $\tau w^2 = 2$ identically. Thus

$|T_2(w)| \leq 0.064$ once $\tau \leq 0.02$, which along the extreme-phase sequence $w = m\pi$, $\tau^{(m)} = 2/(m\pi)^2$, holds for every $m \geq 4$ (there $\tau \leq \tau_4 = 0.012665$ and the bound is 0.0144). Along that sequence

$$1 - \Sigma_1^T|_{w=m\pi} = (-1)^m - T_2(m\pi),$$

which has the sign of $(-1)^m$ for all $m \geq m_0$, with $m_0 = 4$ from Lemma 10 (only $|T_2(m\pi)| < 1$ is used, a far weaker input than the sharp size recorded in Remark 9). By continuity of $q \mapsto \Sigma_1^T(q)$ the intermediate value theorem yields a zero of $1 - \Sigma_1^T$ with $w \in (m\pi, (m+1)\pi)$ for every $m \geq m_0$: infinitely many travel poles, with $\tau \downarrow 0$, i.e. $q \uparrow 1$. Since every travel pole has $q > \frac{1}{10}$ and the pole set is locally finite in $(0, 1)$ (Lemma 74), its only accumulation point is $q = 1$. $\square$

The alternation is in fact immediate: $|T_2(\pi)| \leq 0.027$ by direct evaluation and $|T_2(m\pi)| = O(\tau_m)$ thereafter, so $m_0 = 1$ and there is a travel pole in every window $(m\pi, (m+1)\pi)$, $m \geq 1$. The dominant pole q_1 is the one below this ladder, in $w \in (0, \pi)$. The transport identity and the sign alternation (checked for $m \leq 40$, worst $|T_2(m\pi)| = 0.0267$) are verified in `star_gate_certificate.py`.

Lemma 13 (infinitude of the zeros of the bulk block). *The zero set of S_e in $(0, 1)$, equivalently $\{q \in (0, 1) : S_1(q) = 1\}$, is infinite.*

Proof. Here no transport is needed: $S_e = \cos W - T_2$ at $W = we^{-\tau/2}$ is (5) itself, the series for which both estimates were set up. The map $\tau \mapsto W(\tau) = \sqrt{2/\tau} e^{-\tau/2}$ is continuous and strictly decreasing on $(0, \infty)$ with $W \rightarrow \infty$ as $\tau \downarrow 0$, so each integer $m \geq 4$ gives a unique $\tau^{[m]}$ with $W(\tau^{[m]}) = m\pi$; since $W < w$, $\tau^{[m]} < 2/(m\pi)^2 \leq \tau_4 = 0.012665$, and $\tau W^2 = 2e^{-\tau}$ identically, so Lemma 10 applies at every one of them and only $|T_2| < 1$ is used. Then $S_e = (-1)^m - T_2$ has the sign of $(-1)^m$ along the sequence, and the $\tau^{[m]}$ decrease in m , so the intermediate value theorem applied to the continuous function $q \mapsto S_e(q)$ (Theorem 2, first clause) gives a zero in each of the disjoint intervals $(e^{-\tau^{[m]}}, e^{-\tau^{[m+1]}})$, $m \geq 4$. Finally $S_e = 1 - S_1$ by Proposition 17. $\square$

The zeros of S_e are not the travel poles: those are the zeros of $\cos(Z; q^2)$, which is the same function a half q -step away (Proposition 17), and S_e is bounded away from 0 at every travel pole by Proposition 65. Lemma 13 is used only in Corollary 3. The alternation it asserts is checked directly for $4 \leq m \leq 40$ in `f2_pole_route.py`, at 60 and again at 100 digits: the series for S_e cancels heavily against intermediate terms far larger than its value, and 30 digits already returns wrong signs there.

Remark 14 (the Hubbard–Stratonovich integral representation). Writing $q^{j(j+1)} = q^j e^{-j^2\tau}$ and linearising the Gaussian through $e^{-j^2\tau} = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du$, the denominator block admits the *exact* real integral

$$S_e = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} e^{-u^2/(4\tau)} \Psi(u, q) du, \quad \Psi(u, q) = \sum_{j \geq 0} \frac{(-2(1-q)q e^{iu})^j}{(q; q)_{2j}}, \quad (13)$$

the interchange of sum and integral being justified by absolute convergence ($\sum_j |Z|^j / (q; q)_{2j} < \infty$, $Z = -2(1-q)q e^{iu}$, against the integrable Gaussian). Because the coefficients of Ψ in $\zeta = e^{iu}$ are real, $\Psi(-u, q) = \overline{\Psi(u, q)}$, so (13) is real and may be written over a single variable, $S_e = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} \operatorname{Re} \Psi du$; the exact Gaussian moment underlying it, $\int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du = \sqrt{4\pi\tau} e^{-\tau j^2}$, is machine-checked in Lean 4 (`GaussHS.lean`, theorem `gaussian_moment`; axioms `propepxt`, `Classical.choice`, `Quot.sound`, no sorry).

This representation, though exact and convenient, is *not* an analytic shortcut. The amplitude $\Psi(\cdot, q)$ is entire in ζ , but it is *not* uniformly bounded as $q \rightarrow 1^-$: on any fixed window it grows like $\cosh(w \sin(u/2))$ (numerically $|\Psi|$ already exceeds 300 at $u = 1$, $\tau = 10^{-2}$), and its termwise limit

$\cos(w e^{iu/2})$ differs from Ψ by an $O(1)$ factor at the dominant scale $j \sim \tau^{-1/2}$ (where the subleading part of $(q; q)_{2j}$ is itself $O(1)$; numerically $\Psi/\cos(w e^{iu/2}) \approx 0.62$). The size of S_e is therefore governed by the balance between this exponential growth and the Gaussian; a simple *complex* saddle at $u^* \approx -\sqrt{2}\tau$, of the same order of difficulty as the direct sum, *not* a bounded-amplitude stationary-phase integral with trivial hypotheses [46, Ch. II]. In particular the bounded-variation input of Lemma 7 is *not* eliminated by passing to (13); it, together with the rectangle bound of Lemma 10, remains the analytic core. The value of (13) is representational (one real variable, an entire integrand, an exact Lean-checked moment), not a reduction of hypotheses; the numerical agreement of its saddle value with S_e is recorded in `gaussint_verify.py`.

Remark 15 (an elementary derivation of the coefficients of Remark 9). The leading term and the constant $\sqrt{2}/36$ of Remark 9 can be obtained without the steepest-descent theorem, by an exact algebraic reduction. Writing $1 - e^{-k\tau} = 2e^{-k\tau/2} \sinh(k\tau/2)$ throughout, the j -th term of $S_e = \sum_{j \geq 0} (-2(1-q))^j q^{j(j+1)} / (q; q)_{2j}$ collapses to

$$S_e = \sum_{j \geq 0} (-1)^j e^{-j\tau} \frac{\sinh^j(\tau/2)}{\prod_{k=1}^{2j} \sinh(k\tau/2)} = \sum_{j \geq 0} (-1)^j \frac{(2/\tau)^j}{(2j)!} e^{-j\tau + E_j}, \quad E_j = \frac{j\tau^2}{24} - \frac{\tau^2}{144} (2j)(2j+1)(4j+1) + O(\tau^4 j^5),$$

E_j being the exact $\sinh$ -curvature defect. Two elementary facts finish the extraction. First, the factor $e^{-j\tau}$ resums *in closed form*: $\sum_j (-1)^j (2e^{-\tau}/\tau)^j / (2j)! = \cos W$ with $W = \sqrt{2/\tau} e^{-\tau/2}$, which is exactly the frequency of Remark 9; so the phase correction needs no estimate at all. Second, the entire-function moment identity $\sum_j (-1)^j y^j j^m / (2j)! = (y d/dy)^m \cos \sqrt{y}$ evaluates the curvature term through its leading part $-\frac{\tau^2}{9} (y d/dy)^3 \cos \sqrt{y} = -\frac{\tau^2}{9} \cdot \frac{W^3}{8} \sin W = -\frac{\sqrt{2}}{36} \sqrt{\tau} \sin W$ (using $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau} e^{-3\tau/2}$, the factor $e^{-3\tau/2}$ being absorbed in the $O(\tau)$), giving

$$S_e = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau), \quad W = \sqrt{2/\tau} e^{-\tau/2},$$

all constants explicit and verified to ten digits (`elemcos_verify.py`). Carried to higher moments the same reduction produces every subsequent coefficient as a finite combination of $W^k \sin W$, $W^k \cos W$, so it serves the $O(\tau^2)$ -relative demand of the U -numerator in the same elementary terms. The one input it does *not* render elementary is the remainder estimate; that the alternating expansion is asymptotic with $O(\tau)$ error. That step is a *cancellation* statement: not only does the triangle inequality on $\sum_j (2/\tau)^j / (2j)! |\cdot|$ return $\cosh W \asymp e^{\sqrt{2/\tau}}$, but the partial sums of the $\cos W$ series themselves swing up to $\asymp e^{c\sqrt{2/\tau}}$ before cancelling to $O(1)$ (numerically $\max_J |S_J| = 9, 48, 1459, 7.6 \times 10^4$ at $\tau = 0.1, 0.05, 0.02, 0.01$). Hence *every* partial-summation method (Abel, summation-by-parts, Euler transformation) fails, since each is controlled by these exponentially large partial sums; only a global resummation captures the cancellation. This global resummation is exactly the two-saddle steepest-descent computation carried out self-contained in the proof of Remark 9: the saddles $z_{\pm}^* = \pm iW/2$ supply the cancellation the partial sums cannot. Together, the elementary extraction here and that saddle argument reduce Remark 9 to Cauchy's theorem, Stirling's formula, elementary Gaussian integration, the bounded-variation bound of Lemma 7, and *one* cited estimate: the classical remainder bound for a simple saddle [34, Ch. 4, Thm. 7.1], used once in the proof of Remark 9. It is not independent of that citation, and we do not claim otherwise; Appendix A provides the route that is.

Remark 16 (derivation of the U -numerator amplitude). The sharp picture behind Theorem 80: the asymptotic value $|\sum_k d_k| = \frac{3}{8\sqrt{2}} \tau^{5/2} (1 + O(\sqrt{\tau}))$ at the travel poles, on the one explicit series $P_{12} = \frac{2q^3}{1-q^3} \sum_k d_k$, $d_k = (-2)^k (1-q)^k q^{k^2+3k} / [(q^2; q^2)_k (q^5; q^2)_k]$, which submits to the identical

reduction. Writing $1 - q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$ throughout collapses the terms *exactly* to

$$d_k = (-1)^k e^{-k\tau} \frac{\sinh^{k+1}(\frac{\tau}{2}) \sinh(\frac{3\tau}{2}) \sinh((k+1)\tau)}{\prod_{m=1}^{2k+3} \sinh(\frac{m\tau}{2})},$$

and the elementary expansion (exact linear absorption into the argument, Faulhaber sums for the sinh-curvature, the moment identities $k^m \mapsto (y d/dy)^m$) yields, with $\Phi(y) = (\sin W - W \cos W)/(2W^3)$, $W = \sqrt{y}$, $D = y d/dy$,

$$\sum_k d_k = 6 \left[\Phi + c_3 D^3 \Phi + c_2 D^2 \Phi + \frac{c_3^2}{2} D^6 \Phi + (c_5 + c_2 c_3) D^5 \Phi + \frac{c_3^3}{6} D^9 \Phi \right] (y_*) + O(\tau^3),$$

$y_* = \frac{2}{\tau} e^{-\tau-23\tau^2/36}$, $c_2 = -\frac{5}{12}\tau^2$, $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$; *every constant an exact rational, derived, not fitted* (`elemY3_verify.py`; the error is verified $O(\tau^3)$ over $\tau \in [0.005, 0.08]$ and the formula reproduces the exact series at the poles $m = 7, \dots, 33$ to the displayed accuracy). At the travel poles the expansion gives $|\sum_k d_k|/\tau^{5/2} \rightarrow 3/(8\sqrt{2}) = 0.2652$, i.e. exactly the observed gate value $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ with its fourfold margin: structurally, the pole sits at offset δw from the zero of the displayed expansion, and the constant $3/(8\sqrt{2})$ arises as $(3\tau/2) \delta w / \tau^{5/2}$.

The *phase-matching step* that fixes δw is carried out by the same machinery. The pole equation $\Sigma_1(q_m) = 1$ collapses exactly to $1 - \Sigma_1 = \sum_{j \geq 0} (-1)^j \sinh^j(\frac{\tau}{2}) / \prod_{m=1}^{2j} \sinh(\frac{m\tau}{2})$; the S_e -series stripped of its $e^{-j\tau}$ phase; so the identical expansion applies with $y_* = \frac{2}{\tau} e^{\tau^2/36}$, $c_2 = -\frac{1}{12}\tau^2$ and the *universal* $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$. Solving the two phase conditions perturbatively in the common variable $w = \sqrt{2/\tau}$ near each grid point $(M + \frac{1}{2})\pi$ gives

$$\chi_{\text{pole}} = -\frac{\sqrt{2}}{36} \sqrt{\tau} + \frac{41\sqrt{2}}{2400} \tau^{3/2} + O(\tau^2), \quad \chi_{\text{zero}} = -\frac{\sqrt{2}}{36} \sqrt{\tau} - \frac{259\sqrt{2}}{2400} \tau^{3/2} + O(\tau^2).$$

The $\sqrt{\tau}$ phases agree *identically*; both are produced by the universal cubic curvature c_3 , and the constant $\sqrt{2}/36$ is that of Remark 9; this is precisely why the travel poles track the zeros of $Y_3(1)$. The $\tau^{3/2}$ terms differ by exactly

$$\delta w = \chi_{\text{pole}} - \chi_{\text{zero}} = \frac{\sqrt{2}}{8} \tau^{3/2} + O(\tau^2), \quad \text{whence} \quad \left| \sum_k d_k \right| = \frac{3\tau}{2} \delta w (1 + O(\sqrt{\tau})) = \frac{3}{8\sqrt{2}} \tau^{5/2} (1 + O(\sqrt{\tau})),$$

and therefore $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.17678 < 1/\sqrt{2}$: *the numerator gate, derived, with its fourfold margin*. Verification: the two phase series match the exact pole and zero locations to seven digits at $m = 11, 17, 23, 29$, and $\delta w / \tau^{3/2} = 0.17699, 0.17687, 0.17683, 0.17681 \rightarrow \sqrt{2}/8 = 0.176777$ (`phase_match_verify.py`); the pole-side series independently reproduces the McMahon-type coefficients of the travel-pole expansion. The remainder discipline throughout is that of Remark 9: every coefficient extracted elementarily, the $O(\cdot)$ terms controlled by the same two-saddle argument.

4.1 The travel-pole equation as a q -trigonometric identity

The two series that govern the poles admit an exact identification with the q -cosine of Koornwinder and Swarttouw [23], $\cos(z; q^2) := {}_1\phi_1(0; q; q^2, q^2 z^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j} / (q; q)_{2j}$.

Proposition 17. *Let $z_0 = \sqrt{2(1-q)}$ and $Z = z_0/\sqrt{q}$. Then*

$$1 - S_1 = S_e = \cos(z_0; q^2), \quad 1 - \Sigma_1 = \cos(Z; q^2).$$

In particular the travel poles q_m are exactly the solutions of $\cos(Z; q^2) = 0$, and S_e is the same function evaluated one half q -step below its own zero.

Proof. Collapse the k -recursion (2) on its two odd ladders, the travel one carrying Σ_1 and the bulk one carrying S_1 . Write $1 - q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$ in the sinh-forms of the two series (§4): the products collapse to $1 - \Sigma_1 = \sum_j (-1)^j (2(1 - q))^j q^{j^2} / (q; q)_{2j}$ and $1 - S_1 = \sum_j (-1)^j (2(1 - q))^j q^{j^2+j} / (q; q)_{2j}$, the second being S_e as defined in (1). Comparing with the series for $\cos(z; q^2)$ term by term forces $q^2 z^2 = 2(1 - q)q$, i.e. $z = Z$, in the first case and $q^2 z^2 = 2(1 - q)q^2$, i.e. $z = z_0$, in the second; both series converge for $|q| < 1$, the factor q^{j^2} dominating. Both collapses are recomputed as identities of integer power series, to degree 1160 in `blocks_growth.py` and at two truncation degrees in `f2_pole_route.py`. The per-term identity is machine-checked in Lean 4 (`GramCasoratian.lean`, theorem `sigma1_term`; axioms `propext`, `Classical.choice`, `Quot.sound`, no sorry). $\square$

Remark 18. The identification is consistent with the pole data: at all 58 tabulated poles the computed value of $\cos(z_0; q^2)$ agrees with the stored S_e to at least thirty digits, and $|S_e|/\sqrt{\tau}$ lies in $[0.698, 0.708]$, approaching $1/\sqrt{2}$ (`qtrig_verify.py`). The half-step structure accounts for the size of the gate quantity: $Z - z_0 = z_0(1 - \sqrt{q})/\sqrt{q} = \tau^{3/2}/\sqrt{2} + O(\tau^{5/2})$, the pole condition annihilates $\cos(\cdot; q^2)$ at Z , and the q -derivative rule $(1 - q)D_{q,z} \sin(z; q^2) = \cos(z; q^2)$ of [23] gives the function slope $1/(1 - q) \sim 1/\tau$ near the zero, so the sampled value is of size $(\tau^{3/2}/\sqrt{2})(1/\tau) = \sqrt{\tau/2}$, the observed constant. Turning this observation into a proved lower bound on $|S_e(q_m)|$ independent of Remark 9 requires uniform control of $\cos'(\cdot; q^2)$ between z_0 and Z as $q \rightarrow 1^-$. That control is supplied by Appendix A (Lemma 69 bounds $\max |f''|$ on $[qZ, Z]$, and Proposition 65 converts it into the lower bound), independently of Remark 9.

4.2 The numerators, and why they do not vanish at the poles

Proposition 17 identifies the two *denominators*. The two numerators are the companion q -sine at the same two arguments, and that is what settles their non-vanishing at the travel poles. Throughout, $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$ is the half-step q -sine of Lemma 72, $\sin(z; q^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j+1} / (q; q)_{2j+1}$.

Lemma 19 (the numerators in closed form). *Identically in $0 < q < 1$,*

$$\Sigma_0 = \frac{2q}{Z} \mathfrak{s}(Z), \quad S_0 = \frac{2q}{z_0} \mathfrak{s}(z_0).$$

With Proposition 17 this identifies all four blocks as one q -trigonometric pair sampled at two arguments a half q -step apart: $(\Sigma_0, 1 - \Sigma_1)$ is $(\frac{2q}{u} \mathfrak{s}(u), \cos(u; q^2))$ at $u = Z$, and (S_0, S_e) is the same pair at $u = z_0 = \sqrt{q}Z$. In particular $S_0 = \frac{2q}{1-q} S_o$ with S_o the odd bulk block of Lemma 72, which is the second identity of the dictionary (30) read as a statement about the explicit series, and is therefore independent of the transfer model.

Proof. Collapse the k -recursion (2) on the even ladders, exactly as Proposition 17 does on the odd ones. From $\alpha_k = 2q^{k+1}/(1 - q^{k+1})$,

$$|\gamma_k| = \alpha_k - \alpha_{k+1} = \frac{2(1 - q)q^{k+1}}{(1 - q^{k+1})(1 - q^{k+2})}, \quad |C_k| = \frac{2(1 - q)q^{k+2}}{(1 - q^{k+1})(1 - q^{k+2})},$$

γ_k and C_k themselves being negative on $(0, 1)$, so both even ladders alternate. For the bulk one, $\sum_{i < j} (2i + 1) = j^2$ and the denominators telescope to $(q; q)_{2j}$, giving $\prod_{i < j} |\gamma_{2i}| = [2(1 - q)]^j q^{j^2} / (q; q)_{2j}$ and hence

$$S_0 = \sum_{j \geq 0} (-1)^j \alpha_{2j} \prod_{i < j} |\gamma_{2i}| = 2 \sum_{j \geq 0} (-1)^j \frac{[2(1 - q)]^j q^{(j+1)^2}}{(q; q)_{2j+1}}.$$

Write $q^{(j+1)^2} = q^{j^2+j}q^j$ and $[2(1-q)]^j q^j = (q^{1/2}z_0)^{2j}$: the sum is $(2q/(q^{1/2}z_0)) \sin(q^{1/2}z_0; q^2) = (2q/z_0)\mathfrak{s}(z_0)$. For the travel ladder, $\sum_{i<j}(2i+2) = j^2+j$ and the denominators telescope to $(q; q)_{2j}$ again, so $\prod_{i<j} |C_{2i}| = [2(1-q)]^j q^{j^2+j}/(q; q)_{2j}$ and, with $A_{2j} = 2q/(1-q^{2j+1})$,

$$\Sigma_0 = 2q \sum_{j \geq 0} (-1)^j \frac{[2(1-q)]^j q^{j^2+j}}{(q; q)_{2j+1}} = \frac{2q}{z_0} \sin(z_0; q^2) = \frac{2q^{3/2}}{z_0} \mathfrak{s}(Z) = \frac{2q}{Z} \mathfrak{s}(Z),$$

using $[2(1-q)]^j = z_0^{2j}$ and $z_0 = q^{1/2}Z$. The S_o statement is Lemma 72's $S_o = \frac{z_0}{2}\mathfrak{s}(z_0)$ combined with $z_0^2 = 2(1-q)$. Both collapses are the exponent bookkeeping machine-checked in `QTrigIdentities.lean` and are verified numerically off the poles, hence as identities, in `qnumerator_certificate.py`. $\square$

Lemma 20 (half-step rules and the pole invariant). *Identically in $0 < q < 1$, with $c(u) = \cos(u; q^2)$:*

- (i) $c(qu) = c(u) + q^{3/2}u \mathfrak{s}(q^{1/2}u)$;
- (ii) $\mathfrak{s}(qu) = \mathfrak{s}(u) - u c(q^{1/2}u)$;
- (iii) $c(u) c(q^{1/2}u) + q^{3/2} \mathfrak{s}(u) \mathfrak{s}(q^{1/2}u) = 1$.

Consequently, at every travel pole, where $c(Z) = 0$:

$$q^{3/2} \mathfrak{s}(z_0) \mathfrak{s}(Z) = 1,$$

so neither $\mathfrak{s}(Z)$ nor $\mathfrak{s}(z_0)$ vanishes there.

Proof. (i) and (ii) are termwise. From $(q; q)_{2j} = (q; q)_{2j-1}(1-q^{2j})$, $c(u) - c(qu) = \sum_{j \geq 1} (-1)^j q^{j^2+j} u^{2j} / (q; q)_{2j-1}$; shifting $j \mapsto j+1$ and factoring out $-qu$ leaves $-qu \sin(qu; q^2) = -q^{3/2}u \mathfrak{s}(q^{1/2}u)$, which is (i). From $(q; q)_{2j+1} = (q; q)_{2j}(1-q^{2j+1})$, $\mathfrak{s}(u) - \mathfrak{s}(qu) = q^{-1/2} \sum_{j \geq 0} (-1)^j q^{j^2+j} (q^{1/2}u)^{2j+1} / (q; q)_{2j}$; factoring out $q^{1/2}u$ leaves $u c(q^{1/2}u)$, which is (ii). This is the exponent matching machine-checked as `cshift_exponent` and `sshift_exponent`. For (iii), substitute (i) and (ii) into the Hahn–Exton q -Wronskian $\mathfrak{s}(u)c(qu) - c(u)\mathfrak{s}(qu) = u$ ([45, eq. (4.3.8)]; [33, eq. (4.21)]): the $\mathfrak{s}(u)c(u)$ terms cancel and what is left is $u[c(u)c(q^{1/2}u) + q^{3/2}\mathfrak{s}(u)\mathfrak{s}(q^{1/2}u)]$, so the q -Wronskian and (iii) are equivalent for $u \neq 0$. Identity (iii) is the q -Pythagorean identity of [23]. Taking $u = Z$ and $c(Z) = 0$ gives the pole invariant, which is the step already used inside the proof of Proposition 73. $\square$

Proposition 21 (the numerators do not vanish at the travel poles). *At every travel pole q_m ,*

$$\Sigma_0(q_m) S_0(q_m) = \frac{2q_m}{1-q_m} \neq 0. \quad (14)$$

In particular $\Sigma_0(q_m) \neq 0$ and $S_0(q_m) \neq 0$, at every travel pole, with no asymptotic input and no exceptional set. Moreover $|\Sigma_0(q_m)| |S_0(q_m)| \geq q_m w_m^2$, so at least one of the two numerators exceeds $\sqrt{q_m} w_m$ in modulus.

Proof. By Lemma 19, $\Sigma_0 S_0 = (4q^2/(z_0 Z)) \mathfrak{s}(Z) \mathfrak{s}(z_0)$, and by Lemma 20 the product $\mathfrak{s}(Z) \mathfrak{s}(z_0)$ equals $q^{-3/2}$ at a travel pole. Since $z_0 Z = z_0^2 / \sqrt{q} = 2(1-q) / \sqrt{q}$, the two powers of q combine to $4q^2 \cdot q^{-3/2} \cdot \sqrt{q} / (2(1-q)) = 2q / (1-q)$. The size statement follows from $1-q \leq \tau = 2/w^2$. $\square$

Remark 22. The two numerators are individually of size w_m : $|\Sigma_0(q_m)|/w_m$ and $|S_0(q_m)|/w_m$ rise monotonically to 1 over the first 32 travel poles, from 0.731 and 0.893 at the dominant pole $q_1 = 0.449453\dots$ to 0.99992 and 0.99997 at the last, with Σ_0 strictly alternating in sign (`qnumerator_certificate.py`, `mpmath` at two precisions, worst deviation in (14) $4.5 \cdot 10^{-28}$; the dominant pole is included, it lying below the extreme-phase ladder and being bracketed separately). This matches the asymptotic $\Sigma_0 \sim w \sin w$, the amplitude companion of $1 - \Sigma_1 \sim 1 - \cos w$; but (14) is an identity and the asymptotic is not used anywhere.

5 The transfer model and the block assembly

The travel block and the bulk block are one object seen twice. Both are the resolvent of the semiseparable kernel $q^{\max(a,b)}$ against a geometric diagonal weight, and a single three-term recursion in the section index k covers both: its coefficients are α_k, γ_k in the bulk case and A_k, C_k in the travel case, which is why the four series of (2) are four ladders on one recursion and not four separate constructions. This section develops that into the block assembly the rest of the paper consumes.

Sections 5.1–5.5 are unconditional statements about explicitly given operators, and §5.6 is an unconditional statement about the crossing optimisation that those operators encode. Section 5.7 states the strand-walk input (M) that identifies the one with the other; everything from there on is conditional on it. Section 5.8 isolates the one statement that remains open.

Throughout, $|q| < 1$ and $x = q^{1/2}$. Besides the four blocks of (2) we use the third ladder on the same recursion,

$$T_k = \sum_{j \geq 0} \alpha_{k+2j+1} \prod_{i < j} \gamma_{k+2i} \quad (k \geq 0). \quad (15)$$

Since $\gamma_k = -2q^{k+1}(1-q)/[(1-q^{k+1})(1-q^{k+2})]$ and $C_k = q\gamma_k$, on every compact subset of $|q| < 1$ there is a Ξ with $|\gamma_k|, |C_k| \leq \Xi|q|^{k+1}$, so $|\prod_{i < j} \gamma_{k+2i}| \leq \Xi^j |q|^{j^2+jk}$ and likewise for C ; as α_k and A_k are bounded there, S_k, Σ_k, T_k converge absolutely and locally uniformly and are holomorphic on $|q| < 1$ for every $k \geq 0$.

5.1 The section identity

Definition 23 (section identity). Fix $s_0 \in \{0, 1\}$ and $\vartheta \in \{0, 1\}$ with $s_0 + \vartheta = 1$, and put $e_s = 2q^{\vartheta+s}$ for $s \geq s_0$. Given a bounded *source* $\mathbf{c} = (c_s)_{s \geq s_0}$, a family $\Phi = (\Phi_s)_{s \geq s_0}$ satisfies the *section identity* with data $(s_0, \vartheta, \mathbf{c})$ if

$$\Phi_s = e_s \left(c_s + \sum_{s' \geq s_0} q^{\max(s, s')} \Phi_{s'} \right), \quad s \geq s_0. \quad (16)$$

Its *sections* are $G_k = \sum_{s \geq s_0} q^{ks} \Phi_s$, $k \geq 0$. In operator form $(I - M)\Phi = E$ with

$$M[s, s'] = e_s q^{\max(s, s')}, \quad E_s = e_s c_s. \quad (17)$$

The *bulk* instance is $(s_0, \vartheta) = (1, 0)$ and the *travel* instance is $(s_0, \vartheta) = (0, 1)$; we write M_0 for the bulk operator and $\mathcal{T}$ for the travel one.

Lemma 24 (existence, uniqueness, analyticity). For $|q| < \frac{1}{3}$ and any bounded $\mathbf{c}$, equation (16) has a unique solution in ℓ^1 , namely $\Phi = \sum_{n \geq 0} M^n E$, the series converging in ℓ^1 and coefficientwise in q , one extra power of q per term. One has $|\Phi_s| \leq C|q|^{\vartheta+s}$ with C independent of s , and each section G_k is holomorphic at $q = 0$.

Proof. $\|M\|_{\ell^1 \rightarrow \ell^1} = \sup_{s'} \sum_s |M[s, s']| \leq \sup_{s'} \sum_{s \geq s_0} 2|q|^{\vartheta+s} = 2|q|/(1-|q|) < 1$ for $|q| < \frac{1}{3}$, using $\vartheta + s_0 = 1$ and $|q|^{\max(s, s')} \leq 1$. Hence $I - M$ is invertible on ℓ^1 and the Neumann series converges. Every entry of M is divisible by $q^{\vartheta+s} q^{\max(s, s')}$, hence by q , so $M^n E$ is divisible by q^{n+1} and the series converges coefficientwise. Reading the s -th row of $\Phi = E + M\Phi$ gives $|\Phi_s| \leq 2|q|^{\vartheta+s} (\|\mathbf{c}\|_\infty + \|\Phi\|_{\ell^1})$. Since $|\Phi_s| = O(|q|^s)$, each G_k is a locally uniformly convergent sum of holomorphic functions. $\square$

5.2 One telescoping lemma for both blocks

Lemma 25 (catalytic telescoping). *Let $0 < |q| < 1$ and let $\Phi \in \ell^1$ satisfy (16) with data $(s_0, \vartheta, \mathbf{c})$. Put*

$$a_k = \frac{2q^{\vartheta+(k+1)s_0}}{1-q^{k+1}}, \quad b_k = \frac{2q^{\vartheta+k+2}}{1-q^{k+2}} - \frac{2q^{\vartheta+k+1}}{1-q^{k+1}}, \quad \mathcal{C}_k = \sum_{s \geq s_0} e_s c_s q^{ks},$$

and $\mathbf{a}_k = \sum_{j \geq 0} a_{k+2j} \prod_{i < j} b_{k+2i}$, $\mathbf{c}_k = \sum_{j \geq 0} \mathcal{C}_{k+2j} \prod_{i < j} b_{k+2i}$. Then for every $k \geq 0$

$$G_k = \mathcal{C}_k + a_k G_1 + b_k G_{k+2}, \quad (18)$$

the series $\mathbf{a}_k, \mathbf{c}_k$ converge, $G_k = \mathbf{c}_k + \mathbf{a}_k G_1$, and if $\mathbf{a}_1 \neq 1$ then

$$G_1 = \frac{\mathbf{c}_1}{1-\mathbf{a}_1}, \quad G_k = \mathbf{c}_k + \mathbf{a}_k \frac{\mathbf{c}_1}{1-\mathbf{a}_1} \quad (k \geq 0). \quad (19)$$

In the bulk instance $a_k = \alpha_k$, $b_k = \gamma_k$ and $\mathbf{a}_k = S_k$; in the travel instance $a_k = A_k$, $b_k = C_k$ and $\mathbf{a}_k = \Sigma_k$.

Proof. Split the inner sum of (16) at $s' = s$, where $\max(s, s')$ switches:

$$\Phi_s = e_s c_s + e_s \sum_{s' \geq s} q^{s'} \Phi_{s'} + e_s q^s \sum_{s_0 \leq s' < s} \Phi_{s'}.$$

Multiply by q^{ks} and sum over $s \geq s_0$. All three double sums converge absolutely, since $\Phi \in \ell^1$ and $|q| < 1$, so the orders of summation may be exchanged. The first gives $\mathcal{C}_k$. In the second, with $e_s = 2q^{\vartheta+s}$,

$$\sum_{s \geq s_0} 2q^{\vartheta+(k+1)s} \sum_{s' \geq s} q^{s'} \Phi_{s'} = 2q^{\vartheta} \sum_{s' \geq s_0} q^{s'} \Phi_{s'} \sum_{s=s_0}^{s'} q^{(k+1)s} = \frac{2q^{\vartheta}}{1-q^{k+1}} \sum_{s' \geq s_0} q^{s'} \Phi_{s'} (q^{(k+1)s_0} - q^{(k+1)(s'+1)}),$$

which is $a_k G_1 - \frac{2q^{\vartheta+k+1}}{1-q^{k+1}} G_{k+2}$, because $\sum_{s'} q^{s'} \Phi_{s'} = G_1$ and $\sum_{s'} q^{(k+2)s'} \Phi_{s'} = G_{k+2}$. In the third,

$$\sum_{s \geq s_0} 2q^{\vartheta+(k+2)s} \sum_{s_0 \leq s' < s} \Phi_{s'} = 2q^{\vartheta} \sum_{s' \geq s_0} \Phi_{s'} \sum_{s > s'} q^{(k+2)s} = \frac{2q^{\vartheta+k+2}}{1-q^{k+2}} G_{k+2}.$$

Adding the three gives (18). Iterating it J times,

$$G_k = \sum_{j < J} (\mathcal{C}_{k+2j} + a_{k+2j} G_1) \prod_{i < j} b_{k+2i} + G_{k+2J} \prod_{i < J} b_{k+2i}.$$

For fixed q one has $|b_m| \leq \Xi |q|^{m+1}$ with $\Xi = 4/(1-|q|)$, so $|\prod_{i < J} b_{k+2i}| \leq \Xi^J |q|^{J^2+Jk} \rightarrow 0$; and $|G_m| \leq \|\Phi\|_{\ell^1}$ uniformly in m , so the remainder vanishes as $J \rightarrow \infty$. The same bound, against $\sup_k |a_k| < \infty$ and $\sup_k |\mathcal{C}_k| \leq 2\|\mathbf{c}\|_{\infty}/(1-|q|)$, makes $\mathbf{a}_k$ and $\mathbf{c}_k$ converge. This gives $G_k = \mathbf{c}_k + \mathbf{a}_k G_1$; taking $k = 1$ gives $G_1(1-\mathbf{a}_1) = \mathbf{c}_1$, whence (19). The identifications of a_k, b_k are read off the displays: $(s_0, \vartheta) = (1, 0)$ gives $a_k = 2q^{k+1}/(1-q^{k+1}) = \alpha_k$ and $b_k = \alpha_{k+1} - \alpha_k = \gamma_k$, while $(s_0, \vartheta) = (0, 1)$ gives $a_k = 2q/(1-q^{k+1}) = A_k$ and $b_k = 2q^{k+3}/(1-q^{k+2}) - 2q^{k+2}/(1-q^{k+1}) = C_k$. $\square$

Remark 26 (where each feature of the model is used). Three features enter (18), each exactly once. That the coupling is $q^{\max(s, s')}$ is what makes the split at $s' = s$ produce two *geometric* sums; a symmetric kernel not of the form $\mu^{\max}$ destroys the three-term relation. That e_s is geometric in s is what makes those sums summable in closed form; a non-geometric edge weight leaves G_k coupled to a continuum of sections. That the source is geometric is used only to evaluate $\mathcal{C}_k$; both (18) and (19) hold for any bounded source. The step $k \mapsto k+2$, and with it the two-ladder structure of S_k and Σ_k , comes from the two occurrences of $q^{(k+2)s'}$ above.

Proposition 27 (the travel k -sum telescopes). *Let Φ solve the travel section identity, $(s_0, \vartheta, \mathbf{c}) = (0, 1, \mathbf{1})$. Then, as meromorphic functions on $|q| < 1$,*

$$G_k^T = \frac{\Sigma_k}{1 - \Sigma_1} \quad (k \geq 0), \quad \text{in particular} \quad G_0^T = \sum_{s \geq 0} \Phi_s = \frac{\Sigma_0}{1 - \Sigma_1}.$$

Proof. Lemma 25 with $(s_0, \vartheta) = (0, 1)$ and $c_s \equiv 1$ gives $\mathcal{C}_k = \sum_{s \geq 0} 2q^{1+s} q^{ks} = 2q/(1 - q^{k+1}) = A_k = a_k$, so $\mathbf{c}_k = \mathbf{a}_k = \Sigma_k$ and (19) reads $G_k^T = \Sigma_k + \Sigma_k \Sigma_1 / (1 - \Sigma_1) = \Sigma_k / (1 - \Sigma_1)$. Both sides are holomorphic near $q = 0$ and the right-hand side is meromorphic on $|q| < 1$, so the identity extends by analytic continuation. Combinatorially this is the Neumann series of Lemma 24: $\sum_s \Phi_s = \sum_{n \geq 0} \mathbf{1}^\top \mathcal{T}^n E$, whose n -th term is the generating function of the travel runs with exactly $n + 1$ edges. $\square$

Proposition 28 (the bulk dressing). *Let Φ solve the bulk section identity with $\mathbf{c} = \mathbf{1}$ and let Ψ solve it with $c_s = q^s$. Set*

$$b_0 = \sum_{s \geq 1} \Phi_s, \quad t_0 = \sum_{s \geq 1} q^s \Phi_s, \quad b_1 = \sum_{s \geq 1} \Psi_s, \quad t_1 = \sum_{s \geq 1} q^s \Psi_s.$$

Then, as meromorphic functions on $|q| < 1$,

$$b_0 = \frac{S_0}{1 - S_1}, \quad t_0 = \frac{S_1}{1 - S_1}, \quad t_1 = \frac{T_1}{1 - S_1}, \quad b_1 = T_0 + \frac{S_0 T_1}{1 - S_1}.$$

Proof. Lemma 25 with $(s_0, \vartheta) = (1, 0)$. For $\mathbf{c} = \mathbf{1}$, $\mathcal{C}_k = \sum_{s \geq 1} 2q^{s(k+1)} = \alpha_k = a_k$, so $\mathbf{c}_k = \mathbf{a}_k = S_k$ and $G_k = S_k / (1 - S_1)$; take $k = 0, 1$. For $c_s = q^s$, $\mathcal{C}_k = \sum_{s \geq 1} 2q^{s(k+2)} = \alpha_{k+1}$, so $\mathbf{c}_k = \sum_j \alpha_{k+2j+1} \prod_{i < j} \gamma_{k+2i} = T_k$ by (15), and (19) gives $t_1 = G_1 = T_1 / (1 - S_1)$ and $b_1 = G_0 = T_0 + S_0 T_1 / (1 - S_1)$. Analytic continuation as before. $\square$

With $1 - S_1 = S_e$ (Proposition 17) the first of these is $b_0 = S_0 / S_e$, the third line of the dictionary (30); and b_0 is holomorphic and non-zero at every travel pole, by Propositions 65 and 21 respectively.

5.3 The cocycle, the reciprocity, and the dictionary

The bulk section identity has a second, equivalent, description as a two-point problem for a first-order system. That system is the cocycle of §7.4, and comparing the two descriptions identifies t_1 with P_{12} / S_e .

Corollary 29 (the cocycle is the same object). *Let Φ solve the bulk section identity with source $\mathbf{c}$, and put $L_b = \sum_{a \leq b} \Phi_a$ and $\mathcal{E}_b = -\sum_{a \geq b} q^a \Phi_a$. Then $\Phi_b = 2q^b c_b + 2q^{2b} L_{b-1} - 2q^b \mathcal{E}_b$ and*

$$\begin{pmatrix} L_b \\ \mathcal{E}_{b+1} \end{pmatrix} = \mathcal{M}_b \begin{pmatrix} L_{b-1} \\ \mathcal{E}_b \end{pmatrix} + \begin{pmatrix} 2q^b c_b \\ 2q^{2b} c_b \end{pmatrix}, \quad \mathcal{M}_b = \begin{pmatrix} 1 + 2q^{2b} & -2q^b \\ 2q^{3b} & 1 - 2q^{2b} \end{pmatrix}, \quad (20)$$

with $L_0 = 0$, $\mathcal{E}_1 = -t_0$ or $-t_1$ according to the source, and $\mathcal{E}_b \rightarrow 0$. Let (ℓ, r) and (ℓ', r') be the solutions of the homogeneous recursion ($\mathbf{c} = 0$) with data $(\ell_0, r_1) = (1, 0)$ and $(\ell'_0, r'_1) = (0, 1)$, and put $(P_{11}, P_{12}) = \lim_b (\ell_b, r_{b+1})$ and $(P_{21}, P_{22}) = \lim_b (\ell'_b, r'_{b+1})$, the limits existing because $\sum_b \|\mathcal{M}_b - I\| < \infty$. Eliminating L between the two lines of (20) at $\mathbf{c} = 0$ gives

$$\mathcal{E}_{b+2} = (1 + q^3 - 2(1 - q)q^{2b+2})\mathcal{E}_{b+1} - q^3 \mathcal{E}_b,$$

which is the scalar three-term recursion of §7.4; so r and r' are its two standard solutions and P_{12}, P_{22} their limits, P_{12} being the entry whose closed form is (32). Then, identically in $0 < q < 1$,

$$P_{11} = 1 + T_0, \quad P_{12} = T_1, \quad P_{21} = -S_0, \quad P_{22} = S_e = 1 - S_1.$$

In particular $t_1 = P_{12}/S_e$ is an identity between independently defined objects, not a definition.

Proof. Substituting $L_b = L_{b-1} + \Phi_b$ and $\mathcal{E}_{b+1} = \mathcal{E}_b + q^b \Phi_b$ into $\Phi_b = 2q^b(c_b + q^b L_b - \mathcal{E}_{b+1})$, which is (16) after the split at $s' = b$, the two terms $\pm 2q^{2b} \Phi_b$ cancel and leave $\Phi_b = 2q^b c_b + 2q^{2b} L_{b-1} - 2q^b \mathcal{E}_b$; feeding this back gives (20). The homogeneous solutions form a two-dimensional space, since (20) determines $(L_b, \mathcal{E}_{b+1})$ from $(L_{b-1}, \mathcal{E}_b)$.

Take $\mathbf{c} = \mathbf{1}$, so $L_0 = 0$, $L_b \rightarrow b_0$, $\mathcal{E}_1 = -t_0$ and $\mathcal{E}_{b+1} \rightarrow 0$. Then $(L, \mathcal{E} - 1)$ solves the homogeneous recursion: in the first line the source $2q^b c_b = 2q^b$ is cancelled by $-2q^b \cdot (-1)$, and in the second the source $2q^{2b} c_b = 2q^{2b}$ is cancelled by $(1 - 2q^{2b})(-1) - (-1)$. Its data are $(0, -1 - t_0)$, so $(L, \mathcal{E} - 1) = -(1 + t_0)(\ell', r')$, and letting $b \rightarrow \infty$,

$$b_0 = -(1 + t_0)P_{21}, \quad -1 = -(1 + t_0)P_{22}.$$

By Proposition 28 and $S_e = 1 - S_1$ one has $1 + t_0 = 1 + S_1/S_e = 1/S_e$, whence $P_{22} = S_e$ and $P_{21} = -b_0 S_e = -S_0$.

Take now $c_s = q^s$, with data $L'_0 = 0$, $L'_b \rightarrow b_1$, $\mathcal{E}'_1 = -t_1$, $\mathcal{E}'_{b+1} \rightarrow 0$. Here $(L' + 1, \mathcal{E}')$ solves the homogeneous recursion: in the first line $(1 + 2q^{2b}) \cdot 1$ exceeds 1 by $2q^{2b}$, which is the source $2q^b c_b = 2q^{2b}$, and in the second $2q^{3b} \cdot 1$ is the source $2q^{2b} c_b = 2q^{3b}$. Its data are $(1, -t_1)$, so $(L' + 1, \mathcal{E}') = (\ell, r) - t_1(\ell', r')$, and letting $b \rightarrow \infty$,

$$b_1 + 1 = P_{11} - t_1 P_{21} = P_{11} + t_1 S_0, \quad 0 = P_{12} - t_1 P_{22} = P_{12} - t_1 S_e.$$

The second gives $P_{12} = t_1 S_e = T_1$ by Proposition 28, and the first then gives $P_{11} = b_1 + 1 - t_1 S_0 = T_0 + S_0 T_1/S_e + 1 - (T_1/S_e)S_0 = 1 + T_0$. All four limits are taken for $|q| < \frac{1}{3}$, where Lemma 24 applies, and extend to $0 < q < 1$ by analytic continuation, both sides being meromorphic there. $\square$

Proposition 30 (reciprocity). $t_0 = b_1$ identically in q . Equivalently $T_0 S_e + T_1 S_0 = S_1$, equivalently $P_{11} P_{22} - P_{12} P_{21} = 1$.

Proof. Two independent proofs.

Kernel symmetry. Write $D = \text{diag}(2q^b)_{b \geq 1}$ and $K[b, a] = q^{\max(a, b)}$, so that $M_0 = DK$, $E = D\mathbf{1}$, and $u := (2q^{2b})_b = Dv$ with $v = (q^a)_a$. The two sources of Proposition 28 are E and u , so $\Phi = (I - M_0)^{-1}E$, $\Psi = (I - M_0)^{-1}u$, and

$$t_0 = v^\top (I - M_0)^{-1}E, \quad b_1 = \mathbf{1}^\top (I - M_0)^{-1}u.$$

For $|q| < \frac{1}{3}$, $X := (I - M_0)^{-1}D = \sum_{n \geq 0} (DK)^n D$, and the n -th summand is $DKD \cdots KD$, a palindromic product of the symmetric matrices D and K , hence symmetric; the series converges entrywise. Therefore $t_0 = v^\top X \mathbf{1} = \sum_{a, b} v_a X[a, b] = \sum_{a, b} X[b, a] v_a = \mathbf{1}^\top X v = b_1$, every sum being absolutely convergent. Both sides are meromorphic on $|q| < 1$ by Proposition 28, so the identity extends.

Casoratian. $\det \mathcal{M}_b = (1 + 2q^{2b})(1 - 2q^{2b}) + 4q^{3b}q^b = 1$, so $\ell_b r'_{b+1} - \ell'_b r_{b+1}$ does not depend on b ; at $b = 0$ it is 1, whence $P_{11} P_{22} - P_{21} P_{12} = 1$. Substituting the four entries of Corollary 29 gives $(1 + T_0)S_e + T_1 S_0 = 1$, that is $T_0 S_e + T_1 S_0 = 1 - S_e = S_1$; dividing by S_e and using Proposition 28 turns this into $b_1 = T_0 + S_0 T_1/S_e = S_1/S_e = t_0$. $\square$

The symmetry of K is used and is necessary: replacing $q^{\max(a,b)}$ by the asymmetric $q^{\max(a,b)} + q^{a+2b}$ makes $t_0 - b_1 = 4q^8 + O(q^9)$ (`assembly_certificate.py`, Part 2). The determinant identity $P_{11}P_{22} - P_{12}P_{21} = 1$ is the Casoratian of the cocycle and is automatic; what carries content is the identification of its four entries in Corollary 29, which turns it into the reciprocity.

Corollary 31 (the dictionary). *Identically in $0 < q < 1$,*

$$S_e = 1 - S_1, \quad S_o = \frac{1-q}{2q}S_0, \quad b_0 = \frac{S_0}{S_e}, \quad t_0 = b_1 = \frac{S_1}{S_e}, \quad t_1 = \frac{T_1}{S_e} = \frac{P_{12}}{S_e},$$

and consequently $\kappa := t_0b_1 - b_0t_1 = (S_1^2 - S_0P_{12})/S_e^2$. This is (30) together with its two remaining entries.

Proof. The first line is Proposition 17 and Lemma 19; the rest is Proposition 28 with Proposition 30 and Corollary 29, and κ is substitution. $\square$

5.4 Compactness, the null space, and the singularities

Lemma 32 (compactness and meromorphy). *For $|q| < 1$ the operator M of (17), and its gap-marked version with kernel (21) below, are compact on ℓ^1 . Hence $q \mapsto (I - M)^{-1}$ is meromorphic on $|q| < 1$, with poles of finite order, and at each pole $\ker(I - M) \neq 0$.*

Proof. Let M_N agree with M on rows $s \leq N$ and vanish above. Then $\|M - M_N\|_{\ell^1 \rightarrow \ell^1} \leq \sup_{s'} \sum_{s > N} |e_s| |q|^{\max(s,s')} \leq \sum_{s > N} 2|q|^{\vartheta+s} \rightarrow 0$, so M is a norm limit of finite-rank operators and is compact; the rank-one gap term is compact a fortiori. The family is holomorphic in q on the connected domain $|q| < 1$ and $I - M$ is invertible near $q = 0$ (Lemma 24), so the analytic Fredholm theorem applies. $\square$

Lemma 33 (the null space is at most one-dimensional). *Let $0 < |q| < 1$ and let $R \in \ell^1$ satisfy $R_s = e_s \sum_{s'} q^{\max(s,s')} R_{s'}$ for all $s \geq s_0$. Then its sections satisfy $G_k^R = G_1^R \mathbf{a}_k$ for all $k \geq 0$, and $G_1^R = 0$ forces $R = 0$. In particular $\dim \ker(I - M) \leq 1$, and a non-zero kernel element exists only where $\mathbf{a}_1 = 1$.*

Proof. Lemma 25 with $\mathbf{c} = 0$ gives $\mathcal{C}_k = 0$, hence $\mathbf{c}_k = 0$ and $G_k^R = \mathbf{a}_k G_1^R$; its derivation used only $R \in \ell^1$ and $|q| < 1$. Taking $k = 1$ gives $G_1^R = \mathbf{a}_1 G_1^R$, so $\mathbf{a}_1 = 1$ unless $G_1^R = 0$. If $G_1^R = 0$ then $G_k^R = 0$ for every k . The function $F(t) = \sum_{s \geq s_0} R_s t^s$ is holomorphic on $|t| < 1$ because $R \in \ell^1$, and $F(q^k) = G_k^R = 0$ for all $k \geq 0$; the points q^k accumulate at the interior point $t = 0$, so $F \equiv 0$ and $R = 0$. Hence the kernel is spanned by the class normalised by $G_1^R = 1$. $\square$

Corollary 34 (singularities of the two gapless resolvents). *On $|q| < 1$, the poles of $(I - M_0)^{-1}$ lie in the zero set of S_e and the poles of $(I - \mathcal{T})^{-1}$ lie in the zero set of $1 - \Sigma_1$; conversely every travel pole is a pole of $(I - \mathcal{T})^{-1}$. In particular b_0, b_1, t_0, t_1 are holomorphic at every travel pole q_m , where $S_e(q_m) \neq 0$ by Proposition 65, and $b_0(q_m) = S_0(q_m)/S_e(q_m) \neq 0$ by Proposition 21.*

Proof. By Lemma 32 a pole forces a non-zero kernel element, and by Lemma 33 that forces $\mathbf{a}_1 = 1$, which is $S_1 = 1$, i.e. $S_e = 0$, in the bulk instance and $\Sigma_1 = 1$ in the travel one. Conversely, at a travel pole $\Sigma_0/(1 - \Sigma_1)$ is unbounded because $\Sigma_0(q_m) \neq 0$ (Proposition 21), so by Proposition 27 the travel resolvent is singular there. $\square$

5.5 The gap term is rank one: the Möbius factorisation

Marking the gap bridges changes the bulk kernel by one term. With $\mathfrak{g} = \mathfrak{g}(q, y) = \sum_{L \geq 1} q^L y^{L-1} = q/(1 - qy)$, the gap-marked bulk kernel and system are

$$K_{\mathfrak{g}}(a, b) = q^{\max(a, b)} + q^{a+b} \mathfrak{g}, \quad P_b = 2q^b \left(1 + \sum_{a \geq 1} K_{\mathfrak{g}}(a, b) P_a \right), \quad B(q, y) = \sum_{b \geq 1} P_b. \quad (21)$$

In operator form $(I - M_0 - \mathfrak{g} uv^\top)P = E$ with

$$M_0[b, a] = 2q^b q^{\max(a, b)}, \quad E_b = 2q^b, \quad u_b = 2q^{2b}, \quad v_a = q^a, \quad (22)$$

since $2q^b \cdot q^{a+b} = (2q^{2b})(q^a)$: the gap term is exactly rank one. With the four scalars of Proposition 28,

$$b_0 = \mathbf{1}^\top (I - M_0)^{-1} E, \quad b_1 = \mathbf{1}^\top (I - M_0)^{-1} u, \quad t_0 = v^\top (I - M_0)^{-1} E, \quad t_1 = v^\top (I - M_0)^{-1} u.$$

Proposition 35 (Möbius factorisation). *Let q be such that $I - M_0$ is invertible on ℓ^1 and $1 - \mathfrak{g} t_1 \neq 0$. Then (21) has a unique ℓ^1 solution,*

$$(I - M_0 - \mathfrak{g} uv^\top)^{-1} = R_0 + \frac{\mathfrak{g}}{1 - \mathfrak{g} t_1} R_0 uv^\top R_0, \quad R_0 = (I - M_0)^{-1}, \quad (23)$$

and $B(q, y) = (b_0 + \mathfrak{g} \kappa)/(1 - \mathfrak{g} t_1)$ with $\kappa = t_0 b_1 - b_0 t_1$, which is (27). In particular B is a Möbius function of $\mathfrak{g}$, with a single zero, at $\mathfrak{g}_0 = -b_0/\kappa$, and a single pole, at $\mathfrak{g} = 1/t_1$.

Proof. Sherman–Morrison. Put $\delta = 1 - \mathfrak{g} v^\top R_0 u = 1 - \mathfrak{g} t_1 \neq 0$. Then the right-hand side of (23) is a two-sided inverse of $I - M_0 - \mathfrak{g} uv^\top$:

$$(I - M_0 - \mathfrak{g} uv^\top)(R_0 + \mathfrak{g} \delta^{-1} R_0 uv^\top R_0) = I + \mathfrak{g}(\delta^{-1} - 1 - \mathfrak{g} \delta^{-1} t_1) uv^\top R_0 = I,$$

since $\delta^{-1}(1 - \delta - \mathfrak{g} t_1) = 0$; the other side is identical. All pairings converge absolutely: $u, E \in \ell^1$, $v, \mathbf{1} \in \ell^\infty$, and R_0 is bounded on ℓ^1 . Applying $\mathbf{1}^\top(\cdot)E$,

$$B = \mathbf{1}^\top R_0 E + \mathfrak{g} \delta^{-1} (\mathbf{1}^\top R_0 u)(v^\top R_0 E) = b_0 + \frac{\mathfrak{g} b_1 t_0}{1 - \mathfrak{g} t_1} = \frac{b_0 + \mathfrak{g}(t_0 b_1 - b_0 t_1)}{1 - \mathfrak{g} t_1}. \quad \square$$

Remark 36 (the two hypotheses are the two the paper already carries). Invertibility of $I - M_0$ is $S_e \neq 0$ (Corollary 34), which at the travel poles is Proposition 65. And $1 - \mathfrak{g}_V t_1 \neq 0$ is $s \neq 1$, the gate half $|s| < 1$ of (29), which is Theorem 80 with Lemma 81; $1 - \mathfrak{g}_U t_1 \neq 0$ follows, since $\mathfrak{g}_U = \mathfrak{g}_V/(1 + q) < \mathfrak{g}_V$ and $t_1 > 0$ at the poles. So Proposition 35 consumes no hypothesis that §7.3 was not already carrying. It does not consume $B_V \neq 0$: that is derived from it, not assumed.

Machine verification of the telescoping, the factorisation and the reciprocity

Proposition 35 is formally verified in Lean 4 against Mathlib [27] in `Mobius.lean` and `MobiusL1.lean`. Its hypotheses are its Lean hypotheses: ‘ $I - M_0$ is invertible’ is an operator R_0 with $R_0(I - M_0) = (I - M_0)R_0 = I$, and $1 - \mathfrak{g} t_1 \neq 0$ is carried as stated. With those in hand nothing analytic remains. `Mobius.lean` proves the factorisation for an arbitrary module over an arbitrary field: `opN_opS` and `opS_opN` are the two halves of (23), `solution_unique` is the unique solvability of (21), `B_mobius` is (27), and `B_eq_zero_iff` and `B_pole` are the single zero at $\mathfrak{g}_0 = -b_0/\kappa$ and the single pole at $\mathfrak{g} = 1/t_1$.

`MobiusL1.lean` instantiates that statement at the case the proposition is about, $K = \mathbb{C}$ and $E = \ell^1$ with the concrete data of (22). For $|q| < 1$ a kernel with rows dominated by a summable

sequence defines a $\mathbb{C}$ -linear operator on ℓ^1 (`opOfKer`) and a bounded sequence defines a functional (`funcOfSeq`); `M0`, `uElt`, `EElt`, `vFun` and `oneFun` are M_0 , u , E , $v^\top$ and $\mathbf{1}^\top$ of (22), with the index n standing for site $n+1$. `rank_one` is the identity $2q^b q^{a+b} = (2q^{2b})(q^a)$, `gapSystem_iff` identifies the operator equation $(I - M_0 - \mathbf{g}uv^\top)P = E$ with the display (21) coordinatewise, `l1_solution_unique` is the unique ℓ^1 solvability of (21), `l1_smresolvent_left` and `l1_smresolvent_right` are (23) for those operators, and `l1_B_of_solution` is the conclusion: an ℓ^1 solution of (21) has $B = \sum_b P_b = (b_0 + \mathbf{g}\kappa)/(1 - \mathbf{g}t_1)$.

The first sentence of Proposition 30, $t_0 = b_1$, is verified in `Reciprocity.lean` in the operator form its first proof uses, with E and F kept apart: $D: F \rightarrow E$ and $K: E \rightarrow F$ symmetric for the pairing, $M_0 = DK$, and the two sources $D\mathbf{1}$ and Dv . `pushThrough` is $(I - DK)^{-1}D = D(I - KD)^{-1}$, which replaces the palindromic Neumann series of the paper's proof by an algebraic identity; `X_symm` is the symmetry of X and `t0_eq_b1` the conclusion. The necessity of the symmetry of K is recorded as `K_symm_needed`.

Lemma 25 is verified in `Telescope.lean`. The site $s = n + s_0$ is indexed by $n \in \mathbb{Z}_{\geq 0}$, so e_s is `eCo` and $s_0 + \vartheta = 1$ is the hypothesis `hs`; the source is bounded and $\Phi \in \ell^1$ is carried as summability of $\|\Phi_n\|$, which is what `MobiusL1.summable_norm` supplies for a Φ given as an element of the ℓ^1 space of `MobiusL1.lean` (`l1_three_term`, `l1_G_eq`). `row_sum` is the split of the inner sum at $s' = s$ together with the finite and the infinite geometric sum it produces; `three_term` is (18), the interchange of the two summations being Fubini against the majorant $2|q|^{n+1}\|\Phi_m\|$, which is summable on $\mathbb{Z}_{\geq 0}^2$ as a product; `norm_bTel_le` is the bound $|b_k| \leq \Xi|q|^{k+1}$ with $\Xi = 4/(1 - |q|)$ and `summable_bProd_norm` is the paper's $\Xi^J|q|^{J^2+Jk} \rightarrow 0$, obtained by the ratio test; `summable_aFrak` and `summable_cFrak` are the convergence of $\mathbf{a}_k$ and $\mathbf{c}_k$; `G_eq` is $G_k = \mathbf{c}_k + \mathbf{a}_k G_1$, proved by telescoping $R_j = G_{k+2j} \prod_{i < j} b_{k+2i}$ and letting $R_J \rightarrow 0$; `G_one_solved` and `G_solved` are (19). The last sentence of the lemma, the identification of the two instances, is `aTel_bulk`, `bTel_bulk`, `aTel_travel` and `bTel_travel`.

Proposition 27 and Proposition 28 are verified in `SectionLadders.lean`, which evaluates $\mathcal{C}_k$ at the three geometric sources: `CC_travel` is $\mathcal{C}_k = 2q/(1 - q^{k+1}) = A_k$, `CC_bulk` is $\mathcal{C}_k = \alpha_k$ and `CC_bulk_geo` is $\mathcal{C}_k = \alpha_{k+1}$, whence `cFrak_travel`, `cFrak_bulk` and `cFrak_bulk_geo` identify $\mathbf{c}_k$ with Σ_k , with S_k and with T_k . `travelsum` is Proposition 27 and `bulkdress_b0_t0`, `bulkdress_t1_b1` are the four scalars of Proposition 28. `Sladder_eq`, `SigmaLadder_eq` and `Tladder_eq` are the three ladders written out as in (2) and (15). What is verified is the implication, not the existence of the solution: the ℓ^1 solution is a hypothesis, which is where the paper's $|q| < \frac{1}{3}$ lives, and no analytic continuation step is needed, `G_eq` holding on the whole of $|q| < 1$ for any ℓ^1 solution. Lemma 24 is not formalised.

The first of the two clauses of Proposition 30 that begin “equivalently” is verified as `SectionLadders.recip_ladder` given the two bulk solutions, $t_0 = b_1$ and $T_0 S_e + T_1 S_0 = S_1$ are the same statement, the equivalence being `recip_ladder_iff`. The reciprocity itself enters there as a hypothesis; it is proved separately in `Reciprocity.lean`, in the operator form recorded above, and the two are not joined, that joining being the instantiation of `Reciprocity.lean` at the concrete ℓ^1 resolvent.

Not formalised in this section: Corollary 29, and with it the second “equivalently” clause of Proposition 30, the one in terms of P_{ij} . The obstruction is the existence of the four limits $\lim_b(\ell_b, r_{b+1})$ and $\lim_b(\ell'_b, r'_{b+1})$: Mathlib carries no convergence theorem for an infinite product of matrices $\prod_b \mathcal{M}_b$ with $\sum_b \|\mathcal{M}_b - I\| < \infty$, and supplying one is a statement about products, not about this cocycle. The rest of the corollary, the passage from (16) to (20) and the identification of the four entries once the limits exist, needs only the split already carried out in `Telescope.row_sum`.

Lemma 37 (a positivity segment). *For $0 < q \leq 0.45$ one has $S_e(q) \geq 0.32$ and $|s(q)| = \frac{q}{1-q}|P_{12}|/S_e \leq 0.78$. Since $q_1 = 0.4494536\dots < 0.45$, both hold on $(0, q_1]$.*

Proof. By Proposition 17, $S_e = \sum_{j \geq 0} (-1)^j [2(1-q)]^j q^{j^2+j} / (q; q)_{2j}$, and by (35), $P_{12} = \sum_{k \geq 1} (-1)^{k-1} [2(1-$

$q)]^k q^{k^2+k+1} (1-q^{2k}) / (q; q)_{2k+1}$. For every n , $(q; q)_n \geq (q; q)_\infty \geq 1 - q - q^2$, the last step by truncating Euler's pentagonal series $1 - q - q^2 + q^5 + q^7 - \dots$, whose tail is positive for $q \leq \frac{1}{2}$; on $q \leq 0.45$ this is ≥ 0.3475 . Each term $[2(1-q)]^j q^{j^2+j}$ with $j \geq 1$ increases on $(0, 0.45]$, since $(j^2+j)/q - j/(1-q) > 0$ there, so it is at most its value at $q = 0.45$: successively 0.222750, 0.010048, 0.000092, the remaining terms summing to less than 10^{-5} . Hence $S_e \geq 1 - 0.23290/0.3475 \geq 0.32$. The same estimate applied to $[2(1-q)]^k q^{k^2+k+1}$, likewise increasing, and to $(1 - q^{2k}) \leq 1$, gives 0.100238, 0.004521, 0.000042 and a tail below 10^{-6} , so $|P_{12}| \leq 0.104801/0.3475 \leq 0.3016$ and $|s| \leq \frac{0.45}{0.55} \cdot \frac{0.3016}{0.32} \leq 0.78$. $\square$

5.6 The site cost

The subsections above concern the operators of Definition 23. This one concerns the combinatorial quantity those operators are meant to encode: the per-site cost of the crossing optimisation of [4]. Its value is computed here in closed form, which settles the first half of the model input (M) below and corrects the marker clause of the form in which that input was previously stated.

Definition 38 (the pairing optimisation). Fix marker data $(\varepsilon^*, \delta^*) \in \{\pm 1\} \times \{0, 1\}$, a displacement $k^* \in \mathbb{Z}$, and deposits $d_j \in \mathbb{Z}$ on the integer edges j , edge j joining site j to site $j + 1$. The travel indicator is $f_j = 1$ for $0 \leq j < k^*$, $f_j = -1$ for $k^* \leq j < 0$ and $f_j = 0$ otherwise, and $d_j \equiv f_j \pmod{2}$. A *realisation* consists of

- crossing counts $m_j \geq \max(|d_j|, |f_j|)$ with $m_j \equiv d_j \pmod{2}$, whose support is an interval containing 0; edge j is crossed $u_j = (m_j + f_j)/2$ times upward and $dn_j = (m_j - f_j)/2$ times downward;
- a sign in $\{\pm\}$ on each crossing, with p_j^u of the up-crossings and p_j^d of the down-crossings carrying $+$, subject to $d_j = 2p_j^d - dn_j + u_j - 2p_j^u$;
- at every site s , a bijection between the arrivals and the departures at s .

The arrivals at s are the up-crossings of edge $s - 1$, of class $(L, \pm)$ by their sign, and the down-crossings of edge s , of class $(R, \pm)$; the departures are the down-crossings of edge $s - 1$, of class $(L, \pm)$, and the up-crossings of edge s , of class $(R, \pm)$. In addition there is a virtual arrival of class $(L, +)$ at $s = 0$, and a virtual departure at $s = k^*$, of class (R, ε^*) if $\delta^* = 1$ and (L, ε^*) if $\delta^* = 0$. A matched pair costs 0 on the same side with equal signs, 2 on the same side with opposite signs, and 1 on opposite sides. The cost of a realisation is $\sum_j m_j$ plus the sum of the pairing costs; ℓ_R is the minimum cost, and c is the least number of components other than the one carrying the two virtual events, over the realisations of minimum cost.

Remark 39 (standing of Definition 38). This is the optimisation of [4, Metric formula, as verified], restated. That it computes the relaxed word length is verified there and is not proved. Everything in this subsection is a statement about the optimisation itself and is proved outright; every statement about the growth series that consumes it inherits the standing of the optimisation.

Lemma 40 (the transportation value). *Let $A^\pm, B^\pm, C^\pm, D^\pm$ be non-negative integers with $A^+ + A^- + B^+ + B^- = C^+ + C^- + D^+ + D^-$, and consider the transportation problem with supplies A^+, A^-, B^+, B^- and demands C^+, C^-, D^+, D^- in the four classes $(L, +), (L, -), (R, +), (R, -)$ under the cost of Definition 38. Put*

$$\alpha = (C^+ - C^-) - (A^+ - A^-), \quad \beta = (B^+ - B^-) - (D^+ - D^-), \quad \Phi = (A^+ + A^-) - (C^+ + C^-).$$

Then the minimum cost equals $\max(|\alpha|, |\beta|, |\Phi|)$.

Proof. Write $N_L^a = A^+ + A^-$, $N_L^d = C^+ + C^-$, $N_R^a = B^+ + B^-$, $N_R^d = D^+ + D^-$, so $\Phi = N_L^a - N_L^d = N_R^d - N_R^a$, and $\sigma_L^a = A^+ - A^-$, $\sigma_L^d = C^+ - C^-$, and similarly on the right, so that $\alpha = \sigma_L^d - \sigma_L^a$ and $\beta = \sigma_R^a - \sigma_R^d$.

Let π be the mass sent from left supplies to right demands and π' that from right supplies to left demands. The mass staying on the left is $g = N_L^a - \pi = N_L^d - \pi'$, so $\pi' = \pi - \Phi$; put $P = \pi + \pi' = 2\pi - \Phi$. Cross transports cost 1 each and their cost does not depend on the signs, so for fixed π, π' the cost is $P + 2\text{mis}_L + 2\text{mis}_R$, where mis_L is the least number of opposite-sign pairs among the g same-side pairs on the left, minimised over which supplies stay. If a^+ of the staying supplies and c^+ of the served demands carry $+$, that number is $|a^+ - c^+|$, since $\min(a^+, c^+) + \min(g - a^+, g - c^+) = g - |a^+ - c^+|$ pairs can be matched with equal signs and no more. Here a^+ ranges over the integer interval $[\max(0, g - A^-), \min(A^+, g)]$, which is non-empty because $0 \leq g \leq A^+ + A^-$, and c^+ over $[\max(0, g - C^-), \min(C^+, g)]$, likewise non-empty. The least $|a^+ - c^+|$ over two non-empty integer intervals $[l_1, r_1], [l_2, r_2]$ is $\max(0, l_1 - r_2, l_2 - r_1)$. Here $l_1 - r_2 = \max(0, g - A^-) - \min(C^+, g)$, which is $g - A^- - C^+$ when $g \geq A^-$ and $C^+ \leq g$, and is at most 0 otherwise, so $\max(0, l_1 - r_2) = \max(0, g - A^- - C^+)$; symmetrically $\max(0, l_2 - r_1) = \max(0, g - C^- - A^+)$, and

$$\text{mis}_L = \max(0, g - A^- - C^+, g - C^- - A^+).$$

Substituting $A^\pm = (N_L^a \pm \sigma_L^a)/2$ and $C^\pm = (N_L^d \pm \sigma_L^d)/2$ gives $g - A^- - C^+ = \frac{\Phi - \alpha}{2} - \pi$ and $g - C^- - A^+ = \frac{\Phi + \alpha}{2} - \pi$, hence $2\text{mis}_L = \max(0, |\alpha| + \Phi - 2\pi) = \max(0, |\alpha| - P)$. The same computation on the right gives $2\text{mis}_R = \max(0, |\beta| - \Phi - 2\pi') = \max(0, |\beta| - P)$. Therefore the cost is

$$\phi(P) = P + (|\alpha| - P)^+ + (|\beta| - P)^+,$$

which for $P \geq 0$ is at least $\max(|\alpha|, |\beta|)$, with equality exactly on $\min(|\alpha|, |\beta|) \leq P \leq \max(|\alpha|, |\beta|)$, and which equals P above that range.

It remains to identify the admissible P . From $\pi \geq 0$ and $\pi' \geq 0$ one gets $P \geq |\Phi|$; from $\pi \leq \min(N_L^a, N_R^d)$ and $\pi' \leq \min(N_R^a, N_L^d)$ one gets, using $\Phi = N_L^a - N_L^d = N_R^d - N_R^a$, the single constraint $P \leq \min(N_L^a + N_L^d, N_R^a + N_R^d)$. Finally $P \equiv \Phi \pmod{2}$, and $\alpha \equiv \beta \equiv \Phi \pmod{2}$ because $\sigma \equiv N \pmod{2}$ on each side. Now $|\alpha| \leq |\sigma_L^d| + |\sigma_L^a| \leq N_L^d + N_L^a$ and $|\Phi| \leq N_L^a + N_L^d$, and symmetrically $|\beta|, |\Phi| \leq N_R^a + N_R^d$. Hence $P^* := \max(|\Phi|, \min(|\alpha|, |\beta|))$ is admissible: it has the right parity, it is at least $|\Phi|$, and it is at most $\max(|\Phi|, |\alpha|) \leq N_L^a + N_L^d$ and at most $\max(|\Phi|, |\beta|) \leq N_R^a + N_R^d$. If $|\Phi| \leq \max(|\alpha|, |\beta|)$ then P^* lies in the equality range and $\phi(P^*) = \max(|\alpha|, |\beta|)$, which is optimal since $\phi \geq \max(|\alpha|, |\beta|)$ throughout. If $|\Phi| > \max(|\alpha|, |\beta|)$ then every admissible P satisfies $P \geq |\Phi|$ and $\phi(P) = P$, so the minimum is $|\Phi|$, attained at $P^* = |\Phi|$. In both cases the minimum is $\max(|\alpha|, |\beta|, |\Phi|)$. $\square$

Corollary 41 (the local cost law). *In Definition 38, let $\alpha_s, \beta_s, \Phi_s$ be the three quantities of Lemma 40 formed from the arrivals and departures at site s . Then*

$$\alpha_s = d_{s-1} - [s = 0] + \varepsilon^*[s = k^*][\delta^* = 0], \quad \beta_s = d_s - \varepsilon^*[s = k^*][\delta^* = 1],$$

$$\Phi_s = f_{s-1} + [s = 0] - [s = k^*][\delta^* = 0],$$

one has $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$ in every case, and the minimal pairing cost at s is

$$\text{Site}(s) = \max(|\alpha_s|, |\beta_s|),$$

independently of the crossing counts m_j , of the sign splits p_j^u, p_j^d , and of the travel indicators. In particular at a site carrying no virtual event $\text{Site}(s) = \max(|d_{s-1}|, |d_s|)$.

Proof. The displays are read off Definition 38: the up-crossings of edge $s-1$ contribute σ_L^a , the down-crossings contribute σ_L^d , and $d_{s-1} = 2p_{s-1}^d - \text{dn}_{s-1} + u_{s-1} - 2p_{s-1}^u = \sigma_L^d - \sigma_L^a$, which is α_s before the virtual events; the virtual arrival adds one to A^+ and the virtual departure adds one to C^{ε^*} or to D^{δ^*} . Likewise $d_s = \sigma_R^a - \sigma_R^d$ and $\Phi_s = u_{s-1} - \text{dn}_{s-1} = f_{s-1}$ before the virtual events. Since the sign splits enter only through σ^a, σ^d , and those enter $\alpha_s, \beta_s, \Phi_s$ only through the fixed combinations d_{s-1}, d_s, f_{s-1} , none of the three depends on m_j or on p_j^u, p_j^d .

For $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$: at a site with no virtual event $\Phi_s = f_{s-1} = f_s$, since arrivals and departures balance, and $|d_j| \geq |f_j|$, since a bulk deposit is even with $f = 0$ and a travel deposit is odd with $|f| = 1$. The virtual events sit at $s = 0$ and $s = k^*$, which for $k^* \neq 0$ are exactly the two sites where f jumps, and there are eight remaining cases. For $s = 0 < k^*$: $\Phi_0 = f_{-1} + 1 = 1$, while $\alpha_0 = d_{-1} - 1$ is odd and $\beta_0 = d_0$ is odd. For $s = 0$ and $k^* < 0$: $\Phi_0 = -1 + 1 = 0$. For $s = k^* > 0$: $\Phi = 1 - 1 = 0$ if $\delta^* = 0$, and $\Phi = 1$ with $\alpha = d_{k^*-1}$ odd, $\beta = d_{k^*} - \varepsilon^*$ odd, if $\delta^* = 1$. For $s = k^* < 0$: $\Phi = -1$ with $\alpha = d_{k^*-1} + \varepsilon^*$ odd and $\beta = d_{k^*}$ odd if $\delta^* = 0$, and $\Phi = 0$ if $\delta^* = 1$. For $k^* = 0$, where the two virtual events share the site: $\Phi_0 = 0$ if $\delta^* = 0$, and $\Phi_0 = 1$ with $\alpha_0 = d_{-1} - 1$ and $\beta_0 = d_0 - \varepsilon^*$ both odd if $\delta^* = 1$. In each case $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$, so Lemma 40 gives $\text{Site}(s) = \max(|\alpha_s|, |\beta_s|)$. $\square$

Corollary 42 (the closed form for ℓ_R , and rigidity of the crossing counts). *Let the span be the least interval of edges containing 0, all j with $d_j \neq 0$ and all j with $f_j \neq 0$. Every realisation of minimum cost has support exactly the span and crossing counts*

$$m_j = \max(|d_j|, |f_j|), \quad \text{forced to 2 when that value is 0,}$$

and

$$\ell_R = \sum_{j \in \text{span}} m_j + \sum_s \max(|\alpha_s|, |\beta_s|),$$

the second sum over the sites of the span.

Proof. By Corollary 41 the pairing cost at every site is independent of the m_j , so the total cost is $\sum_j m_j$ plus a quantity that does not depend on the m_j ; hence a minimum-cost realisation minimises $\sum_j m_j$ alone, subject to $m_j \geq \max(|d_j|, |f_j|)$, $m_j \equiv d_j$, and the support being an interval containing 0. Since $d_j \equiv f_j$, the value $\max(|d_j|, |f_j|)$ has the parity of d_j and is the least admissible value on the span except where it vanishes, where the support condition forces $m_j \geq 2$. Enlarging the support beyond the span adds an edge with $m \geq 2$ and creates one further site, of cost $\max(0, 0) = 0$, without changing any other site cost, so it adds 2 and never helps. $\square$

Corollary 41 is the first clause of (M1) below, proved. The marker clause needs the sign, and the form previously used does not carry it.

Corollary 43 (the marker junctions). *Let $k^* > 0$, let $d_L = d_{-1}$ be the last bulk deposit (even) and $d_R = d_0$ the first travel deposit (odd). Then the cost of the junction at site 0 is*

$$\text{Site}_0(d_L, d_R) = \max(|d_L - 1|, |d_R|),$$

for all four marker data. At the far junction, site k^* , with $d_L = d_{k^*-1}$ odd and $d_R = d_{k^*}$ even, the cost is $\max(|d_L + \varepsilon^*|, |d_R|)$ if $\delta^* = 0$ and $\max(|d_L|, |d_R - \varepsilon^*|)$ if $\delta^* = 1$; it is not independent of $(\varepsilon^*, \delta^*)$ and it is not the mirror image of Site_0 .

Proof. Corollary 41. At site 0 neither ε^* nor δ^* occurs. $\square$

Remark 44 (the earlier marker form is false). Earlier versions of this paper, and [6], carried $\text{Site}_0(d_L, d_R) = \max(|d_L| - 1, |d_R|)$. That agrees with Corollary 43 exactly on $d_L \geq 0$ and fails on every cell with $d_L \leq -2$ and $|d_R| \leq |d_L|$: the smallest is $d_L = -2$, $d_R = 1$, where the true cost is 3 and the earlier form gives 1. The verification on which the earlier form rested covered $0 \leq d_L \leq 12$ only. The junction cost therefore depends on the *sign* of the last bulk deposit, not on its magnitude alone.

The correction is local. Every bulk edge cost and every interior bulk site cost depends on the deposit only through its magnitude (Corollary 41), so reversing the sign of the junction-adjacent deposit of a bulk run is a weight-preserving involution on bulk runs; writing B_σ for the generating function of the bulk runs whose junction-adjacent deposit has magnitude σ , the two signs carry $B_\sigma/2$ each, and the junction-dressed vector of (M3) is

$$\mu_s^{(y)} = B_0 x^{2s+1} + \sum_{\sigma \geq 2 \text{ even}} \frac{B_\sigma}{2} \left(x^{\max(\sigma-1, 2s+1)} + x^{\max(\sigma+1, 2s+1)} \right), \quad (24)$$

in place of $\sum_\sigma B_\sigma x^{\max(\sigma-1, 2s+1)}$. This changes no statement below: Proposition 49 concludes that the junction is not of rank one, and the symmetrised matrix (24) has rank 6 on the block $\sigma \in \{0, 2, 4, 6, 8, 10\}$, $2s+1 \in \{1, 3, 5, 7, 9, 11\}$, by exact elimination over $\mathbb{Q}$ at $x = \frac{1}{2}$, against rank 5 for the earlier form on the same block.

Proposition 45 (the cut criterion and the shield lower bound). *Call a site s of the span cut if $\alpha_s = \beta_s = \Phi_s = 0$, and let Z be the set of cut sites interior to the span. At a cut site every minimum-cost pairing matches each arrival with a departure on its own side, so no strand crosses s ; at any other interior site s some minimum-cost pairing lets a strand cross. Consequently every minimum-cost realisation has at least $|Z| + 1$ components and*

$$c \geq |Z|.$$

A maximal run of L gap edges of the bulk ($d_j = f_j = 0$) flanked by bulk deposits and containing no virtual event contributes exactly its $L - 1$ interior sites to Z .

Proof. At a cut site $\phi(P) = P$ in the notation of Lemma 40, so the unique minimiser is $P = 0$, which is the assertion. At any other interior site $\max(|\alpha_s|, |\beta_s|) > 0$, since $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$; the equality range of ϕ is $[\min(|\alpha_s|, |\beta_s|), \max(|\alpha_s|, |\beta_s|)]$ and contains a value $P > 0$ admissible for the caps of Lemma 40: namely $\min(|\alpha_s|, |\beta_s|)$ when that is positive, which is the choice P^* of Lemma 40; and 2 otherwise, since $\min(|\alpha_s|, |\beta_s|) = 0$ forces $\Phi_s = 0$, hence $\alpha_s \equiv \beta_s \equiv 0 \pmod{2}$ and $\max(|\alpha_s|, |\beta_s|) \geq 2$. The two caps of Lemma 40 are m_{s-1} and m_s , each augmented by 1 for a virtual event of s on that side, and both are at least 2 when $\Phi_s = 0$: an edge with $f = 0$ has $m \geq 2$ on the span by Corollary 42, while $\Phi_s = f_{s-1} + [s = 0] - [s = k^*][\delta^* = 0] = f_s + [s = k^*][\delta^* = 1]$ shows that an adjacent edge with $|f| = 1$ forces a virtual event of s on its side, which supplies the missing unit. Removing the $|Z|$ cut sites disconnects the crossings of the span into at least $|Z| + 1$ classes, no two of which can lie in the same component; one component carries the virtual events, so at least $|Z|$ are isolated cycles. For the last sentence, an interior site of such a run has $d_{s-1} = d_s = 0$ and $f_{s-1} = 0$ and no virtual event, hence is cut, while the two sites at the ends of the run have $|\alpha|$ resp. $|\beta|$ equal to the flanking bulk deposit, which is even and non-zero. $\square$

Remark 46 (what the shield law still owes). Proposition 45 gives $c \geq |Z|$. The reverse inequality $c \leq |Z|$ asks for one minimum-cost realisation whose components are exactly the $|Z| + 1$ classes cut out by Z , that is, for pairings that connect all crossings within each class. It is not proved here. It is verified by exhaustive enumeration, over every sign split and every site bijection, on all bulk

configurations with $k^* = 0$, up to 8 edges, even deposits of magnitude at most 4 and both signs, and all four marker data: 1 048 544 configurations, gap runs up to length 6, 0 exceptions to both $c = |Z|$ and the closed form of Corollary 42 (`code/zeta_probe/tools/sitecost`, mode `shield`). The closed form $|Z|$ also recovers, by inspection, the boundary shield $\text{sh}_R = \mathbf{1}[\varepsilon^* = -1 \text{ or } \delta^* = 1]$ that [4, the explicit transducer] records at $k^* = 0$: by Corollary 41 the marker site is cut exactly when $d_{-1} = 0$, $\delta^* = 0$ and $\varepsilon^* = 1$, which is exactly where that indicator vanishes.

Machine verification of the site cost

Lemma 40, Corollary 41, Corollary 42, Corollary 43, the counterexample of Remark 44 and all of Proposition 45 except its component count are formally verified in Lean 4 against Mathlib [27], in six files: `SiteCost.lean` (the transportation problem at one site), `MarkedSite.lean` (the virtual events and the local cost law), `PairingMatrix.lean` (bijections against multiplicity matrices), `Realisation.lean` (the whole edge path), `CutCross.lean` (the middle sentence of Proposition 45) and `CutComponents.lean` (the counting step of the component bound, over an abstract graph).

A pairing at a site is the structure `Plan`: sixteen non-negative integers with the eight matching equations of Definition 38; `Plan.cost` is its cost and `Plan.cross` the number of pairs on opposite sides. The minimum of Lemma 40 is formalised as `IsLeast`, so both halves are proved separately: `cost_lower_bound` holds for every pairing and `exists_plan_cost_eq` exhibits one of exactly that cost. Definition 38 asks for a *bijection* between the arrivals and the departures rather than for a matrix, and the two are reconciled: `planOfBij` sends a bijection to its multiplicity matrix without changing the cost, and `bijOfMatrix` realises every matrix with the right row and column sums by a bijection, so `bij_min` is Lemma 40 stated over bijections and `MarkedSite.site_cost_bij` is Corollary 41 stated over bijections. A site is the structure `MarkedSite`: the displacement k^* , the site s , the marker data, and the crossing data of the two adjacent edges, constrained to realise the travel indicator `travel`. A realisation of a whole path is the structure `Realisation` over a `PathData`, which carries k^* , the marker data, the deposits and the span $[A, B]$; `PathData.emptyPath` is a witness that the hypotheses are consistent.

Paper step	Lean declaration	axioms
$\text{cost} \geq \max(\alpha , \beta , \Phi)$, six duals	<code>cost_lower_bound</code>	standard
a pairing attaining it	<code>exists_plan_cost_eq</code>	standard
Lemma 40, the minimum	<code>transport_min</code>	standard
a bijection gives a matrix of equal cost	<code>planOfBij_cost</code>	standard
a matrix gives a bijection of equal cost	<code>bijOfMatrix_cost</code>	standard
Lemma 40 over bijections	<code>bij_min</code>	standard
collapse to $\max(\alpha , \beta)$	<code>siteCost_eq</code>	standard
$\alpha_s = d_{s-1} - [s=0] + \varepsilon^*[s=k^*][\delta^*=0]$	<code>MarkedSite.alpha_eq</code>	standard
$\beta_s = d_s - \varepsilon^*[s=k^*][\delta^*=1]$	<code>MarkedSite.beta_eq</code>	standard
$\Phi_s = f_{s-1} + [s=0] - [s=k^*][\delta^*=0]$	<code>MarkedSite.Phi_eq</code>	standard
$ \Phi_s \leq \min(\alpha_s , \beta_s)$, all site types	<code>MarkedSite.Phi_le_min</code>	standard
Corollary 41, the minimum at any site	<code>MarkedSite.site_cost</code>	standard
Corollary 41 over bijections	<code>MarkedSite.site_cost_bij</code>	standard
independence of m_j and of p_j^u, p_j^d	<code>MarkedSite.cost_indep</code>	propext
interior site cost $\max(d_{s-1} , d_s)$	<code>MarkedSite.cost_interior</code>	propext
Corollary 43, near junction	<code>PathData.marker_near</code>	propext
Corollary 43, far junction, $\delta^*=0$	<code>PathData.marker_far_left</code>	propext
Corollary 43, far junction, $\delta^*=1$	<code>PathData.marker_far_right</code>	propext
and it is neither $(\varepsilon^*, \delta^*)$ -free nor the mirror of Site_0	<code>PathData.marker_far_depends</code> (the same declaration)	none
off the span every site is empty	<code>Realisation.pair_cost_zero_outside</code>	standard
Corollary 42, the minimum	<code>lR_closed</code>	standard
Corollary 42, rigidity of m_j	<code>Realisation.rigidity</code>	standard
Proposition 45, cut site: no crossing	<code>cut_forces_no_cross</code>	standard
the same over a realisation	<code>Realisation.cut_no_cross</code>	standard
the transportation value at a prescribed cross mass	<code>exists_plan_of_cross</code>	standard
the two caps $m_{s-1}, m_s \geq 2$ where $\Phi_s = 0$	<code>Realisation.capL_two</code>	standard
	<code>Realisation.capR_two</code>	standard
Proposition 45, a non-cut site: a crossing	<code>Realisation.exists_cross_of_not_cut</code>	standard
Proposition 45, a gap run's cut sites	<code>PathData.gap_run_cut</code>	standard
Z interior to the span	<code>PathData.cutSet_lt, cutSet_le</code>	standard
every edge of the span carries a crossing	<code>Realisation.m_pos_of_min</code>	standard
blk constant on components	<code>CutComponents.blk_reachable</code>	propext
blk attains $0, \dots, Z $ on the span	<code>CutComponents.exists_pos_with_gz</code>	standard
at least $ Z + 1$ components, given the graph	<code>CutComponents.exists_injective_components</code>	standard
and $c \geq Z $ from it	<code>..._avoiding</code>	standard
the two marker forms agree on $d_L \geq 0$	<code>marker_forms_agree</code>	standard
and differ at $(d_L, d_R) = (-2, 1)$	<code>marker_forms_differ</code>	none

“standard” means `propext`, `Classical.choice`, `Quot.sound`, or a subset of them; several of the rows above in fact report only `propext` and `Quot.sound`. The last row is closed by `decide` and depends on no axiom at all, as do `PathData.marker_far_depends` and `Realisation.site_cost_eq`. The cost of a realisation is summed over the sites of its support interval, which is legitimate because both virtual events lie there (`Realisation.virtual_in_window`) and every site outside it is empty.

The middle sentence of Proposition 45, that at an interior site which is not cut some minimum-cost pairing lets a strand cross, is verified in the form the proof uses. `exists_plan_cost_eq` settles every case except $\min(|\alpha_s|, |\beta_s|) = 0$ with $\Phi_s = 0$; there `exists_plan_of_cross` runs the construction of Lemma 40 at the prescribed cross mass $P = 2$, which is admissible because $\Phi_s = 0$ forces both caps to be at least 2, and that is a path-level fact, proved as `Realisation.capL_two` and `Realisation.capR_two` from $m_j \geq \max(|d_j|, |f_j|)$ on the span together with $\Phi_s = f_{s-1} + [s=0] - [s=k^*][\delta^*=0] = f_s + [s=k^*][\delta^*=1]$.

The counting step of the component bound is verified in `CutComponents.lean`, over an abstract

graph. Write $\text{blk}(v)$ for the number of cut sites at or below the edge carrying the crossing v . `blk_adj` and `blk_reachable` are that `blk` is constant on connected components, given the locality hypothesis **Local**: an edge of the graph joins crossings on the two edges adjacent to one site, and joins crossings on *different* edges only at a site that is not cut. `exists_pos_with_gz` is the discrete intermediate value argument, that `blk` takes every value in $0, \dots, |Z|$ on the span, and `exists_injective_components` concludes with an injection of $\{0, \dots, |Z|\}$ into the connected components, which is “at least $|Z| + 1$ components” with no finiteness hypothesis. `exists_injective_components_avoiding` is the passage to $c \geq |Z|$ once the two virtual events are known to share a component. Two of the three inputs are supplied against the structures of `Realisation.lean`: `PathData.cutSet` is Z with its two bounds (`cutSet_lt`, `cutSet_le`), and `Realisation.m_pos_of_min` is the occupancy, every edge of the span of a minimum-cost realisation carrying a crossing.

Two things in this subsection are *not* formalised. First, that the optimisation of Definition 38 computes the relaxed word length (Remark 39, an input of [4], verified there and not proved). Second, the strand graph itself, and with it the component count $c \geq |Z|$. `Realisation.pair` stores the multiplicity matrix of the site pairing rather than the bijection, so the vertex set and the incidence relation that `CutComponents.lean` quantifies over are not definable against the present structure and would need a further definition. `Realisation.cut_no_cross` already carries the mathematical content of the locality hypothesis at the level of the matrix; what is missing is the passage from a matrix with vanishing cross mass to an incidence relation on individual crossings.

5.7 The model (M), and the assembly

Everything above is a statement about the explicit operators of Definition 23 and (21), or about the optimisation of Definition 38. What connects those to the growth series is the strand-walk input of Route B, which we state once and name.

Hypothesis 47 (the transfer model (M)). (M1) *Local cost law*. The site cost between adjacent active edges carrying deposits d, d' is $x^{\max(|d|, |d'|)}$, independent of the crossing counts and of the travel indicator. Bulk deposits are even, $|d| = 2s$, and travel deposits odd, $|d| = 2s + 1$, so this cost is $q^{\max(s, s')}$ in the bulk and $x^{2\max(s, s') + 1}$ on the travel side, and the edge weights are $2q^s$ and $2x^{2s+1}$ respectively: Definition 23 in its two instances. At the near marker site the cost is $x^{\text{Site}_0(d_L, d_R)}$ with $\text{Site}_0(d_L, d_R) = \max(|d_L - 1|, |d_R|)$, d_L the last bulk deposit and d_R the first travel deposit, for all four marker data $(\varepsilon^*, \delta^*)$; at the far marker site it is the $(\varepsilon^*, \delta^*)$ -dependent cost of Corollary 43. (M2) *Shield law*. A maximal interior run of L forced gap edges costs q^L and carries $L - 1$ isolated cycles; hence the gap-bridge kernel is $\mathfrak{g}(q, y) = q/(1 - qy)$ and the gap-marked bulk kernel is the $K_{\mathfrak{g}}$ of (21). (M3) *Decomposition*. The bivariate series of §7.1 is

$$\mathcal{W}(x, y) = \sum_{\varepsilon^*, \delta^*} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} \mu^{(y)} \rangle + \mathcal{W}_0, \quad (25)$$

with $\mathcal{T}$ the travel operator of Definition 23, $\mathcal{W}_0$ the $k = 0$ sector, and $\lambda^{(y)}, \mu^{(y)}$ the junction-dressed bulk boundary vectors, $\mu^{(y)}$ given by (24) and $\lambda^{(y)}$ by the far-junction cost of Corollary 43, both formed from the generating functions $B_\sigma(q, y)$ of the bulk runs with the gap-marked kernel (21) whose junction-adjacent deposit has magnitude σ , so that $\sum_\sigma B_\sigma = B(q, y)$. The travel assignment $e_s = 2q^{1+s}$ of Definition 23 charges each travel edge one half-site, so a run of $|k|$ edges, which carries $|k| - 1$ interior sites, is weighted x times its relaxed weight; that single factor of x is a normalisation of the travel block, absorbed into $\lambda^{(y)}$, and affects no pole, no order and no non-vanishing statement below.

What §5.6 settles, and what it does not, is this. Inside the optimisation of Definition 38, (M1) is Corollary 41 together with Corollary 43, and it is proved; the marker clause is proved in the corrected form, the earlier form being false (Remark 44). Of (M2), the cost q^L and the inequality $c \geq L - 1$ per interior run are proved (Corollaries 41, 42 and Proposition 45); the reverse inequality $c \leq L - 1$ is verified and not proved (Remark 46). (M3) is not proved. And that the optimisation of Definition 38 computes the relaxed word length at all is the verified, unproved input of [4] (Remark 39). So (M) as a statement about the growth series remains verified and not proved, and its weakest links are now named: the metric formula, the reverse shield inequality, and (M3). *Every statement from here to the end of this section, and every statement elsewhere in this paper that cites one of them, is conditional on (M); this is not repeated in the individual statements.* Nothing in §§2–4.2, §4, §§5.1–5.6 above, or Appendices A–B uses it: those speak only about the explicit series of (2), about P_{12} , about the operators just defined, and about the optimisation of Definition 38.

Under (M), $V = \mathcal{W}(x, 1)$ and $U = \mathcal{W}(x, q)$ (§7.1), the two evaluations differing only through $\mathbf{g}_V = \mathbf{g}(q, 1) = \frac{q}{1-q}$ and $\mathbf{g}_U = \mathbf{g}(q, q) = \frac{q}{1-q^2}$; and $\lambda^{(y)}, \mu^{(y)}, \mathcal{W}_0$ are holomorphic at a travel pole q_m , by (23), as soon as $S_e(q_m) \neq 0$ and $1 - \mathbf{g}t_1 \neq 0$ there, which is Proposition 65 and Remark 36.

Proposition 48 (the shape of the local form). *Let q_m be a travel pole, let $p \geq 1$ be the order of the zero of $1 - \Sigma_1$ there (finite, Lemma 74), and let π be the order of the pole of $(I - \mathcal{T})^{-1}$ there (finite, Lemma 32). Then $\pi \geq p \geq 1$ and, for each y ,*

$$\mathcal{W}(x, y) = \frac{\mathcal{N}_y}{(q - q_m)^\pi} + (\text{poles of order } < \pi) + H_m, \quad H_m \text{ holomorphic at } q_m,$$

with the leading numerator the rank-one pairing

$$\mathcal{N}_y = c_\star \Pi_y(q_m), \quad \Pi_y(q_m) := \sum_{\varepsilon^*, \delta^*} \langle \lambda^{(y)}, R \rangle \langle L, \mu^{(y)} \rangle, \quad c_\star \neq 0, \quad (26)$$

where R spans $\ker(I - \mathcal{T}(q_m))$, normalised by its sections,

$$\sum_{s \geq 0} q^{ks} R_s = \Sigma_k(q_m) \quad (k \geq 0), \quad \text{in particular } \sum_s R_s = \Sigma_0(q_m), \quad \sum_s q^s R_s = \Sigma_1(q_m) = 1,$$

and L spans $\ker(I - \mathcal{T}(q_m)^\top)$, explicitly $L_s = R_s/e_s$. In particular $\Pi_y(q_m) \neq 0$ implies that q_m is a pole of $\mathcal{W}(\cdot, y)$, of order exactly π .

Proof. Write $\mathcal{R}(q) = (I - \mathcal{T}(q))^{-1} = \sum_{j=1}^\pi A_j(q - q_m)^{-j} + \text{hol}$, with $A_\pi \neq 0$. Multiplying $(I - \mathcal{T}(q))\mathcal{R}(q) = I$ by $(q - q_m)^\pi$ and letting $q \rightarrow q_m$ gives $(I - \mathcal{T}(q_m))A_\pi = 0$, so $\text{ran } A_\pi \subseteq \ker(I - \mathcal{T}(q_m))$, which is one-dimensional by Lemma 33; the same step on $\mathcal{R}(q)(I - \mathcal{T}(q)) = I$ gives $A_\pi(I - \mathcal{T}(q_m)) = 0$. Hence $A_\pi x = c_\star \langle L, x \rangle R$ with $c_\star \neq 0$, R spanning the kernel and L the cokernel. The normalisation of R is Lemma 33: $G_k^R = G_1^R \Sigma_k$ and $\Sigma_1(q_m) = 1$, so $G_1^R = 1$ gives $G_k^R = \Sigma_k(q_m)$. That $L_s = R_s/e_s$ spans the cokernel follows from $\mathcal{T} = D_e K$ with $D_e = \text{diag}(e_s)$ and K symmetric: from $D_e K R = R$ one gets $K D_e (D_e^{-1} R) = K R = D_e^{-1} R$, that is $\mathcal{T}^\top L = L$, and $L \in \ell^\infty$ because $|R_s| \leq |e_s| \|R\|_{\ell^1}$; the cokernel is one-dimensional by Fredholm. Substituting the Laurent expansion into (25), and using that $\lambda^{(y)}, \mu^{(y)}, \mathcal{W}_0$ are holomorphic at q_m , gives the displayed local form with $\mathcal{N}_y$ as in (26); the coefficient of $(q - q_m)^{-\pi}$ is non-zero exactly when $\Pi_y(q_m) \neq 0$, and then q_m is a pole of order π . Finally $\langle \mathbf{1}, \mathcal{R}E \rangle = \Sigma_0/(1 - \Sigma_1)$ (Proposition 27) has a pole of order exactly p at q_m , since $\Sigma_0(q_m) \neq 0$ (Proposition 21), while a matrix element of $\mathcal{R}$ has pole order at most π ; hence $\pi \geq p$. $\square$

Proposition 49 (the marker junction is not rank one). *The junction matrix $J[\sigma, s]$ of (24) has rank at least 3. Already the one-sign matrix $[x^{\max(\sigma-1, 2s+1)}]$ has rank at least 3: on $\sigma \in \{0, 4, 8\}$, $2s+1 \in \{1, 5, 9\}$ it is*

$$\begin{pmatrix} x & x^5 & x^9 \\ x^3 & x^5 & x^9 \\ x^7 & x^7 & x^9 \end{pmatrix}, \quad \det = x^9(x-x^3)(x^5-x^7) \neq 0 \quad (0 < x < 1),$$

its rank on $\sigma \in \{0, 2, 4, 6, 8\}$, $2s+1 \in \{1, 3, 5, 7, 9\}$ is 4, and the symmetrised matrix (24) has rank 6 on $\sigma \in \{0, 2, 4, 6, 8, 10\}$, $2s+1 \in \{1, 3, 5, 7, 9, 11\}$. By contrast the gap-bridge term q^{a+b} of (21) is rank one, which is why (27) factorises. Consequently $\mu^{(y)}$, and with it the pairing $\Pi_y(q_m)$, is not a function of the total bulk sum $B(q, y) = \sum_{\sigma} B_{\sigma}(q, y)$ alone: the product form $\mathcal{N}_y = \Sigma_0(q_m) \cdot B(q_m, y) \cdot J$, with J a junction factor independent of the bulk data, does not hold, and $\Pi_y(q_m) \neq 0$ does not follow from $\Sigma_0(q_m) \neq 0$ together with $B(q_m, y) \neq 0$.

Proof. The displayed block is the case $\sigma-1 \in \{-1, 3, 7\}$, $2s+1 \in \{1, 5, 9\}$, which interleaves as $-1 < 1 < 3 < 5 < 7 < 9$; subtracting the second row from the first gives $(x-x^3, 0, 0)$, and expanding along it leaves $(x-x^3)(x^5x^9-x^9x^7) = x^9(x-x^3)(x^5-x^7)$. The two rank statements are exact Gaussian elimination over $\mathbb{Q}$, at $q = \frac{1}{3}$ and at $x = \frac{1}{2}$ respectively (`assembly_certificate.py`, Part 4). For the last sentence: by (M3), $\mu^{(y)}$ is the image of the vector $(B_{\sigma})_{\sigma}$ under the junction matrix. That matrix has rank at least 3, so its rows span a space of dimension at least 3, and two vectors (B_{σ}) with equal sums $\sum_{\sigma} B_{\sigma}$ have different images. A product form would in particular make $\mu^{(y)}$ a function of that sum. $\square$

Theorem 50 (the lifting reduction (L)). *At every travel pole q_m the factorisation (27) holds with $\kappa = t_0b_1 - b_0t_1$ and $t_0 = b_1$, and the dictionary (30) holds identically in $0 < q < 1$, in the completed form of Corollary 31. Consequently the clearing (28),*

$$B_U(q_m) = 0 \iff (1-s)B_V(q_m) + q_mb_0 = 0, \quad s = \frac{q}{1-q}t_1,$$

holds, and both $B_V(q_m) > 0$ and $B_U(q_m) > 0$ at every travel pole.

Proof. Proposition 35, whose two hypotheses are supplied by Corollary 34 with Proposition 65 and by Remark 36; Proposition 30; Corollary 31. The clearing is the algebra of §7.3, which uses nothing beyond (27) and $t_0 = b_1$. For the signs: substituting $\kappa = b_1^2 - b_0t_1$ and $qt_1 = s(1-q)$ into (27) at $\mathbf{g} = \mathbf{g}_V$ gives

$$B_V = \frac{b_0(1-q)(1-s) + qb_1^2}{(1-q)(1-s)},$$

whose numerator and denominator are both positive at a travel pole, since $b_0 > 0$ and $s < 1$ (Corollary 82) and b_1 is real. Then $(1-s)B_V + q_mb_0 > 0$, so $B_U(q_m) \neq 0$ by the clearing, and $B_U > 0$ by the same substitution at $\mathbf{g} = \mathbf{g}_U$. $\square$

Remark 51 (what (L) still owes). That the bulk transfer is the operator of (22), that is, (M1) and (M2). Of these, (M1) is now Corollary 41, the exponent $q^{\max(s, s')}$ of the bulk kernel being $x^{\max(|d|, |d'|)}$ at even deposits; and of (M2) the cost q^L of a gap run and the inequality $c \geq L-1$ are Corollary 42 and Proposition 45, while $c \leq L-1$ is verified and not proved. Both rest on the optimisation of Definition 38 being the relaxed word length, which is the standing input of [4]. Everything the paper previously listed as unproved *downstream* of that, namely the factorisation (27), the reciprocity $t_0 = b_1$, the entry $t_1 = P_{12}/S_e$, and the resolvent lines of (30), is proved above. What Theorem 50 does *not* give is the passage from $B_U(q_m) \neq 0$ to $\mathcal{N}_q \neq 0$: that passage is the product form, and it fails by Proposition 49.

5.8 The one statement that remains open

Hypothesis 52 (the junction pairing (R-J)). With R, L as in Proposition 48 and $\lambda^{(y)}, \mu^{(y)}$ as in (M3),

$$\Pi_y(q_m) = \sum_{\varepsilon^*, \delta^*} \langle \lambda^{(y)}, R \rangle \langle L, \mu^{(y)} \rangle \neq 0 \quad \text{at infinitely many travel poles } q_m,$$

for $y = 1$ and for $y = q$.

This replaces the block assembly (R) as it stood in the previous version of Theorem 56. It is strictly weaker: (R) asserted in addition a product form for the pairing, and that product form is unavailable by Proposition 49. Three remarks on its shape.

(i) *The shape of the pairing.* A marker junction with a left bulk deposit of magnitude σ and sign $\pm$ contributes, in the travel variable s ($|d_R| = 2s + 1$), the vector $\lambda_s = x^{\max(\sigma \mp 1, 2s+1)}$, which equals xq^s for $2s + 1 \geq \sigma \mp 1$ and the constant $x^{\sigma \mp 1}$ below that. Hence, using $\sum_s q^s R_s = \Sigma_1(q_m) = 1$,

$$\langle \lambda, R \rangle = x \sum_s q^s R_s + \sum_{2s+1 < \sigma \mp 1} (x^{\sigma \mp 1} - xq^s) R_s = x + \Delta_\sigma^\pm,$$

a bounded explicit term plus a correction involving only the components R_s with $2s + 1 < \sigma \mp 1$. Summing over σ and the two signs against the bulk-run weights makes every component enter, but the structure “constant plus correction” is the natural target.

(ii) *It is not a statement about Σ_0 and S_0 .* Both of those are non-zero at every travel pole (Proposition 21), and that proposition is used, here and in Theorem 55, for what it does prove. What it does not give is $\Pi_y(q_m) \neq 0$, by Proposition 49.

(iii) *No counterexample is known*, and the pairing is positive at the dominant pole.

Proposition 53 ((R-J) at the dominant pole). $\Pi_1(q_1) > 0$ and $\Pi_q(q_1) > 0$. In particular q_1 is a pole of $V = \mathcal{W}(\cdot, 1)$ and of $\mathcal{W}(\cdot, q)$, the latter being U under the bridge of §7.1.

Proof. By Lemma 37, $S_e > 0$ and $1 - \mathfrak{g}_V t_1 > 0$ on $(0, q_1]$, hence also $1 - \mathfrak{g}_U t_1 > 0$ there. By Corollary 34 and (23) every entry of the gap-marked bulk resolvent at $y \in \{1, q\}$ is therefore holomorphic on a neighbourhood of $[0, q_1]$. Each such entry is a power series in q with non-negative coefficients, so by Pringsheim’s theorem its radius of convergence is a singularity on the positive axis and hence exceeds q_1 ; its value at q_1 is a convergent sum of non-negative terms, positive because the term $2q^b$ is. So $B_\sigma(q_1, y) \geq 0$ with at least one σ strictly positive, and $\lambda^{(y)}, \mu^{(y)}$ have strictly positive entries at q_1 , for $y = 1$ and $y = q$.

For the null vectors, the same argument applied to $\mathcal{T}$ gives $\Phi(q) = (I - \mathcal{T}(q))^{-1} E \geq 0$ entrywise for $0 < q < q_1$, q_1 being the smallest zero of $1 - \Sigma_1$. Put $N(q) = \sum_s \Phi_s(q) = \Sigma_0/(1 - \Sigma_1)$, which tends to $+\infty$ as $q \uparrow q_1$ because $\Sigma_0(q_1) \neq 0$ (Proposition 21), and set $\hat{\Phi} = \Phi/N$, so $\hat{\Phi} \geq 0$ and $\|\hat{\Phi}\|_{\ell^1} = 1$. From $\hat{\Phi} = \mathcal{T}(q)\hat{\Phi} + E/N$ and the norm convergence $\mathcal{T}(q) \rightarrow \mathcal{T}(q_1)$ of compact operators, the family $\{\mathcal{T}(q)\hat{\Phi}(q)\}$ is relatively compact in ℓ^1 , so some sequence $q \uparrow q_1$ has $\hat{\Phi} \rightarrow R$ in ℓ^1 with $R \geq 0$, $\|R\|_{\ell^1} = 1$ and $R = \mathcal{T}(q_1)R$. By Lemma 33 this R spans the kernel, and $L_s = R_s/e_s \geq 0$ with $L \neq 0$. Each of the four summands of $\Pi_y(q_1)$ is therefore a product of two convergent sums of non-negative terms, each containing a strictly positive one, so $\Pi_y(q_1) > 0$. The last sentence is Proposition 48. $\square$

The difficulty in (R-J) is therefore uniformity in m , not any single m . The positivity argument above fails as soon as q_m lies above a zero of S_e , which happens for all large m : by Proposition 17 the zeros of S_e and of $1 - \Sigma_1$ are the zeros of one q -cosine sampled a half q -step apart, so they interlace.

5.9 Certificate

Remark 54 (the computational certificate). `code/zeta_probe/assembly_certificate.py` re-derives the statements of this section in two independent arithmetic models and exits 0 with every check passing. Numerical agreement is evidence and is not a proof; the proofs above are what carry the statements. What the certificate adds is an independent arithmetic model and the hypothesis deletions.

Part 1, exact arithmetic in $\mathbb{Z}[[q]]/(q^{N+1})$ at the two truncation orders $N = 28$ and $N = 40$, no floating point: Propositions 27, 28 and 30 and Corollaries 29 and 31, coefficient by coefficient; the bulk block reproduces the bulk-run counts 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 706, 594 and the travel block the travel-run counts 2, 6, 10, 26, 54, 114, 274, 582, 1298; $P_{11} = 1 + T_0$, $P_{12} = T_1$, $P_{21} = -S_0$, $P_{22} = S_e = 1 - S_1$, $P_{11}S_e + P_{12}S_0 = 1$, the Hahn–Exton form (32), and $S_o = \frac{1-q}{2q}S_0$.

Part 2, exact arithmetic in $\mathbb{Q}[\mathfrak{g}][[q]]/(q^{19})$: the factorisation of Proposition 35 holds identically in the free parameter $\mathfrak{g}$, and three hypothesis deletions each break it, with the first surviving order recorded. A rank-two gap term fails at q^6 ; a gap term q^{a+2b} in place of q^{a+b} fails at q^4 ; and the asymmetric gapless kernel $q^{\max(a,b)} + q^{a+2b}$ breaks $t_0 = b_1$ at q^8 .

Part 3, two precisions, 40 and 70 significant decimal digits: at the first ten travel poles, $q_1 = 0.4494536306$ through $q_{10} = 0.997756895$, the dictionary $b_0 = S_0/S_e$, $t_0 = b_1 = S_1/S_e$, $t_1 = P_{12}/S_e$ holds to relative error $2.5 \cdot 10^{-26}$ at 40 digits and $8.8 \cdot 10^{-57}$ at 70, the loss of about 14 digits being the cancellation in the alternating ladders at the highest pole; the two precisions agree on the pole locations to $1.6 \cdot 10^{-31}$. The resolvent is evaluated there by the $O(N)$ two-point solve of Corollary 29, that is by the analytic continuation, the defining Neumann series not converging at those q . The Möbius value is cross-checked against an independent dense solve of the truncated system at $q = 0.5, 0.7, q_1$ and $y = 1, q$, agreeing to $6.2 \cdot 10^{-28}$. The gate signs $b_0 > 0$ and $s < 1$ and the lifting increment $(1-s)B_V + qb_0 > 0$ hold at all ten, with $b_0\tau \rightarrow 2$ and $s \rightarrow \frac{1}{4}$.

Part 4, exact rational arithmetic: the one-sign junction matrix $[x^{\max(\sigma-1, 2s+1)}]$ has rank 4 over $\mathbb{Q}$ at $q = \frac{1}{3}$ on the 5×5 block, its interleaved 3×3 minor equals $9/524288$ at $x = \frac{1}{2}$, the symmetrised junction matrix (24) has rank 6 on the 6×6 block at $x = \frac{1}{2}$, and the gap-bridge matrix $[q^{a+b}]$ has rank 1.

Part 5, exact integer arithmetic, `code/zeta_probe/tools/sitecost`: the site cost is computed by two independent exact solvers, a min-cost flow and a transportation dynamic program, which agree on all 1 012 664 supply-demand pairs with entries at most 7; on the same range the closed form $\max(|\alpha|, |\beta|, |\Phi|)$ of Lemma 40 is exact, with 0 misses. Corollary 41 is confirmed on 4 532 157 site configurations, each evaluated by both solvers, namely every crossing count from the minimum to the minimum plus 10, every sign split, both signs of both deposits with $|d| \leq 24$, every travel indicator, all four marker data and all eight site types (bulk interior, travel interior in both directions, the near and far junctions for $k^* > 0$ and $k^* < 0$, and the coincident junction at $k^* = 0$): 0 exceptions, with the cost independent of the crossing counts and of the sign splits in every case. Deleting any one of the three pairing costs is witnessed necessary: replacing the sign-flip bounce cost 2 by 0, 1, 3, 4, the pass cost 1 by 0, 2, 3, or the bounce cost 0 by 1, 2 each breaks Corollary 41, with 2544 to 7392 exceptions out of 7524 configurations and the smallest counterexample recorded. The marker closed forms of Corollary 43 hold with 0 exceptions on the full signed range, and the form $\max(|d_L| - 1, |d_R|)$ fails on 42 of 156 cells at the near junction and on 36 of 156 at its $k^* < 0$ counterpart, all with $d_L < 0$. Corollaries 42 and the shield count are confirmed as recorded in Remark 46.

6 The relaxed series V

The relaxed series is the travel resolvent dressed by the bulk. The resolvent itself is settled unconditionally; passing from it to V costs one structural hypothesis, which is stated here and not elsewhere.

Theorem 55 (the travel resolvent). *The travel resolvent $G_0^T = \Sigma_0/(1 - \Sigma_1)$ is transcendental over $\mathbb{Q}(x)$. This is unconditional: it uses no growth hypothesis, no asymptotic theorem, and no statement about the strand-walk model. Its one numerical input is the contour bound of Lemma 10, of which it consumes only $|T_2(m\pi)| < 1$, against a computed worst value 0.027; that bound is a verified enclosure over the parameter region, not a sampled one, and its arithmetic model is stated with the lemma.*

Proof. Σ_0 and $1 - \Sigma_1$ are holomorphic on $|q| < 1$ (Theorem 2, first clause) and $1 - \Sigma_1 \neq 0$. The zero set of $1 - \Sigma_1$ in $(0, 1)$ is infinite and accumulates exactly at $q = 1$ (Lemma 12, which consumes from Lemma 10 only $|T_2| < 1$, together with Lemma 74). At each such q_m the numerator does not vanish, by Proposition 21. Hence every q_m is a pole of G_0^T , of order the vanishing order of $1 - \Sigma_1$ there, which is finite. So G_0^T has infinitely many poles in $|q| < 1$ accumulating at $|q| = 1$; under $q = x^2$ these are infinitely many poles in $|x| < 1$ accumulating at $|x| = 1$. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$, so no algebraic function has such a pole set; hence G_0^T is transcendental. (The same argument shows it is not D -finite.) $\square$

Theorem 56. *Assume the transfer model (M) of Hypothesis 47 and the structural input*

(R-J) *the junction pairing does not vanish: $\Pi_1(q_m) \neq 0$ at infinitely many travel poles q_m , which is Hypothesis 52 at $y = 1$.*

Then the relaxed growth series V is transcendental over $\mathbb{Q}(x)$.

Proof. By Proposition 48, at a travel pole q_m the relaxed series has the local form $V = \mathcal{N}_1(q - q_m)^{-\pi} + (\text{poles of order } < \pi) + H_m$ with H_m holomorphic, $\pi \geq 1$, and $\mathcal{N}_1 = c_\star \Pi_1(q_m)$ with $c_\star \neq 0$. That proposition consumes two non-vanishing statements, and they are separate: $S_e(q_m) \neq 0$, in the quantitative form $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$ of Proposition 65, which makes the bulk boundary vectors $\lambda^{(1)}, \mu^{(1)}$ holomorphic at q_m and says nothing about vanishing; and $\Sigma_0(q_m) \neq 0$, from Proposition 21, which pins the pole order $\pi \geq p \geq 1$. What neither supplies is $\Pi_1(q_m) \neq 0$, by Proposition 49, and that is exactly (R-J).

Under (R-J) the leading Laurent coefficient is non-zero at infinitely many travel poles, so V has a pole at each of those. The travel poles are infinite in number and accumulate at the boundary of the disc of convergence (Lemma 12) and are locally finite in $(0, 1)$ (Lemma 74), so an infinite subset of them accumulates at $q = 1$ and nowhere else. Hence V has infinitely many poles in $|q| < 1$ accumulating at $|q| = 1$; under $q = x^2$ these are infinitely many poles in $|x| < 1$ accumulating at $|x| = 1$. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$, so an infinite pole set accumulating at the boundary is impossible for one; hence V is transcendental. No comparison of T_2 against $\cos W$ is made or needed, and none is available at this strength, the proved bound on $|T_2|$ being $0.17\tau^{1/4}$ (Lemma 10), which exceeds $\sqrt{\tau}/2$ as $\tau \rightarrow 0$; Remark 9 sharpens the size of T_2 and is not used. $\square$

Remark 57 (the standing of (R-J), and what became of (R)). *What (R) was.* Route B decomposes a relaxed element into two bulk runs, one travel interval of length $|k|$ and four marker junctions; (R) was that decomposition read as a local form for V at a travel pole, *together with* a product form for its numerator, $\mathcal{N}_V = \Sigma_0 \cdot S_0/(1 - S_1) \cdot J$ with J a junction factor.

What is now proved. The k -sum telescoping (Proposition 27), the bulk dressing (Proposition 28), and the shape of the local form, with a rank-one numerator paired against the null vector of the travel transfer (Proposition 48). These are the whole of (R) except the product form.

What is refuted. The product form. The marker junction cost is $\max(|a_L| - 1, |a_R|)$, a max-coupling of rank at least 3 with an explicit non-zero 3×3 minor (Proposition 49), so the numerator is not a multiple of the total bulk sum and the step “ $\mathcal{N}_V(q_m) \neq 0$ because $\Sigma_0(q_m) \neq 0$ and $S_0(q_m) \neq 0$ ” is not available. Proposition 21 is unaffected and is used, in Theorem 55 and inside Proposition 48; what fails is the deduction that leant on it here.

What is open. Exactly one statement, (R-J): that the junction pairing Π_1 is non-zero at infinitely many travel poles. It holds at the dominant pole (Proposition 53) and no counterexample is known; the difficulty is uniformity in m . It is strictly weaker than (R).

Where it is used. Only in Theorem 56 and, at $y = q$, in Theorem 61. Theorem 55, Propositions 21 and 65, Corollary 58, §4 and Appendices A–B do not touch it. All of §5 additionally rests on the transfer model (M) (Hypothesis 47), which is verified and not proved.

Evidence. The blocks reproduce their own combinatorial series exactly: $S_0/(1 - S_1)$ against the bulk-run counts 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 706, 594 and $\Sigma_0/(1 - \Sigma_1)$ against the travel-run counts 2, 6, 10, 26, 54, 114, 274, 582, 1298 (`blocks_growth.py`, and `assembly_certificate.py` Part 1 in exact integer arithmetic at two truncation orders). V itself is computed to 131 exact terms by an independent catalytic transfer, not from the blocks.

Corollary 58. *The exponential growth rate of v_n is $\beta_2 = 1.4916177871\dots$, with β_2^2 the reciprocal of the smallest travel pole q_1 ; the value $\frac{3}{2}$ is excluded. It needs only the location of the dominant travel pole: not Remark 9, not Appendix A, and not Appendix B. It does share with Theorem 56 the transfer model (M), which is what places q_1 on V rather than on G_0^T alone, but it does not need (R-J): at the dominant pole the junction pairing is a sum of non-negative terms and is positive (Proposition 53). The lower bound β_2 for u_n and v_n is free even of (M), coming from the pure-travel subcount [4]. The same rate for u_n follows from Proposition 1 (hypothesis (T) of Theorem 61) together with the same positivity at q_1 ; $u_n \leq v_n$ gives the upper bound unconditionally.*

Whether the number β_2 is transcendental is a separate question, open, and with a clean reduction to a value statement for a Hahn–Exton q -Bessel function; it is treated in Appendix C, which is independent of everything else in this paper.

7 The true series U

Transcendence of U needs two things that the analysis of §4 and Appendix B does not supply: that U has the travel poles at all, and that the residue of U at each of them is non-zero. The first is the combinatorial hypothesis (T), which is not proved here or in [6]. The second was previously the hypothesis (L); its algebraic content is now a theorem (§5.5, Theorem 50), and what survives of it is the junction non-vanishing (R-J) at $y = q$, the same statement that Theorem 56 consumes at $y = 1$. This section states (T) in full, with its dependencies, with every place it is used, and with what would close it, and records what the lifting reduction now does and does not give, before turning to the analytic content. A reader should not have to open [6] to see what Theorem 61 rests on.

7.1 The bivariate bridge

With $c(g) \in \mathbb{Z}_{\geq 0}$ the cycle count of §1, set

$$\mathcal{W}(x, y) = \sum_{n, c \geq 0} w_{n, c} x^n y^c, \quad w_{n, c} = \#\{g : \ell_R(g) = n, c(g) = c\},$$

the grading being by *relaxed* length. Then $V(x) = \mathcal{W}(x, 1)$ and, since $\ell_T = \ell_R + 2c$ and $q = x^2$, $U(x) = \mathcal{W}(x, x^2) = \mathcal{W}(x, q)$: the two series are two evaluations of one bivariate object, at $y = 1$ and at $y = q$, and everything separating them is carried by the marker y . The second identity is the one place where the metric identity $\ell_T = \ell_R + 2c$ is used, and it is that identity whose lower bound is open (§7.2). Both evaluations are validated against enumeration to the radius of the enumeration, $u_0, \dots, u_{22}$ and $v_0, \dots, v_{14}$ exactly [6].

7.2 The first input (T): the travel block carries no marker

(T) is Proposition 1: the denominator $1 - \Sigma_1$ is y -free, hence literally the same series in $\mathcal{W}(x, 1)$ and in $\mathcal{W}(x, q)$, so U has the travel poles q_m of §2.

Where it is used. In the proof of Theorem 61, to transport the pole set of Lemma 12 from V to U , and in the same way in Corollary 58 for the growth rate of u_n . Nothing in §4, Appendix A or Appendix B touches it.

What it rests on. Two unproved statements. First, the localisation of cycles: a pure-travel element, one whose lamp support lies inside the travel interval, has $c = 0$. This is checked exhaustively, 0 violations on the 3869 pure-travel elements of true length ≤ 16 , and is expected because the travel interval is the forced single backbone of the Eulerian trail while an isolated cycle is a closed bulk excursion separated from that backbone by a reachability gap. Second, the metric identity $\ell_T = \ell_R + 2c$, whose inequality $\ell_T \leq \ell_R + 2c$ is proved in [4] and whose reverse inequality is open there and verified on the 275 823 elements of the radius-24 ball. Granting the first, pure-travel elements have $\ell_T = \ell_R$, the two gradings of the pure-travel generating function agree termwise, and the travel recursion (2), assembled from pure-travel runs alone, is computed inside a $c \equiv 0$ sector, so the marker cannot enter it.

What would close it. A proof of $\ell_T \geq \ell_R + 2c$ together with a proof of the localisation. Both are finite-state statements about strand walks; neither is analytic.

Evidence. The pure-travel generating function computed in both gradings to radius 18 is identical coefficient by coefficient, 1, 3, 5, 8, 12, 17, 24, 35, 52, 78, 116, 173, 256, 383, 572, 858, 1276, 1907, 2832, with ratio tending to β_2 .

7.3 The lifting reduction (L), and what it now gives

The marker enters the block assembly through exactly one factor, the gap-bridge kernel $\mathfrak{g}(q, y) = q/(1 - qy)$ of (M2), summing a maximal run of L forced gap edges, each of weight q and each an isolated cycle except the one shielded by the inner block's turn-around. Its two evaluations, $\mathfrak{g}_V = \mathfrak{g}(q, 1) = \frac{q}{1-q}$ and $\mathfrak{g}_U = \mathfrak{g}(q, q) = \frac{q}{1-q^2} = \frac{\mathfrak{g}_V}{1+q}$, are holomorphic and non-zero on $(0, 1)$, hence at every travel pole.

The bulk resolvent is semiseparable, $M_0[b, a] = 2q^b q^{\max(a, b)}$ plus a rank-one gap term of weight $\mathfrak{g}$, and its Riccati solution is a Möbius function of $\mathfrak{g}$:

$$B(q, y) = \frac{b_0 + \mathfrak{g} \kappa}{1 - \mathfrak{g} t_1}, \quad \kappa = t_0 b_1 - b_0 t_1, \quad (27)$$

with b_0, b_1, t_0, t_1 the y -free gapless-resolvent sums of Proposition 28, which satisfy the reciprocity $t_0 = b_1$ (Proposition 30). The factorisation (27) is proved as Proposition 35; its two hypotheses are $S_e \neq 0$ and $s \neq 1$, which the paper was carrying already (Remark 36). Write $s = \mathfrak{g}_V t_1$, so that $q t_1 = s(1 - q)$. Clearing denominators in (27) at $\mathfrak{g} = \mathfrak{g}_V$ and at $\mathfrak{g} = \mathfrak{g}_U$ gives the two genuineness polynomials

$$B_V(q_m) = 0 \iff b_0(1 - q)(1 - s) + q b_1^2 = 0, \quad B_U(q_m) = 0 \iff b_0(1 - q)(1 + q - s) + q b_1^2 = 0,$$

which differ by exactly $q(1-q)b_0$. The first equals $(1-q)(1-s)B_V$, so on dividing by $1-q \neq 0$,

$$B_U(q_m) = 0 \iff (1-s)B_V(q_m) + q_m b_0 = 0. \quad (28)$$

Hence if at a travel pole $s < 1$, $b_0 > 0$ and $B_V(q_m) > 0$, the left-hand side of (28) is strictly positive and the gap-marked bulk sum $B_U(q_m)$ is non-zero: B_U is B_V plus the strictly positive increment $q_m b_0/(1-s)$. The factor $B_V(q_m) \neq 0$ is the genuineness of V 's own bulk sum at q_m ; it is *not* imported from Theorem 56, which would be circular, but proved directly in Theorem 50 from (27), $b_0(q_m) > 0$ and $s(q_m) < 1$, using at bottom that $S_0(q_m) \neq 0$ (Proposition 21) and $S_e(q_m) \neq 0$ (Proposition 65) hold at every travel pole. What remains is the *gate*

$$b_0 > 0 \quad \text{and} \quad s := \frac{q}{1-q} t_1 < 1, \quad t_1 = \frac{P_{12}}{S_e}, \quad (29)$$

on the off-diagonal cocycle entry P_{12} , together with the dictionary identifying the entries of (27) with the blocks of (1),

$$S_e = 1 - S_1, \quad S_o = \frac{1-q}{2q} S_0, \quad b_0 = \frac{S_0}{1-S_1} = \frac{2q}{1-q} \cdot \frac{S_o}{S_e}, \quad (30)$$

S_o being identified in Lemma 72 and the two resolvent lines in Corollary 31. The remaining entry $t_1 = P_{12}/S_e$ is an identity between independently defined objects (Corollary 29), not a definition, so all four lines of (30) now stand on the same footing.

What is proved. The pair on which (L) rested, namely the factorisation (27) and the dictionary (30), is Theorem 50. Everything from (27) to (29) is then algebra, and Theorem 50 carries it through to $B_V(q_m) > 0$ and $B_U(q_m) > 0$ at every travel pole. Three ingredients previously listed as verified rather than proved are now proved: the reciprocity $t_0 = b_1$ (Proposition 30), the entry $t_1 = P_{12}/S_e$ (Corollary 29), and the identification of the resolvent quantities b_0, b_1, t_0 with the blocks (Proposition 28). What remains unproved beneath them is the transfer model (M) itself: that the bulk transfer is $M_0[b, a] = 2q^b q^{\max(a,b)}$ with the rank-one gap term, which is (M1) and (M2). On the numerical side, the kernel $K(a, b) = q^{\max(a,b)}$ on active half-sizes reproduces $S_0/(1-S_1) = 2q + 2q^2 + 6q^3 + 2q^4 + 18q^5 + 6q^6 + 42q^7 + \dots$ (`blocks_growth.py`), and the whole dictionary is re-derived in exact integer arithmetic and at two precisions in `assembly_certificate.py`.

Where it is used. In the proof of Theorem 61, to reach $B_U(q_m) \neq 0$; and inside Corollary 82, where the half $b_0 > 0$ is proved from a sign chain running through the closed form in (30). The other half of the gate, $|s| < 1$, uses neither: it is Theorem 80 with Lemma 81, and it is unconditional.

What it does not give. The passage from $B_U(q_m) \neq 0$ to a non-zero residue of U at q_m . That passage is the product form for the assembled numerator, and it fails: the marker junction is a coupling of rank at least 3 (Proposition 49), so the residue is the junction pairing $\Pi_q(q_m)$ of (26) and not a multiple of $B_U(q_m)$. Its non-vanishing is (R-J) at $y = q$, Hypothesis 52, the same statement Theorem 56 consumes at $y = 1$.

An equivalent earlier form, recorded because it is how the reduction was first isolated: $B_U(q_m) = 0$ iff $\mathbf{g}_U$ meets the Möbius zero $\mathbf{g}_0 = -b_0/\kappa$ of (27). Since $\mathbf{g}_U = q/(1-q^2) \rightarrow \infty$ along the poles, B_U is non-zero as soon as $\mathbf{g}_0$ stays bounded, which the travel-pole limits $b_0\tau \rightarrow 2$, $t_1 \sim \tau/4$, $b_1^2 \sim 2/\tau$ deliver as $\mathbf{g}_0 \rightarrow -1$. The sign form (28) is sharper and is what is used.

7.4 The gate block in closed form

Both columns of the cocycle satisfy the scalar three-term recursion $y_{n+1} = (1+q^3-2(1-q)q^{2n+2})y_n - q^3y_{n-1}$; setting $x = q^n$ turns it into the linear q -difference equation

$$Y(qx) + q^3Y(x/q) = (1+q^3-2(1-q)q^2x^2)Y(x), \quad (31)$$

whose regular solution is, in closed form, a Hahn–Exton q -Bessel function [17, 14, 23],

$$Y_3(x) = \sum_{k \geq 0} d_k x^{2k+3}, \quad d_k = \frac{(-2)^k (1-q)^k q^{k^2+3k}}{(q^2; q^2)_k (q^5; q^2)_k} = J_{3/2}^{(3)} \text{ normalised}, \quad (32)$$

and $P_{12} = \frac{2q^3}{1-q^3} Y_3(1)$. The denominator S_e is the block of (1), controlled by Lemma 81. It remains to bound the numerator residual against its confluent limit; this is where the two-Bessel identity enters.

7.5 The amplitude atom, formally verified

As $q \rightarrow 1$, equation (31) degenerates to $\tau Y'' - (2\tau/x)Y' + 2Y = 0$, solved by $x^{3/2} J_{3/2}(Wx)$. The q -corrected connection coefficient across the turning point $X = \nu = \frac{3}{2}$ is governed by the following identity, whose point is that the confluence is *entire*; it passes *through* the turning point, bypassing the Liouville–Green singularity where asymptotic methods stall.

Lemma 59 (two-Bessel confluence; amplitude normalisation). *The regular solution of (31) has the confluent form*

$$Y_3(x) = \gamma_{\text{cl}} x^{3/2} \left[J_{3/2}(X) + \frac{\tau X}{2} J_{5/2}(X) \right] + O(\tau^2), \quad X = wx, \quad \gamma_{\text{cl}} = \frac{3\sqrt{\pi}}{4} \left(\frac{w}{2} \right)^{-3/2},$$

and its single-solution envelope satisfies $A/A_{\text{cl}} = 1 + \tau h(X) + O(\tau^2)$ with the explicit rational

$$h(X) = \frac{1 + 3/(2X^2)}{1 + 1/X^2} \in [1, \tfrac{3}{2}] \quad (X > 0).$$

Consequently $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$ with explicit constant $\frac{3}{2}$.

Proof. From the q -Pochhammer confluence $d_k = \tau^{-k} D_k (1 - k\tau + O(\tau^2))$ with $D_k = (-1)^k / [2^k k! (\frac{5}{2})_k]$, the leading sum is $\gamma_{\text{cl}} x^{3/2} J_{3/2}(X)$ by the Bessel representation of ${}_0F_1$, and the $-k\tau$ correction contributes $\frac{\tau X}{2} \gamma_{\text{cl}} x^{3/2} J_{5/2}(X)$ via ${}_0F_1(\cdot; \frac{7}{2}; \cdot)$. Both series are entire in X . The envelope of $J_{3/2} + \frac{\tau X}{2} J_{5/2}$ follows from $A^2 = M_{3/2}^2 + (\frac{\tau X}{2})^2 M_{5/2}^2 + \tau X (J_{3/2} J_{5/2} + Y_{3/2} Y_{5/2})$ and the closed half-integer cross-term identity

$$J_{3/2}(X) J_{5/2}(X) + Y_{3/2}(X) Y_{5/2}(X) = \frac{4}{\pi X^2} \left(1 + \frac{3}{2X^2} \right),$$

which with $M_{3/2}^2 = \frac{2}{\pi X} (1 + \frac{1}{X^2})$ gives $h(X) = (1 + \frac{3}{2X^2}) / (1 + \frac{1}{X^2})$; this is monotone decreasing from $h(0^+) = \frac{3}{2}$ to $h(\infty) = 1$, so $1 \leq h \leq \frac{3}{2}$. $\square$

Remark 60 (machine verification). The cross-term identity and the bound $1 \leq h(X) \leq \frac{3}{2}$ have been formalised and checked in Lean 4 against Mathlib (file `AtomN.lean`); the proof terms depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The identity is proved by a single `linear_combination` off $\sin^2 + \cos^2 = 1$, and the bound by `nlinarith`.

Theorem 61. *Assume the transfer model (M) of Hypothesis 47 and the two structural inputs, neither of which this paper proves:*

- (T) *Proposition 1: $1 - \Sigma_1$ is the identical, cycle-marker-free denominator in U and in V , so U inherits the travel poles q_m (§7.2);*

(R-J) Hypothesis 52 at $y = q$: the junction pairing $\Pi_q(q_m)$ of (26) is non-zero at infinitely many travel poles.

Then the true growth series U is transcendental over $\mathbb{Q}(x)$.

The hypothesis (L) that earlier versions of this theorem carried is gone: its content, the factorisation (27) with the dictionary (30) and the reciprocity $t_0 = b_1$, is Theorem 50, and it yields $B_U(q_m) \neq 0$ at every travel pole. What it does not yield is the residue, because the marker junction is not rank one (Proposition 49); that is why (R-J) appears here as it does in Theorem 56.

Unconditionally, and independently of (M), (T) and (R-J), the half $|s| < 1$ of the gate (29) holds at every travel pole. The numerator amplitude estimate $(\star)$ (at every travel pole q_m , $|P_{12}(q_m)| w_m^3 < 2$) is Theorem 80 (Appendix B): $|P_{12}| \leq 0.619 \tau^{3/2}$ for $\tau \leq 5 \cdot 10^{-3}$ and $|P_{12}| w^3 \leq 0.75$ at the six poles above that threshold. The passage to $|s| < 1$ is Corollary 82: $|s| \leq 0.983$ on the uniform range and $|s| \leq 0.296$ at the six exceptional poles, with the denominator lower bound $|S_e| \geq 0.63\sqrt{\tau}$ supplied by Lemma 81. The remaining half $b_0 > 0$ is proved in the same corollary from a sign chain through the closed form for b_0 , which is now Corollary 31 and therefore rests on (M) alone. The leading-order picture of Remark 16 (the phase-matched zero giving $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the threshold $1/\sqrt{2}$) is consistent with, and explained by, the second-difference identity of Proposition 73.

Proof. By (T), U inherits the travel poles. By Proposition 48 at $y = q$, at each travel pole U has the local form $\mathcal{N}_q(q - q_m)^{-\pi} + (\text{lower order}) + H_m$ with $\mathcal{N}_q = c_\star \Pi_q(q_m)$ and $c_\star \neq 0$; the boundary vectors are holomorphic there because the gate (29) holds at every travel pole. Indeed, on the uniform range $\tau \leq 5 \cdot 10^{-3}$, $|P_{12}| \leq 0.619 \tau^{3/2}$ (Theorem 80) and $|S_e| \geq 0.63\sqrt{\tau}$ (Lemma 81) give $|s| = \frac{q}{1-q} |P_{12}|/|S_e| \leq \frac{\tau}{e^\tau - 1} \cdot \frac{0.619}{0.63} \leq 0.983 < 1$, and $b_0 > 0$ follows from the sign chain of the same lemmas through the closed form of Corollary 31; the six exceptional poles are checked directly (Corollary 82). By (R-J) the leading coefficient $\mathcal{N}_q$ is non-zero at infinitely many q_m , so U has an uncanceled pole at each of those; these accumulate at $q = 1$ (Lemmas 12 and 74), so U has infinitely many poles in the unit disc accumulating at its boundary and is transcendental over $\mathbb{Q}(x)$, as in Theorem 56. Transcendence does not consume the exceptional-pole enumeration of Theorem 80: the uniform range $\tau \leq 5 \cdot 10^{-3}$ alone carries infinitely many poles (Lemma 12 produces them at $\tau \downarrow 0$); the enumeration only upgrades “at infinitely many travel poles” to “at every travel pole” in the gate. For orientation: the computed values over $m = 5, \dots, 71$ are $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.1768$ and $s \rightarrow \frac{1}{4}$, well inside every bound above. $\square$

Remark 62 (the assembly is machine-checked). The composition just performed (the gate arithmetic, the two halves of (29), the passage to the pole set, and the step from infinitely many poles to non-algebraicity) is verified in Lean 4 (`UAssembly.lean`, seven theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`), each analytic input entering as an explicit hypothesis: Theorem 80, Lemma 81, Corollary 82, Lemma 12, the reduction (L) (the Lean hypothesis `hred`), and the finiteness of an algebraic function’s pole set in the disc. The file formalises *no* analysis: no q -series, no steepest descent, no numerics; its content is exactly that those hypotheses *compose*, so its hypothesis list is an audit of what Theorem 61 rests on, not a verification of any of them. In particular `hred`, the hypothesis “the gate implies a non-zero residue”, is assumed there and not derived; Theorem 50 proves its first half, that the gate implies $B_U(q_m) \neq 0$, and Proposition 49 shows that its second half is not the identity the Lean file’s phrasing suggests but the junction non-vanishing (R-J). The machine check therefore still says nothing about the standing of that passage. The quantitative step is isolated as `gate_ratio_lt_one`, whose hypothesis $K < c$ (numerator constant strictly below denominator constant) is what the gate needs and what a numerator bound alone cannot supply: with $K = 0.619$ from Theorem 80 and the constant 0.35 of Proposition 65 it is *false*,

which is precisely why Lemma 81's 0.63 is required.

Remark 63 (what is proved, and what (T) and (R-J) still owe). The unconditional content of this section is the gate: at every travel pole, $|P_{12}| \leq 0.619\tau^{3/2}$ (Theorem 80), $|S_e| \geq 0.63\sqrt{\tau}$ (Lemma 81) and hence $|s| \leq 0.983 < 1$ (Corollary 82), together with the infinitude and accumulation of the travel poles (Lemma 12) and the fact that these facts compose to non-algebraicity (Remark 62). Conditional on the transfer model (M) alone, the lifting reduction is now proved: the factorisation (27), the reciprocity $t_0 = b_1$, the full dictionary (30), the half $b_0 > 0$ of the gate, and $B_U(q_m) \neq 0$ at every travel pole (Theorem 50).

What is *not* proved is (T), and (R-J), and (M) beneath both. (T) is stated in full in §7.2 with its dependencies and with what would close it; it rests on the localisation of cycles and on the open lower bound in $\ell_T = \ell_R + 2c$, and its standing is that of a verified conjecture. (R-J) is Hypothesis 52: it holds at the dominant pole (Proposition 53) and no counterexample is known, and it is what the refutation of the product form (Proposition 49) leaves behind. (M) is Hypothesis 47, exhaustively verified and unproved. Accordingly the standing of Theorem 61 is the weakest of these three, and the word “unconditionally” is not applied to U anywhere in this paper. Making Theorem 61 unconditional now requires exactly three things: a proof of Proposition 1, a proof of (R-J) at $y = q$, and a proof of (M). None is an analytic obstacle of the kind Appendix B addresses; all three are combinatorial, and all three are absent.

Remark 64 (what the turning-point atom does and does not give). Lemma 59 discharges the amplitude *through the turning point* $X = \frac{3}{2}$, which is the obstruction to any Liouville–Green treatment. It does *not* supply the amplitude at the *pole* $X = w$ to the $O(\tau^2)$ -relative accuracy the numerator gate requires. There the two-Bessel form carries a relative defect whose $O(1)$ part is exactly $\frac{1}{36}$; i.e. the classical turning-point-to-pole connection factor $\frac{36}{35}$, which is itself derivable and, once applied, matches $Y_3(1/q)$ to $O(\tau)$ -relative; and whose *residual* $O(\tau)$ part (the amplitude drift from the turning point to the pole) is not supplied by this route. Because $P_{12} = \frac{2q^3}{3(1-q^3)}Y_3(1/q) - \frac{2}{3}S_e$ is an $O(\sqrt{\tau})$ cancellation, that residual is amplified to the size of P_{12} itself. This residual is closed in Appendix B, not by resumming the amplitude at all, but by the second-difference identity of Proposition 73: P_{12} is the symmetric second difference of the q -cosine at its own zero, so the gate needs only $\max |\psi''| \leq 3.5w$ on a window of width τ , which Theorem 80 proves; Lemma 81 and Corollary 82 then close the remaining halves of the gate ($|S_e| \geq 0.63\sqrt{\tau}$ and $b_0 > 0$). The two off-diagonal cocycle entries obey the Abel relation $x_N Y_N - X_N y_N = -1$ (the discrete q -Wronskian, conserved because $\det M_n \equiv 1$), whence

$$P_{12} = \frac{X_N S_e - 1}{x_N} \quad \text{exactly,}$$

x_N, X_N being the leading components of the same three-term recursion. Hence P_{12} is fixed by the *same* family of Hahn–Exton blocks that controls S_e : one analytic class; that of Remark 9; not two. In this precise sense U sits on V 's footing. What U additionally requires is the pole asymptotics of x_N, X_N ; again Hahn–Exton, again governed by the exact q -Wronskian of Olde Daalhuis [33], $W_q(J_\nu, Y_\nu) = q^{\nu(\nu-1)}(1-q^2)/\Gamma_{q^2}(\frac{1}{2})^2$ for the numerically satisfactory pair, whose $q \rightarrow 1$ expansion is explicit via Moak's q -Stirling formula [28] (e.g. $\Gamma_{q^2}(\frac{1}{2}) = \sqrt{\pi}(1 - \frac{3}{8}\tau + O(\tau^2))$). The block-asymptotic assembly is carried out *elementarily* in Remark 16: the numerator amplitude has a derived closed-form expansion with exact rational constants, and the phase-matching of the travel-pole equation against the amplitude's zero (both collapsing to the same sinh-product family) derives the gate value $1/(4\sqrt{2})$. U 's analytic inputs thereby coincide with V 's: the further pieces (($\star$) and the closure of the gate) are proved in Appendix B. (The orthogonal arithmetic route of §8 would settle U without any of this analysis, but is itself open.)

8 An orthogonal arithmetic route

For each prime p , $U \bmod p \in \mathbb{F}_p[[q]]$ is algebraic over $\mathbb{F}_p(q)$ iff its coefficient sequence is p -automatic (Christol [9, 10]), iff its p -kernel is finite. We find the p -kernel to be *maximally* non-automatic: through the accessible depth its size tracks the full p -ary tree with *no* deficit. (An apparent deficit of 1 at level 3; 1, 4, 13, 39 vs. the full 1, 4, 13, 40 at $p = 3$; is a *comparison-length artifact*: a level- k decimation subsequence retains only $\sim N/p^k$ terms, so at a short prefix two distinct decimations can coincide; the deficit disappears once the comparison length reaches 9.) Two instances are machine-checked in Lean 4; both proofs use `native_decide`, so besides the three standard axioms they depend on the compiler-trust axiom `_native.native_decide.ax_1_1` of the declaration concerned. For the true series itself, the 3-kernel of $(u_n \bmod 3)$ through level 3 is the full ternary tree, deficit 0: all $1 + 3 + 9 + 27 = 40$ decimation words of length ≤ 3 are pairwise distinct: at comparison lengths 9 and 10 alike; on the 271-term prefix $n \leq 270$ (`UKernel.lean`, from the validated `tools/u_modp_rust` output, whose u_n are self-checked against the 43 known terms and byte-identical to the earlier validated $n \leq 180$ run on the overlap). For the sibling block Σ_1 , transcendental over $\mathbb{Q}(q)$ by Corollary 3, the same level-3 full tree is certified in `SigmaKernel.lean` (built from the A/C recursion over $\mathbb{F}_3$), with the short-prefix deficit-1 confirmed spurious. The Berlekamp–Massey [26] linear complexity of $(u_n \bmod 3)$ is $\approx N/2$. This is the strongest finite-data evidence for non-automaticity, and hence for transcendence over $\mathbb{Q}(x)$, independent of all of §§4–7. A proof for U would require a non-automaticity criterion valid for *dense*-support series (the coefficients carry no lacunary gaps), which the standard gap/Mahler methods do not provide.

9 Conclusion

The arithmetic of A396406 separates cleanly. The Pólya–Carlson dichotomy for the catalytic building blocks is unconditional (Theorem 2), and so is the lower bound β_2 for the growth rate; the matching upper bound, and with it the identification of β_2 as the rate, consumes the transfer model (M) and nothing else (Corollary 58). For the two denominators Σ_1, S_1 the exclusion of the rational alternative needs no growth hypothesis: it follows from the infinitude of their level sets in $(0, 1)$, hence from the same Lemma 10 that the rest of the paper consumes (Corollary 3). For the two numerators it rests on Hypothesis 5, verified out to degree 1156 and unproved, except for the joint statement that Σ_0 and S_0 are not both rational, which the pole route supplies. **The travel resolvent $\Sigma_0/(1 - \Sigma_1)$ is transcendental over $\mathbb{Q}(x)$, unconditionally** (Theorem 55), and it needs no growth hypothesis. Its non-cancellation inputs are two and are separate. The denominator input is Proposition 65, proved in Appendix A by an elementary chain with explicit numerical constants: an annulus representation, an Euler bound, a Poisson theta envelope, and a q -Wronskian invariant. No asymptotic theorem is cited on that path. The numerator input is Proposition 21, the identity $\Sigma_0(q_m)S_0(q_m) = 2q_m/(1 - q_m)$, which follows from the closed forms $\Sigma_0 = \frac{2q}{Z}\mathfrak{s}(Z)$, $S_0 = \frac{2q}{z_0}\mathfrak{s}(z_0)$ and the q -Pythagorean identity at a zero of the q -cosine; it is exact, it holds at every pole with no exceptional set, and it consumes no asymptotics. Remark 9 gives the denominator conclusion by a different and shorter route (the saddles $z_\pm^* = \pm iW/2$ with all constants derived explicitly, the two saddles supplying the global cancellation that no partial-summation method can), but that route is a sketch only (Remark 11), and no theorem uses it. Both V and U take their pole set from Lemma 12, which consumes only the rectangle bound $|T_2| \rightarrow 0$ of Lemma 10. The block assembly is developed in §5, on the observation that the travel and the bulk block are the same object, the resolvent of $q^{\max(a,b)}$ against a geometric weight: this proves the travel telescoping, the bulk dressing, the reciprocity $t_0 = b_1$ with its cocycle form $\det P = 1$, the identity $t_1 = P_{12}/S_e$,

the Möbius factorisation (27), and the shape of V 's local form at a travel pole. It also refutes the product form of the numerator: the marker junction has rank at least 3 (Proposition 49). V is **transcendental over $\mathbb{Q}(x)$ conditional on the junction non-vanishing (R-J)** of Theorem 56, which holds at the dominant pole and is open in general (Remark 57). The true series U stands on the same *analytic* footing: its further analytic inputs are proved in Appendix B: the gate block is the symmetric second difference of the q -cosine at its own zero (Proposition 73), the two-interval uniform certificate of Theorem 80 gives $|P_{12}| \leq 0.619 \tau^{3/2} < \tau^{3/2}/\sqrt{2}$ for $\tau \leq 5 \cdot 10^{-3}$ with the six poles above that threshold checked directly, and the exact identity $S_e = \frac{qZ}{2} \mathfrak{s}(Z) - P_{12}$ turns the same certificate into the denominator bound that closes the gate ($|s| \leq 0.983$; Corollary 82), consistent with the phase-matched leading values $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ and $s \rightarrow \frac{1}{4}$. U is **transcendental over $\mathbb{Q}(x)$ conditional on (T) and (R-J)** of Theorem 61: the invariance of the travel block under the cycle marker, and the junction non-vanishing at $y = q$. Both are stated in full, (T) in §7.2 and (R-J) in §5.8; both are combinatorial; and neither is proved. The hypothesis (L) that earlier versions carried has become Theorem 50, and it yields $B_U(q_m) \neq 0$ at every travel pole. Every statement in §5 beyond the operator algebra, and with it Theorems 56 and 61 and Corollary 58, rests in addition on the transfer model (M) (Hypothesis 47), which is exhaustively verified and not proved. No analytic input separates U from V ; what separates them is (T), which V does not use. The turning-point amplitude atom on the U -side is the elementary rational bound of Lemma 59, formally verified in Lean 4. The independent mod- p route of §8 would settle U by a completely different mechanism, if a dense-support non-automaticity criterion can be found. Whether the *number* β_2 is transcendental is untouched by all of this and remains open; Appendix C reduces it to a value statement for a Hahn–Exton q -Bessel function and shows the obstruction to be the arithmetic ratio $\rho = \frac{1}{2}$.

A An explicit lower bound for S_e at the travel poles

Throughout $z_0 = \sqrt{2(1-q)}$, $Z = z_0/\sqrt{q}$, $w = \sqrt{2/\tau}$, $f(z) = \cos(z; q^2) = C(z^2)$, $C(y) = \sum_{j \geq 0} (-1)^j q^{j^2+j} y^j / (q; q)_{2j}$, and $s(z) = \sin(z; q^2)$; by Proposition 17, $S_e = f(z_0)$ and the travel poles are the solutions of $f(Z) = 0$.

Proposition 65. *At every travel pole q_m with $\tau_m \leq 1.29 \cdot 10^{-4}$ one has $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$. The finitely many (Lemma 74) poles with $\tau_m > 1.29 \cdot 10^{-4}$ (the dominant pole $q_1 = 0.449453 \dots$ ($|S_e|/\sqrt{\tau_1} = 0.581$; `star_gate_certificate.py`, Corollary 82) and the thirty-nine tabulated below it, $\tau_{41} = 1.2354 \cdot 10^{-4}$ being the first below the threshold) satisfy the same bound by direct high-precision computation.*

The proof occupies the rest of the appendix: an annulus representation turns all derivative bounds into cancellation-free integrals, and the pole condition supplies the value-level input.

Lemma 66. *For $0 < r < 1$, $C(y) = \frac{1}{2\pi} \int_0^{2\pi} \Theta(y/x^2)(x; q)_\infty^{-1} d\theta$ with $x = re^{i\theta}$ and $\Theta(t) = \sum_{j \in \mathbb{Z}} (-1)^j q^{j^2+j} t^j = (q^2; q^2)_\infty (q^2 t; q^2)_\infty (1/t; q^2)_\infty$.*

Proof. Euler's identity gives $1/(q; q)_{2j} = [x^{2j}](x; q)_\infty^{-1}$; extract the coefficient by Cauchy's formula on $|x| = r$. For $j \leq -1$ the integrand $x^{-2j}(x; q)_\infty^{-1}$ is analytic in $|x| < 1$ (the zeros of $(x; q)_\infty$ lie at $x = q^{-k}$, $|q^{-k}| \geq 1$), so the negative-index terms vanish and the sum completes to the bilateral one, which is the triple product. Equivalently this is the $p = q^2$ case of the q -Borel- q -Laplace inversion of [47] and [29, Lemma 1], the q^2 -Borel transform of C being the even part of $(x; q)_\infty^{-1}$. $\square$

Lemma 67. *For $0 < r \leq \frac{1}{2}$: $(x; q)_\infty^{-1} \leq \exp(w_r \cos \theta + K_r)$ in modulus, with $w_r = r/(1-q)$ and $K_r = r^2/(2(1-r)(1-q^2))$.*

Proof. $-\log|1-u| = \operatorname{Re} u + \operatorname{Re} \sum_{n \geq 2} u^n/n \leq \operatorname{Re} u + \frac{1}{2}|u|^2/(1-|u|)$ for $|u| < 1$; summing over $u = xq^j$ gives $\sum_j \operatorname{Re}(xq^j) = w_r \cos \theta$ and $\sum_j r^2 q^{2j}/(2(1-rq^j)) \leq r^2/(2(1-r)(1-q^2)) = K_r$. $\square$

Lemma 68. *Let $p = q^2$, $t = \rho e^{i\varphi}$, $a = (\log \rho - \tau) + i\tilde{\varphi}$ with $\tilde{\varphi} \in (-\pi, \pi]$ representing $\varphi + \pi$. For $n \in \{0, 1, 2\}$,*

$$|\Theta^{(n)}(t)| \leq \sqrt{\pi/\tau} e^{(\log \rho - \tau)^2/(4\tau)} \rho^{-n} (|a|/(2\tau) + \tau^{-1/2})^n G(\tilde{\varphi}), \quad G(\tilde{\varphi}) = \sum_{k \in \mathbb{Z}} e^{-\pi^2(k - \tilde{\varphi}/2\pi)^2/\tau}.$$

Proof. $\Theta(t) = \theta_p(-pt)$ with $\theta_p(y) = \sum_n p^{n(n-1)/2} y^n$. Poisson summation gives the *exact* identity

$$S(a) := \sum_{j \in \mathbb{Z}} e^{-\tau j^2 + aj} = \sqrt{\pi/\tau} \sum_{k \in \mathbb{Z}} e^{a_k^2/(4\tau)}, \quad a_k := a - 2\pi i k. \quad (33)$$

Since $\operatorname{Re}(a_k^2) = (\log \rho - \tau)^2 - (\tilde{\varphi} - 2\pi k)^2$, the imaginary part of a contributes *decay*, and $\sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)} = e^{(\log \rho - \tau)^2/(4\tau)} G(\tilde{\varphi})$; no cancellation is involved, the term $-(\tilde{\varphi} - 2\pi k)^2$ being precisely what constitutes G . Differentiating (33) in a , the j - and $j(j-1)$ -weighted sums are S' and $S'' - S'$, and $\partial_a e^{a_k^2/(4\tau)} = (a_k/2\tau) e^{a_k^2/(4\tau)}$, so the k -th image carries the weight

$$W_1(a_k) = \frac{|a_k|}{2\tau}, \quad W_2(a_k) = \frac{|a_k|^2}{4\tau^2} + \frac{1}{2\tau} + \frac{|a_k|}{2\tau}.$$

At the radius $\rho = e^\tau$ of Lemma 69 one has $a_k = i(\tilde{\varphi} - 2\pi k)$ exactly, so $|a_k| = |\tilde{\varphi} - 2\pi k|$. To pull the weight out of the k -sum, note that for $k \neq 0$ the excess $|a_k| - |a_0|$ is paid for by the Gaussian factor, which is smaller by $e^{-((\tilde{\varphi} - 2\pi k)^2 - \tilde{\varphi}^2)/(4\tau)}$; the aggregate excess is maximised at the nearest image and is at most $\max_{\delta > 0} 2\delta e^{-\pi\delta/\tau} = 2\tau/(\pi e) \leq 0.2342\tau$, which is below $\tau^{-1/2}$ for $\tau \leq 0.0198$. Hence $\sum_k W_n(a_k) e^{\operatorname{Re}(a_k^2)/(4\tau)} \leq (|a|/(2\tau) + \tau^{-1/2})^n \sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)}$, which is the stated bound. $\square$

Lemma 69. *For $\tau \leq 0.0198$ and every $\xi \in [qZ, Z]$, with $r^2 = \xi^2 e^{-\tau}$: $M_2 := \max_{[qZ, Z]} |f''| \leq 7w^4$.*

Proof. $f''(z) = 2C'(z^2) + 4z^2 C''(z^2)$, and differentiating Lemma 66 under the integral, $|C^{(n)}(y)| \leq (2\pi r^{2n})^{-1} \int_0^{2\pi} |\Theta^{(n)}(t)| (x; q)_\infty^{-1} d\theta$ at $\rho = y/r^2 = e^\tau$, which zeroes $(\log \rho - \tau)^2$. The key step is the exact unfolding: as θ runs over $[0, 2\pi)$ the variable $u = \pi - 2\theta$ runs over two periods, consecutive branches carrying opposite signs of $\sin(u/2)$, so

$$\frac{1}{2\pi} \int_0^{2\pi} e^{w_r \cos \theta} G(\pi - 2\theta) d\theta = \frac{1}{2\pi} \int_{\mathbb{R}} \cosh(w_r \sin \frac{u}{2}) e^{-u^2/(4\tau)} du,$$

an identity; the derivative weights unfold with it, becoming polynomials in u , and $|\sin(u/2)| \leq |u|/2$. The even weight gives an exact Gaussian moment, $\int v^2 e^{-(v-s)^2/(4\tau)} dv = \sqrt{4\pi\tau}(2\tau + s^2)$ with $s = \tau w_r \leq \sqrt{2\tau} e^{\tau/2}$, and $\sqrt{\pi/\tau} \cdot \sqrt{4\pi\tau} = 2\pi$ cancels the prefactors; the odd weight does *not*, giving instead $\int |v| e^{-(v-s)^2/(4\tau)} dv = \sqrt{4\pi\tau} s \operatorname{erf}(s/(2\sqrt{\tau})) + 4\tau e^{-s^2/(4\tau)}$, and the error function and the exponential are kept, not discarded. Writing $s = \tau w_r$, the two assembled factors are

$$Q_1 = \tau^{-1/2} + \frac{s \operatorname{erf}(s/(2\sqrt{\tau})) + 2\sqrt{\tau/\pi} e^{-s^2/(4\tau)}}{2\tau}, \quad Q_2 = \frac{2\tau + s^2}{4\tau^2} + \tau^{-3/2} [\text{same bracket}] + \frac{1}{\tau},$$

the prefactor $\sqrt{\pi/\tau}$ of Lemma 68 cancelling against $\sqrt{4\pi\tau}$ in every term. On the whole range, $\sup \sqrt{\tau} Q_1 = 1.8273$ and $\sup \tau Q_2 = 3.6596$, both attained at $\tau = 0.0198$, $\xi = Z$. With the weights of Lemma 68 this assembles, for $\tau \leq 0.0198$ (so $r \leq 0.1981 < \frac{1}{2}$, $K_r \leq 0.631$, $e^{\tau w_r^2/4} \leq e^{0.506}$), to

$$|C'(y)| \leq e^{K_r - \tau + \tau w_r^2/4} (1.83 \tau^{-1/2}) r^{-2}, \quad |C''(y)| \leq e^{K_r - 2\tau + \tau w_r^2/4} (3.670/\tau) r^{-4}.$$

With $4\xi^2/r^4 \leq (2/\tau)e^{3\tau}(1+2\tau)$ the total $M_2\tau^2$ is at most 25.04 at $\tau = 0.0198$, decreasing to 20.2 at $\tau = 1.29 \cdot 10^{-4}$, in both cases below the target $7w^4\tau^2 = 28$. Those two numbers come from pairing the largest K_r and the largest $\tau w_r^2/4$, both attained at $\xi = Z$, with the largest r^{-2} , attained at $\xi = qZ$; taking instead the supremum over $\xi \in [qZ, Z]$ of the assembled bound at a single ξ gives 23.33 at $\tau = 0.0198$, 20.06 at $\tau = 1.29 \cdot 10^{-4}$ and 19.84 as $\tau \rightarrow 0$, so the true margin against 28 is 16.7% rather than the 10.6% the factored form reports. Every constant in this proof, and the two ξ -suprema, are regenerated by `annulus_certificate.py` at two precisions; see Remark 70. $\square$

Remark 70 (what the certificate checks, and what it found). `annulus_certificate.py` recomputes, at 30 and again at 60 digits with zero disagreement: the window geometry ($\sup r = 0.19802$, $\sup K_r = 0.62963$, $\sup \tau w_r^2/4 = 0.50497$ against the quoted $\frac{1}{2}$, 0.631, 0.506); the excess constant $2/(\pi e) = 0.234199$ of Lemma 68; Lemma 67 against the true $|(x; q)_\infty^{-1}|$ on a (τ, ξ, θ) grid; the unfolding identity of this proof against numerical quadrature; the two moment suprema above; the assembled $M_2\tau^2$ on a 400-point τ grid with the ξ -supremum taken by search rather than at an endpoint; and the chain of Proposition 65 below. It also checks that the bound dominates the true $\max |f''|$ where both are computable, the ratio $\max |f''|/(\text{bound})$ coming out between 0.023 and 0.045 at $\tau = 0.0198$, 0.0127, $5 \cdot 10^{-3}$, 10^{-3} , consistent with Remark 71. Two constants of an earlier version of this proof did not survive it and are corrected above: $\sup \sqrt{\tau} Q_1 = 1.82733$ is not bounded by 1.825, and $2/(\pi e) = 0.234199$ is not bounded by 0.234. Both corrections move $M_2\tau^2$ by less than 0.01 and neither affects any conclusion. The arithmetic model is `mpmath mpf` at the two stated precisions, not interval arithmetic.

Remark 71 (how lossy Lemma 69 is, and why one sample cannot say). The ratio M_2/w^4 does not converge: f is the q -cosine, so $f''(z) \approx -(W/z_0)^2 \cos(zW/z_0)$ with $(W/z_0)^2 = e^{-\tau}/(\tau(1-q)) \sim w^4/4$, while the window $[qZ, Z]$ spans a phase of only $(1-q)Z/\tau \sim \sqrt{2\tau}$ radians. As τ varies the window slides through the cosine and M_2/w^4 sweeps $[0, \frac{1}{4}]$; a single value of τ therefore reports an arbitrary point of that sweep and says nothing about the maximum. The maximum is attained where the window covers an extremum of the phase, i.e. along the extreme-phase points $w = m\pi$, and there $M_2/w^4 \uparrow \frac{1}{4}$: measured 0.2499993 at $w = 120\pi$, against 0.22599 at the top of the range $\tau = 0.0198$ and 0.02337 at $\tau = 0.0163$. Over the 165 values of τ scanned, $M_2 < \frac{1}{4}w^4$ (`m2_certificate.py`, `mpmath` at two precisions, agreeing to ten digits at every point; a scan, not a supremum proof). Lemma 69 is therefore lossy by a factor of about 28 at the top of the range, and its target $7w^4$ clears the measured values by about 31.

Proof of Proposition 65. At a pole $f(Z) = 0$, so the Koornwinder–Swarttouw rules give the exact facts $f(qZ) = qZ s(Z)$ and $s(qZ) = s(Z)$; the mean value theorem on $[qZ, Z]$ then pins $f'(\zeta) = -q s(Z)/(1-q)$ at some $\zeta \in (qZ, Z)$, and a second mean value on $[z_0, Z] \subset [qZ, Z]$ gives

$$|S_e| = (Z - z_0) |f'(\tilde{\xi})| \geq \frac{q(1 - \sqrt{q})}{1 - q} Z |s(Z)| - (1 - \sqrt{q})(1 - q)Z^2 M_2.$$

The sine factor is bounded below by an invariant of Casoratian (q -Wronskian) type for the Hahn–Exton pair; the closed-form q -Wronskian is due to Swarttouw [45] (eq. (4.3.8)); see also [33, eq. (4.21)]. Concretely, with $c_n = \cos(Zq^n; q^2)$, $s_n = \sin(Zq^n; q^2)$ and $F_n = c_n^2 - Zq^{n+1}c_n s_n + q s_n^2$, the rules give the exact telescoping $F_n - F_{n+1} = -Zq^{n+1}(1-q)c_{n+1}s_{n+1}$ with $F_n \rightarrow 1$; the form has smallest eigenvalue $\lambda_n \geq q - Z/2$ (by $\sqrt{a^2 + b^2} \leq a + b$), $\sum_{n \geq 0} Zq^{n+1}(1-q) = qZ$, and the discrete Gronwall gives $F_0 \geq e^{-0.61Z} \geq 1 - 0.61Z$ for $Z \leq \min(0.2, q/3)$; at the pole $F_0 = q s(Z)^2$, so $|s(Z)| \geq \sqrt{1 - 0.61Z}$. (The Gronwall sum is $qZ/(2q - Z)$, whose supremum over the range is 0.5568 Z , so 0.61 has room; $Z \leq \min(0.2, q/3)$ holds throughout, $Z = 0.19999$ at $\tau = 0.0198$ and $Z = 0.016063$ at $\tau = 1.29 \cdot 10^{-4}$.) Inserting Lemma 69, the error fraction is

$28\sqrt{2}(1.02)\sqrt{\tau} \leq 0.46$ for $\tau \leq 1.29 \cdot 10^{-4}$, a range containing every pole with $m \geq 41$, whence $|S_e(q_m)| \geq (1 - 0.02 - 0.46)\sqrt{\tau_m/2} \geq 0.35\sqrt{\tau_m}$ there. Evaluated rather than rounded, the same chain gives $0.3856\sqrt{\tau_m}$ at the threshold and more below it, the exact error fraction being 0.4520 against the quoted 0.46 (`annulus_certificate.py`). The tabulated poles are verified numerically at 60+ digits (`qtrig_verify.py`, `mpmath`, no interval arithmetic: $|S_e|/\sqrt{\tau} \in [0.698, 0.708]$ as asserted by the script; the tabulated set is poles $2, \dots, 60$ in the indexing of §2 (the dominant pole is not among them, its value 0.581 coming from `star_gate_certificate.py`) and the table omits poles 53 and 60 (rows 52, 59), whose $\tau < 7.7 \cdot 10^{-5}$ lies inside the proved range), and the two ranges overlap on every tabulated pole with $m \geq 41$; only poles $2, \dots, 40$ (table rows $1, \dots, 39$) are load-bearing. The polynomial identities underlying the telescoping and the pole reduction are machine-checked (`GramCasoratian.lean`: `F_shift`, `F_drift`, `pole_qsine`, `sigma1_term`; each audits inside the standard three, the first two on `propext`, `Quot.sound` alone; no sorry). $\square$

On the intermediate window $1.29 \cdot 10^{-4} < \tau_m \leq 5 \cdot 10^{-3}$ the bound also follows, with margin, from Lemma 81 ($0.63 > 0.35$), independently of any enumeration of the poles there.

B Proof of the amplitude estimate: the gate holds at every travel pole

This appendix proves the estimate $(\star)$ of Theorem 61 (at every travel pole, $|P_{12}(q_m)| w_m^3 < 2$, $w_m = \sqrt{2/\tau_m}$; quantitatively $|P_{12}| \leq \frac{3.5\sqrt{2}}{8} \tau^{3/2} < 0.619 \tau^{3/2}$ for $\tau \leq 5 \cdot 10^{-3}$, and $|P_{12}| w^3 \leq 0.75$ at the six poles above that threshold) and then closes the full gate (29): the same certificate yields the denominator bound $|S_e| \geq 0.63\sqrt{\tau}$ (Lemma 81), hence $s < 1$ at every travel pole, and, granting the transfer model (M) for the closed form of b_0 (Corollary 31), also $b_0 > 0$ and $B_U(q_m) \neq 0$ there (Corollary 82 with Theorem 50). Nothing in this appendix depends on (M) except the two statements just named as depending on it. Throughout, $h = \tau/2$, $\sigma^2 = \tau/2$, $\psi(t) = \cos(Ze^t; q^2)$; by Proposition 73, $P_{12} = -\frac{1}{2}[\psi(h) + \psi(-h)]$ and $\psi(0) = 0$, so

$$|P_{12}| \leq \frac{h^2}{2} \max_{|t| \leq h} |\psi''(t)|. \quad (34)$$

The gate block is a second difference at a zero

The following exact identity is not needed for Theorem 56, but it is the sharpest available description of the object the U -gate must bound, and it makes the size $\tau^{3/2}$ structural rather than numerical.

Lemma 72 (the gate block in closed form). *Identically in $0 < q < 1$, with $c(u) = \cos(u; q^2)$ and $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$:*

$$P_{12} = \frac{qZ}{2} \mathfrak{s}(Z) - c(z_0), \quad S_o := \sum_{j \geq 0} \frac{(-1)^j [2(1-q)]^j q^{j^2+2j} (1-q)}{(q; q)_{2j+1}} = \frac{z_0}{2} \mathfrak{s}(z_0),$$

S_o being the odd bulk block of the closed form $b_0 = \frac{2q}{1-q} S_o/S_e$ of Corollary 31.

Proof. From (32), $P_{12} = \frac{2q^3}{1-q^3} \sum_k d_k$ with $d_k = (-2)^k (1-q)^k q^{k^2+3k} / [(q^2; q^2)_k (q^5; q^2)_k]$; since $(q; q)_{2k+3} = (q; q^2)_{k+2} (q^2; q^2)_{k+1}$ and $(q; q^2)_{k+2} = (1-q)(1-q^3)(q^5; q^2)_k$, this is

$$P_{12} = \sum_{k \geq 1} (-1)^{k-1} \frac{[2(1-q)]^k q^{k^2+k+1} (1-q^{2k})}{(q; q)_{2k+1}}. \quad (35)$$

Since $q^{1/2}Z = z_0$ and $z_0^2 = 2(1 - q)$, the j -th summand of $\frac{qZ}{2}\mathfrak{s}(Z) = \frac{z_0}{2}\sin(z_0; q^2)$ is $(-1)^j(1 - q)[2(1 - q)]^j q^{j^2+j}/(q; q)_{2j+1}$, and that of $c(z_0)$ is $(-1)^j[2(1 - q)]^j q^{j^2+j}/(q; q)_{2j}$. Their difference carries the factor $(1 - q) - (1 - q^{2j+1}) = q^{2j+1} - q$: the $j = 0$ terms cancel, and what remains is (35) term by term. The S_o identity is the same computation one half-step down: the j -th summand of $\frac{z_0}{2}\mathfrak{s}(z_0)$ is $(-1)^j(1 - q)[2(1 - q)]^j q^{j^2+2j}/(q; q)_{2j+1}$, which is S_o 's summand verbatim. All three matchings are exponent bookkeeping of the kind machine-checked in `QTrigIdentities.lean`; the identities are verified off-pole (hence as identities) and at the six exceptional poles to 10^{-100} (`star_gate_certificate.py`); the collapse identities behind Proposition 73 are further verified at the 58 tabulated poles (`halfstep_verify.py`). $\square$

Proposition 73. *At every travel pole, with $Z = z_0/\sqrt{q}$ the corresponding zero of $\cos(\cdot; q^2)$,*

$$P_{12} = -\frac{1}{2}[\cos(z_0; q^2) + \cos(z_0/q; q^2)].$$

Since $z_0 = Ze^{-\tau/2}$ and $z_0/q = Ze^{+\tau/2}$, and since $\cos(Z; q^2) = 0$, the right-hand side is exactly the symmetric second difference of $\cos(\cdot; q^2)$ at its own zero, taken in the variable $\log u$ with step $\tau/2$.

Proof. Write $c(u) = \cos(u; q^2)$ and $\mathfrak{s}(u) = q^{-1/2}\sin(q^{1/2}u; q^2)$. The shift rules and the pole invariant are Lemma 20: $c(qu) = c(u) + q^{3/2}u\mathfrak{s}(\sqrt{q}u)$, $\mathfrak{s}(qu) = \mathfrak{s}(u) - uc(\sqrt{q}u)$, and $q^{3/2}\mathfrak{s}(z_0)\mathfrak{s}(Z) = 1$ at every travel pole. Taking $u = Z/\sqrt{q}$ in the first shift rule gives $c(z_0) - c(z_0/q) = qZ\mathfrak{s}(Z)$. Taking $u = Z$ in the Hahn–Exton q -Wronskian $\mathfrak{s}(u)c(qu) - c(u)\mathfrak{s}(qu) = u$ and in the first shift rule, and using $c(Z) = 0$ twice, gives $\mathfrak{s}(Z)c(qZ) = Z$ and $c(qZ) = q^{3/2}Z\mathfrak{s}(z_0)$, whence again the pole invariant and $c(qZ) = Z/\mathfrak{s}(Z)$. Substituting the closed form $P_{12} = \frac{qZ}{2}\mathfrak{s}(Z) - c(z_0)$ of Lemma 72 into $c(z_0) + c(z_0/q) = 2c(z_0) - qZ\mathfrak{s}(Z)$ yields the boxed identity. $\square$

The two algebraic steps of the proof are machine-checked (`SelowerSkeleton.lean`): `shift_at_pole` is the instantiation of the half-step rule at $u = z_0/q$, written so that the index bookkeeping is explicit, and `P12_second_difference` is the elimination; both are stated in doubled form over an arbitrary commutative ring and depend only on `propext`. The identity is *verified* at all 58 tabulated poles to a worst relative deviation of $2 \cdot 10^{-44}$ (`halfstep_verify.py`). The consequence, (34) above, is that the gate is a bound on a second derivative on a window of width τ , not an asymptotic evaluation.

Lemma 74 (The travel poles are locally finite). *For every $\tau_\star > 0$ there are only finitely many travel poles with $\tau_m > \tau_\star$.*

Proof. By Proposition 17 the travel poles are the zeros in $q \in (0, 1)$ of $g(q) := \cos(Z; q^2) = C(Z^2)$ with $Z^2 = 2(1 - q)/q$, i.e. of

$$g(q) = \sum_{j \geq 0} (-1)^j \frac{2^j q^{j^2} (1 - q)^j}{(q; q)_{2j}}.$$

First, g has no zero near $q = 0$. For $0 < q \leq \frac{1}{10}$ one has $(q; q)_{2j} \geq \prod_{i \geq 1} (1 - q^i) \geq 1 - \frac{q}{1-q} \geq \frac{8}{9}$ and $(1 - q)^j \leq 1$, so $|g(q) - 1| \leq \frac{9}{8} \sum_{j \geq 1} 2^j q^{j^2} \leq \frac{9}{8} (2q + 4q^4/(1 - q)) \leq 0.24 < 1$, whence $g(q) > 0$ there.

Second, g is real-analytic on $(0, 1)$: the series converges locally uniformly, since on $q \leq 1 - \delta$ the j -th term is bounded by $2^j q^{j^2}/(q; q)_\infty$ and q^{j^2} dominates 2^j . Hence the zeros of g are isolated in $(0, 1)$, and any zero set with an accumulation point in the *open* interval would force $g \equiv 0$, contradicting $g(\frac{1}{10}) > 0$.

Now fix $\tau_\star > 0$. The poles with $\tau_m > \tau_\star$ lie in $q \in [\frac{1}{10}, e^{-\tau_\star}]$, a compact subinterval of $(0, 1)$. An analytic function on a domain whose zeros are isolated has only finitely many zeros on any compact subset; hence there are finitely many such poles. $\square$

Remark 75. Only local finiteness is proved here; that the poles *do* accumulate at $q = 1$ (equivalently $\tau_m \downarrow 0$) is Lemma 12. The present lemma says merely that they cannot accumulate anywhere inside $(0, 1)$, so that any threshold τ_* leaves a finite set to be checked directly. It is used four times: to make the splice in Proposition 65 well posed, to license the phrase “for all large m ” in Theorem 56, to make the exceptional set of Theorem 80 finite, and in Lemma 12’s accumulation clause (its proof also shows $g > 0$ on $(0, \frac{1}{10}]$, so every travel pole has $q_m > \frac{1}{10}$).

Lemma 76. *For $0 < \rho < 1$, with $\theta \sim N(\pi/2, \tau/2)$: $\mathbb{E}[(\rho e^{i\theta}; q)_\infty^{-1}] = \sum_{M \geq 0} q^{M^2/4} (i\rho)^M / (q; q)_M =: \Phi(\rho)$, and $\text{Re } \Phi(\rho) = \cos(\rho/\sqrt{q}; q^2)$. The sum is the q -Airy function of Kajiwara–Masuda–Noumi–Ohta–Yamada [21] at base $\sqrt{q}$, and the representation is Zhang’s q -Laplace transform of the second kind at level $\frac{1}{2}$ [47]; only the probabilistic packaging is ours.*

Proof. Euler’s identity $(x; q)_\infty^{-1} = \sum_M x^M / (q; q)_M$ integrated termwise (absolute uniform convergence on $|x| = \rho < 1$), with $\mathbb{E}[e^{iM\theta}] = i^M q^{M^2/4}$; splitting M by parity and matching exponents $((2j+1)^2 = 4(j^2 + j) + 1$; Lean `parity_exponent`) identifies $\text{Re } \Phi$ with the cosine series of Proposition 17. $\square$

Lemma 77. *Write $A(\xi) = (\rho e^{i(\pi/2+\xi)}; q)_\infty^{-1}$. Then $\log A(\xi) = \sum_{k \geq 1} (i\rho)^k e^{ik\xi} / (k(1-q^k))$; the cumulants $\Lambda_n = i^n \sum_k k^{n-1} (i\rho)^k / (1-q^k)$ satisfy $|\Lambda_n| \leq (n-1)! \rho / ((1-q)(1-\rho)^n)$; Gaussian integration by parts gives, at radius $\rho(t) = z_0 e^t$,*

$$\psi'(t) = \frac{2}{\tau} \text{Im } \mathbb{E}[\xi A], \quad \psi''(t) = \frac{4}{\tau^2} \text{Re } \mathbb{E}[(\frac{\tau}{2} - \xi^2)A];$$

and for complex a, b with $\text{Re}(1 - b\sigma^2) > 0$, $\mathbb{E}[\xi^m e^{a\xi + b\xi^2/2}] = \beta^{-1/2} e^{a^2\sigma^2/(2\beta)} M_m$, $\beta = 1 - b\sigma^2$, $M_0 = 1$, $M_1 = \mu = a\sigma^2/\beta$, $M_m = \mu M_{m-1} + (m-1)(\sigma^2/\beta) M_{m-2}$.

Proof. $-\log(1-u) = \sum u^m/m$ summed over $u = \rho e^{i\theta} q^j$; $1 - q^k \geq 1 - q$ and a binomial estimate; $\rho \partial_\rho A = -i \partial_\theta A$ with $\mathbb{E}[f'] = \mathbb{E}[f\xi/\sigma^2]$ once and twice; completion of the square and analytic continuation in (a, b) . The exponent bookkeeping is machine-checked (`QTrigIdentities.lean`). $\square$

Lemma 78. *At every travel pole with $\tau \leq 5 \cdot 10^{-3}$ and every $t \in [-\tau, h]$: $|\psi'(t)| \leq 1.10 w$.*

Lemma 79. *At every travel pole with $\tau \leq 5 \cdot 10^{-3}$, writing $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$: $\cos(qZ; q^2) = Z/\mathfrak{s}(Z)$, and $|\mathfrak{s}(Z)| \geq Z/(\tau \max_{-\tau \leq t \leq 0} |\psi'(t)|) \geq 0.90$. Moreover, for $t \in [-\tau, 0]$, $|\Phi(\rho(t))| \geq q^{3/4}(|\mathfrak{s}(Z)| - 0.49\tau)$.*

Proof of Lemmas 78 and 79. The identity is the pole reduction underlying Proposition 73 (the Hahn–Exton q -Wronskian at the zero, cited there). The bound in Lemma 79 is the mean value theorem on $[-\tau, 0]$, where $qZ = Ze^{-\tau}$ exactly and all radii satisfy $\rho(t) \leq z_0$. For the last bound: $\text{Im } \Phi(\rho) = q^{3/4} \mathfrak{s}(\rho/\sqrt{q})$ (the odd half of the parity split in Lemma 76), and the drift of $\mathfrak{s}(\rho(t)/\sqrt{q})$ over $t \in [-\tau, 0]$ is the even/odd G_4 term of the certificate, $\leq 0.49\tau$ (`budget.py`, `G4_coef`). For Lemma 78: the principal term is the $m = 1$ tilted moment of Lemma 77 at $a = \Lambda_1$, $b = \Lambda_2$, at most $e^{c_1\sqrt{\tau}}|\beta|^{-1/2} L_1 (\text{Re } \beta)^{-1} \cdot w \leq 1.088 w$, where L_1 is the interval constant bounding $|\Lambda_1|/w$ (Lemma 77, $n = 1$) and $c_1 \leq 0.177$ bounds the exponent residual per $\sqrt{\tau}$ (the two leading halves of the exponent cancel exactly; every residual carries a geometric tail); the tail (linear orders 3..10 with ξ -weighted moments, the product convolution, and the degree- ≥ 11 remainder) is at most 0.131 *absolute*, i.e. $\leq 0.0066 w$ at the top endpoint $w = 20$. All constants are produced by the committed certificate `code/zeta_probe/budget.py` (function `uniform_stack`), which works in 30-digit floating point with the interval constants realised as endpoint evaluations. That evaluation is *re-certified in directed-rounding interval arithmetic* by `budget_iv.py`: the same stack is evaluated with τ entering

as an *interval*, over 160 cells covering $[1.2 \cdot 10^{-4}, 5 \cdot 10^{-3}]$ and 160 covering $[10^{-8}, 1.2 \cdot 10^{-4}]$ (contiguous, with both outer endpoints clamped exactly), giving enclosures valid for *every* τ in each cell; so on those ranges neither the floating-point arithmetic nor the monotonicity of any atom is assumed. The certified maxima are 3.349 and 2.650, against the budget 3.5. On the remaining range $(0, 10^{-8}]$, which no finite set of cells can cover, the power-counting argument below (every term a positive power of τ) is what proves the bound. The companion `star_gate_certificate.py` works in the floating-point model at 45–260 digits with explicit truncation control. $\square$

Theorem 80 (the amplitude estimate $(\star)$). *At every travel pole, $|P_{12}(q_m)| w_m^3 < 2$. Quantitatively: for $\tau \leq 5 \cdot 10^{-3}$, $\max_{|t| \leq h} |\psi''(t)| \leq 3.5 w$, so (34) gives $|P_{12}| \leq \frac{3.5\sqrt{2}}{8} \tau^{3/2} < 0.619 \tau^{3/2} < \tau^{3/2}/\sqrt{2}$; and the six poles with $\tau > 5 \cdot 10^{-3}$,*

$$\tau_1 = 0.79972, \quad \tau_2 = 9.049 \cdot 10^{-2}, \quad \tau_3 = 3.248 \cdot 10^{-2}, \quad \tau_4 = 1.656 \cdot 10^{-2}, \quad \tau_5 = 1.001 \cdot 10^{-2}, \quad \tau_6 = 6.701 \cdot 10^{-3},$$

satisfy $|P_{12}| w^3 = 0.745, 0.530, 0.510, 0.505, 0.503, 0.502$ by direct evaluation at 120-digit precision with explicit truncation control (`star_gate_certificate.py`; the underlying identities are independently verified by `halfstep_verify.py` at the 58 tabulated poles, 47–54 digits on the five it shares with this list: the dominant pole is covered by `star_gate_certificate.py` alone). The first is the dominant pole $q_1 = 1/\beta_2^2$ of Corollary 58. Enumeration of the exceptional set: every travel pole has $q_m > \frac{1}{10}$ and the set is finite (Lemma 74); the count of six is established by a derivative-majorant subdivision for $\tau \geq 0.03$ (three poles; in the arithmetic model disclosed above) and needs no enumeration for $\tau \leq 5 \cdot 10^{-3}$, where the uniform certificate covers every pole; on the window $5 \cdot 10^{-3} < \tau < 0.03$ the three listed poles are located by a $\Delta w \approx 5 \cdot 10^{-4}$ scan (they sit on the $(k + \frac{1}{2})\pi$ ladder to three decimals, and $|\cos(Z; q^2)|$ swings to 1.00 between them with minimum 0.019 outside the root cells). The scan is the weakest step of the enumeration and is stated as such; transcendence in Theorem 61 does not consume it.

Proof. By Lemma 77, $\psi'' = \frac{4}{\tau^2} \operatorname{Re}[-e^{\Lambda_0} G_Q(\Lambda_2 \sigma^4 / \beta + \mu^2) + e^{\Lambda_0} \mathbb{E}[(\frac{\tau}{2} - \xi^2) e^Q (T - 1)]]$, three terms, with $Q = \Lambda_1 \xi + \Lambda_2 \xi^2 / 2$, $G_Q = \mathbb{E}[e^Q] = \beta^{-1/2} e^{\Lambda_1^2 \sigma^2 / (2\beta)}$, $\beta = 1 - \Lambda_2 \sigma^2$, $\mu = \Lambda_1 \sigma^2 / \beta$ (Lemma 77 at $m = 0, 1$), and $T = e^\Omega$, $\Omega = \sum_{n \geq 3} \Lambda_n \xi^n / n!$, the exponential of the higher cumulants. The μ^2 term's phase is pinned by the pole: $|\cos \arg \Phi(\rho(t))| = |\psi(t)| / |\Phi(\rho(t))| \leq \max |\psi'| |t| / |\Phi|$ with $\psi(0) = 0$ (Lemma 78) and $|\Phi| \geq q^{3/4} (|\mathfrak{s}(Z)| - 0.49\tau)$ (Lemma 79); its argument's sine obeys $|\sin 2 \arg \mu| \leq 13.3 \tau^{3/2}$, the two $O(1/w)$ tilts of Λ_1 and β cancelling to $O(\tau)$ because they are the same Lambert data shifted by one index. Every remaining quantity is bounded by an interval constant in scaled variables, each resting on a one-line monotonicity ($(1 - e^{-\tau})/\tau$ decreasing; $e^{c\tau}$, $\rho^2 = 2(1 - q)e^\tau$ increasing; w, q decreasing), with tails via $1 - q^k \geq (1 - q)kq^{k-1}$ and the signed cancellations via the alternating-series bound $\tau^2/6 - \tau^3/12 \leq 1 + q - 2(1 - q)/\tau \leq \tau^2/6$. The degree- ≥ 11 remainder is handled on two intervals: on $[1.2 \cdot 10^{-4}, 5 \cdot 10^{-3}]$ by an exponential majorant whose profile $e^{0.097/\sqrt{\tau}} \tau^{4.5}$ is increasing exactly there (threshold $1.16 \cdot 10^{-4}$), total $\leq 3.03 w$; on $(0, 1.2 \cdot 10^{-4}]$ with no exponential device (linear remainder geometric with ratio $\frac{1}{2}$ on the cutoff region; all products by $|e^\Omega - 1 - \Omega| \leq \frac{1}{2} |\Omega|^2 e^{|\Omega|}$ with $|\Omega| \leq 0.45$ on $|\xi| \leq t_0 \kappa^{-1/3}$; the intermediate zone suppressed by the Gaussian; here $t_0 = 1 - \rho$ and $\kappa = \rho/(1 - q)$), total $\leq 2.10 w$. Both are endpoint evaluations of quantities proved monotone; on $[10^{-8}, 5 \cdot 10^{-3}]$ the same conclusion is obtained without any monotonicity assumption by the interval-arithmetic certification `budget_iv.py` described above, and the monotonicity argument is what carries the remaining range $(0, 10^{-8}]$. The assembly is the committed script `budget.py`. $\square$

Lemma 81 (the denominator at the pole). *At every travel pole, exactly: $S_e = \frac{qZ}{2} \mathfrak{s}(Z) - P_{12}$. Consequently, for $\tau \leq 5 \cdot 10^{-3}$,*

$$|S_e| \geq 0.90 \cdot \frac{qZ}{2} - |P_{12}| \geq \left[\frac{0.90}{\sqrt{2}} \sqrt{qv} - 0.619 \tau \right] \sqrt{\tau} \geq 0.63 \sqrt{\tau}, \quad v = \frac{1-q}{\tau},$$

and S_e has the sign of $\mathfrak{s}(Z)$.

Proof. The identity is Lemma 72 rearranged, since $S_e = \cos(z_0; q^2)$ (Proposition 17). For the bound: $|\mathfrak{s}(Z)| \geq 0.90$ (Lemma 79), $\frac{qZ}{2} = \sqrt{q(1-q)/2} = \sqrt{\tau/2} \sqrt{qv}$, $|P_{12}| \leq 0.619 \tau^{3/2}$ (Theorem 80), and $\sqrt{qv} \geq \sqrt{(1-\tau)(1-\tau/2)} \geq 1 - \frac{4}{5}\tau$ for $\tau \leq \frac{1}{2}$ (since $(1 - \frac{4}{5}\tau)^2 \leq (1-\tau)(1 - \frac{\tau}{2})$ there); at $\tau = 5 \cdot 10^{-3}$ the bracket is ≥ 0.630 (exact endpoint value 0.6309) and it increases as τ decreases. The sign statement holds because $0.90 \cdot \frac{qZ}{2} > |P_{12}|$ with the same margin. $\square$

Corollary 82 (the gate closes). *At every travel pole the gate (29) holds; with Theorem 50 this gives $B_U(q_m) \neq 0$ (the enumeration standing of the six-pole list is that of Theorem 80). The half $|s| < 1$ is proved here outright; the half $b_0 > 0$ consumes the closed form $b_0 = \frac{2q}{1-q} S_o/S_e$, which is Corollary 31 and therefore rests on the transfer model (M). Quantitatively, for $\tau \leq 5 \cdot 10^{-3}$:*

$$|s| = \frac{q}{1-q} \frac{|P_{12}|}{|S_e|} \leq \frac{\tau}{e^\tau - 1} \cdot \frac{0.619}{0.63} \leq 0.983 < 1,$$

and $b_0 > 0$: with the closed form $b_0 = \frac{2q}{1-q} S_o/S_e$ of Corollary 31 and $S_o = \frac{z_0}{2} \mathfrak{s}(z_0)$ (Lemma 72), the pole invariant $q^{3/2} \mathfrak{s}(z_0) \mathfrak{s}(Z) = 1$ (Proposition 73's proof) gives $\text{sign } S_o = \text{sign } \mathfrak{s}(Z) = \text{sign } S_e$ (Lemma 81), so $S_o/S_e > 0$. At the six poles with $\tau > 5 \cdot 10^{-3}$ the gate quantities are evaluated directly (*star_gate_certificate.py*): $|s| = 0.296, 0.256, 0.252, 0.251, 0.251, 0.250$ and $b_0\tau = 2.174, 2.001, 2.000, 2.000, 2.000, 2.000$; moreover $|S_e|/\sqrt{\tau} \geq 0.58$ at all six, so the bound $0.35\sqrt{\tau}$ of Proposition 65 holds at the dominant pole, as its statement records.

Machine verification of the q -trigonometric layer

Every series identity of the q -trigonometric sections and of this appendix is proved by matching the j -th summand; the matching is exponent bookkeeping, which is where such arguments fail in practice. That bookkeeping, and the Casoratian constancy behind the half-step invariant, are machine-checked in Lean 4 (`QTrigIdentities.lean`; the last four rows of the table below live in `SelowerSkeleton.lean`); convergence and the interchange of summation are analytic and are not formalised.

Paper step	Lean declaration	axioms
half-step rule for c , exponent match	<code>cshift_exponent</code>	<code>propext</code>
half-step rule for s , exponent match	<code>sshift_exponent</code>	<code>propext</code>
parity split of Φ	<code>parity_exponent</code>	<code>propext</code>
functional equation of F	<code>qairy_exponent</code>	<code>propext</code>
P_{12} k -sum split, low branch	<code>P12_split_lo</code>	<code>propext</code>
P_{12} k -sum split, high branch	<code>P12_split_hi</code>	<code>propext</code>
Casoratian constancy	<code>casoratian_re_const</code>	<code>standard</code>
constancy $\Rightarrow$ every rung	<code>casoratian_re_eq_zero</code>	<code>standard</code>
shift rule at the pole	<code>shift_at_pole</code>	<code>propext</code>
P_{12} as a second difference	<code>P12_second_difference</code>	<code>propext</code>
scaling $\tau^2 w = \sqrt{2} \tau^{3/2}$	<code>tau_sq_mul_w_sq</code>	<code>standard</code>
gate threshold $2/w^3 = \tau^{3/2}/\sqrt{2}$	<code>gate_threshold</code>	<code>standard</code>

The algebraic skeleton of Proposition 65 is likewise machine-checked (`SelowerSkeleton.lean`): `pole_facts` derives the two exact facts $s(qZ) = s(Z)$ and $c(qZ) = qZ s(Z)$ from the two shift rules together with $c(Z) = 0$; `F_at_pole` collapses the invariant to $q s(Z)^2$ at a pole; `form_lower_bound` is the quadratic-form estimate behind $\lambda_n \geq q - Z/2$; and `prod_one_sub_ge` is the Weierstrass bound $\prod(1 - \delta_i) \geq 1 - \sum \delta_i$ used in the discrete Gronwall step. The analytic inputs (the annulus representation, the Poisson envelope and the mean value theorem) are hypotheses there, not formalised.

“standard” means `propext`, `Classical.choice`, `Quot.sound`. Every file named in this paper lives in the Mathlib-based package of the accompanying development; none of its 62 files contains `sorry`, each ends with a `#print axioms` audit, and every declaration cited anywhere in this paper appears in one of those audits. (The audits are selective in the other direction: a file may prove auxiliary lemmas that its audit block does not list.) Some declarations report “does not depend on any axioms”, which is the legitimate output for a goal closed by `decide` or `rfl` (for instance `OverDetermination.linear_order_19_ok`) and is also, character for character, what an audit prints for a declaration that failed to elaborate; the audits quoted here were therefore read from a cold elaboration of each file one at a time (`lake env lean`, exit status checked), not from a warm `lake build`, which does not replay the output for an up-to-date library. Three declarations cited in this paper are proved by `native_decide` and carry, besides the standard axioms, the compiler-trust axiom named at their point of use (Remark 6; §8). The realness of the ladder coefficient is what makes the added term in `casoratian_re_const` purely imaginary, and is carried as an explicit hypothesis.

The boundary of the formalisation: the analytic layer

The combinatorial layer of this paper is formalised (§5.6), the analytic layer is not, and the obstruction is one and the same for every statement in it: Mathlib carries no q -Pochhammer symbol $(a; q)_n$ or $(a; q)_\infty$, no product form of a theta function, no Bessel or q -Bessel function, no q -difference equations, and no steepest-descent, stationary-phase or van der Corput machinery. It does carry the two-variable Jacobi theta series with its summability, analyticity and modular functional equation, and the Mellin transform; it does not carry the Jacobi triple product, which is the identity that connects the two. The list below records, for each analytic statement of this paper, the object whose absence blocks a formalisation. The statements are at PROVED, not at VERIFIED, and this table is the reason.

Statement	missing prerequisite
Lemma 67	$(x; q)_\infty$ and its logarithm
Lemma 72	$(q; q)_n$; the q -cosine and q -sine
Proposition 73	$(q; q)_n$; the q -cosine
Lemma 81	$(q; q)_\infty$; the q -sine
Lemma 66	the Jacobi triple product; $(q^2; q^2)_\infty$
Lemma 74	analyticity of a q -series and its zero set
Lemma 12	the same, plus a limit at $q \rightarrow 1$
Lemma 13	the same
Proposition 65	$(q; q)_\infty$; quantitative q -trigonometric bounds
Lemma 68	a Gaussian majorant for a theta series
Lemma 69	second derivatives of a q -series
Lemma 78	derivative bounds for the q -sine
Lemma 79	the q -sine and q -cosine at a zero
Theorem 80	a quantitative Taylor bound on a q -phase
Lemma 76	the q -Airy function; a Gaussian expectation
Lemma 77	cumulant bounds; Gaussian integration by parts
Lemma 7	$\log \Gamma$ with an explicit error term; $\log(q; q)_\infty$
Lemma 10	Mellin–Barnes contour shifting; a verified contour-integral enclosure
Lemma 59, confluence	Bessel J_ν at half-integer order

The first five rows would be reached by a q -Pochhammer layer alone, together with the Jacobi triple product; the next four would be reached by that layer together with Mathlib’s existing complex analysis; the remainder need asymptotic machinery that is a larger absence than the q -series one. The half of Lemma 59 that is an explicit rational inequality *is* verified (`AtomN.lean`); the confluence half is the row above (Remark 60).

C The number β_2 : value transcendence

This appendix concerns the *number* $\beta_2 = 1/\sqrt{q^*}$ of Corollary 58, not the function V . Nothing in §§3–9 or in Appendices A–B depends on any statement made here; the material is recorded because it locates the obstruction to value transcendence precisely, at the arithmetic ratio $\rho = \frac{1}{2}$, one notch beyond the proven $\rho \geq 1$ theory.

Remark 83 (the number β_2 : a value-transcendence question). Whether the *number* β_2 is transcendental is open, but the question has a clean reduction. Since $\beta_2 = 1/\sqrt{q^*}$ with $\Sigma_1^T(q^*) = 1$, the number β_2 is algebraic iff q^* is; if q^* were algebraic, the series Σ_1^T would take the *rational* value 1 at an algebraic interior point of its disc. For the Jacobi-theta class this is impossible: by Nesterenko’s theorem on Eisenstein values [30], $\sum_n \alpha^{n^2}$ is transcendental at *every* algebraic α with $0 < |\alpha| < 1$ [13]. The blocks here are *theta-like*; their coefficients ride the squares $q^{(j+1)^2}$, which is the source of the natural boundary of Theorem 2; but their Euler-product denominators $(q; q)_{2k+1}$ spread the support densely, and no analogue of [30] is known off the modular class.

Concretely the reduction localizes on a single confluent basic hypergeometric function. Writing $\Sigma_1^T(q) = 1 - S(q)$,

$$S(q) = \sum_{k \geq 0} \frac{(-2(1-q))^k q^{k^2}}{(q; q)_{2k}} = {}_1\phi_1(0; q; q^2, 2q(1-q)),$$

a q -Bessel-class function (using $(q; q)_{2k} = (q; q^2)_k (q^2; q^2)_k$); β_2 is algebraic iff the least positive zero $q^* = 0.449453630\dots$ of S is. Certified exclusions at 430 digits (residual 10^{-442} ; PSLQ with post-verification of every candidate): neither q^* nor β_2 satisfies an integer polynomial of degree ≤ 8

with height 10^{20} , degree ≤ 12 with height 10^{18} , or degree ≤ 16 with height 10^{14} ; no bilinear relation ties q^* to the fold point q_c of the zero curve (height 10^{12}). The continued fraction of q^* (383 certified partial quotients, two-precision guard) is Gauss–Kuzmin generic: digit frequencies match, Khinchin mean 2.63 (2.685), Lévy exponent 3.21 (3.276), largest quotient $a_{170} = 8250$ (probability $\approx 6\%$ over 383 draws, consistent with chance). The number offers no arithmetic foothold of any tested kind. We record what the natural tools give here, since it locates the difficulty precisely. First, S is *neither* q -holonomic *nor* a Mahler function in q (verified to $O(q^{48})$): the diagonal specialization $z = 2q(1 - q)$ ties the argument to the base and destroys the single-variable q -difference structure, so the methods of Mahler–Nishioka [31] and of holonomy do not apply. Second, two elementary refinements fail *decisively*, not for want of effort: an ultrametric argument is void; the v -adic and archimedean sums of $S(\alpha)$ are unrelated numbers, so $|S(\alpha)|_v = 1$ does not bound the real value; and a Liouville/height bound is beaten by the q -Pochhammer denominators, whose growth over the conjugates of modulus > 1 is cubic ($\sim K^3$) against the merely quadratic ($\sim K^2$) tail, Kronecker-sharply (equality would need Mahler measure 1, i.e. a root of unity). A more promising reformulation uses two variables. Set $G(q, z) = {}_1\phi_1(0; q; q^2, z) = \sum_k (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$, so $S(q) = G(q, 2q(1 - q))$; unlike S , the function G is q -holonomic in z , satisfying the second-order q -difference equation

$$qG(q, z) - (q + 1 - qz)G(q, q^2z) + G(q, q^4z) = 0.$$

One checks that $z^* := 2q^*(1 - q^*) = 0.4948901\dots$ is the *smallest positive zero* of $G(q^*, \cdot)$. Hence β_2 algebraic would force an *algebraic zero of a q -Bessel-class function at an algebraic base*; and the classical analogue is false: the zeros of the Bessel function J_ν are transcendental (Siegel [41]). A q -analogue of Siegel’s theorem for the zeros of ${}_1\phi_1$ would therefore settle β_2 . The right framework is arithmetic. Writing the equation as a system $Y(q^2z) = A(z)Y(z)$ with $Y = (G(z), G(q^2z))^\top$, the companion matrix

$$A(z) = \begin{pmatrix} 0 & 1 \\ -q & q + 1 - qz \end{pmatrix}$$

is *polynomial* in z with $\mathbb{Z}[q]$ entries and $\det A = q$, so the iterated matrices carry no denominators and the Casoratian obeys $W(q^2z) = qW(z)$, i.e. $W(z) = cz^{1/2}$: the two solutions form a q -Bessel pair with Frobenius exponents $0, \frac{1}{2}$. The second solution is explicit, $\tilde{G}(z) = z^{1/2}H(z)$ with $H(z) = \sum_k (-1)^k q^{k^2} (1 - q)z^k / (q; q)_{2k+1}$, and the pair satisfies the exact q -Wronskian identity

$$qG(z)H(q^2z) - G(q^2z)H(z) = q - 1,$$

so at the zero z^* one has the clean relation $G(q^2z^*)H(z^*) = 1 - q$. Moreover $G(q, \cdot)$ is *entire of order zero*; a q - E -function, not a q - G -function. Hence β_2 transcendental would follow from a q -analogue of the Siegel–Shidlovskii theorem for this q - E -function: such a theorem gives $G(q, \alpha)$ transcendental at every algebraic $\alpha \neq 0$ off the singular locus, whereas $G(q^*, z^*) = 0$ is algebraic; forcing q^* , and with it β_2 , transcendental (and by the shared q -connection core, likewise U). All the structural hypotheses hold here (polynomial companion matrix, constant determinant, entire order-zero solution); the sole open point is whether the available q -Siegel–Shidlovskii results (André [2]; Di Vizio–Hardouin [11]; and the Siegel-method theory of [1]) extend to a ${}_1\phi_1$ that is *irregular* at $z = \infty$; a precise, expert-checkable question, not an absence of theory. (Note the contrast with the *function-level* results of this paper, which say nothing about individual interior values.)

For *irrationality* (weaker than transcendence) the classical architecture transports verbatim. G and H are Hahn–Exton (third Jackson) q -Bessel functions: $G(q, z) \propto J_{-1/2}(\sqrt{z}/q; q^2)$ and $H(q, z) \propto J_{+1/2}(\sqrt{z}/q; q^2)$ (verified; the zeros correspond exactly), i.e. the q -analogues of $\cos$ and $\sin$; and z^* is the q -analogue of $(\pi/2)^2$. Niven’s irrationality proof for π runs on the Lommel-polynomial ladder $J_{m+1/2}(r) = R_{m,1/2}(r)J_{1/2}(r) - R_{m-1,3/2}(r)J_{-1/2}(r)$ evaluated at the zero of

$J_{-1/2} = \sqrt{2/\pi r} \cos$; the exact q -analogue of every ingredient exists (Koelink–Swarttouw [22]): the ladder $J_{\nu+1}(x; q^2) = ((1 - q^{2\nu})/x + x)J_\nu - J_{\nu-1}$, the q -Lommel connection $J_{\nu+m} = R_{m,\nu}J_\nu - R_{m-1,\nu+1}J_{\nu-1}$ with $x^m R_{m,\nu} \in \mathbb{Z}[q, x^2]$, and the second-kind functions carrying $q^{m^2/2}$ prefactors; all verified numerically for our pair. Moreover, since a rational zero would make $G(q^*, z^*) = 0$ a *rational value* at rational base and argument, β_2 is irrational as soon as this Hahn–Exton function omits the value 0 at rational points: a plain *value-irrationality* statement, strictly weaker than every transcendence target above. What blocks even this is now a single quantified invariant: the *arithmetic ratio* ρ of the series (b -adic decay exponent of the terms over b -adic growth exponent of the cleared denominators, $q = s/t$ in lowest terms). Every proven q -irrationality theorem (Tschakaloff, q -exponential, the Bézivin–Bundschuh–Väänänen classes) lives at $\rho \geq 1$, where the Pochhammer denominator itself supplies the decay; our G has an independent q^{k^2} numerator against the double Pochhammer $(q; q)_{2k}$, giving $\rho = \frac{1}{2}$. All three classical constructions fail by exactly this margin, measured: the z -Padé remainder $q^{3.5n^2}$ against height q -degree $\approx 9n^2$; the z -direction continued fraction (convergents in $\mathbb{Z}[q, z]$, Pincherle limit the H -ratio) converges only geometrically against t^{n^2} heights; and the ν -ladder at the zero has $J_{m+1/2}(x_0) = R_{m,1/2}(x_0)J_{1/2}(x_0) \sim Cx_0^m$; geometric, because the q -factorial is geometric where Niven’s $m!$ is super-exponential; against t^{m^2} . The ν -ladder is in fact fully costed: the q -Lommel coefficients satisfy $\deg_q p_m = m^2$ *exactly* (immediate from the recurrence, verified symbolically), so their denominators at $q = s/t$ are the pure power t^{m^2} : carrying *no* cyclotomic structure, so the lcm-compression that powers the little- q -Legendre proofs has nothing to act on; while by Poincaré–Perron [37, 36] the m -recurrence, whose coefficient tends to $x_0 + 1/x_0$ with *distinct* characteristic roots $x_0, 1/x_0$, admits *only geometric* solutions: no theta-type q^{m^2} -decaying solution exists in the ν -direction for either kind (the second kind’s $q^{\nu^2/2}$ prefactor is exactly cancelled by the $(q^{-\nu+1}; q)_\infty$ blow-up of its shifted argument). Hence the cleared ν -ladder forms grow like $t^{m^2}x_0^{-m} \rightarrow \infty$ for *every* rational $q = s/t$, including $s = 1$. The $b \leftrightarrow z$ transformation ${}_1\phi_1(0; q; q^2, z) = \frac{(z; q^2)_\infty}{(q; q^2)_\infty} {}_1\phi_1(0; z; q^2, q)$ redistributes but conserves ρ . The structural summary: the theta-type decay lives only in the z -direction, where the equation is *irregular* at $z = \infty$ (Newton polygon slopes $\{0, 1\}$) and every integral structure carries the double-Pochhammer weight $\rho = \frac{1}{2}$; the ν -direction is *regular* (Poincaré class) with clean $\mathbb{Z}[q, X]$ arithmetic but only geometric decay. Decay and arithmetic live in different directions and never meet; the same irregularity that obstructs q -Siegel–Shidlovskii creates the arithmetic deficit. The precise open door remains a q -irrationality criterion at $\rho = \frac{1}{2}$; one notch beyond the proven $\rho \geq 1$ theory.

Remark 84 (the β_2 family). q^* is the first member of an infinite family. On $(0, 1)$ the parabola $w(q) = 2q(1 - q)$ falls like $1 - q$ while every zero branch falls like $(1 - q)^2$ (the confluent limit), so w overtakes each branch exactly once: S has infinitely many zeros $q_1^* < q_2^* < \dots \rightarrow 1$, the j -th given by $w(q_j^*) = z_j(q_j^*)$ (verified at $j = 2$ to 43 digits: exactly one zero of $G(q_2^*, \cdot)$ lies below w). The accumulation obeys the classical cosine-zero law

$$1 - q_j^* \sim \frac{8}{\pi^2(2j - 1)^2}, \quad \text{measured } (1 - q_j^*) \frac{(2j - 1)^2 \pi^2}{8} = 0.679, 0.961, 0.986, 0.993, 0.9956, 0.9970, 0.9979, 0.998$$

The second member is $q_2^* = 0.913486638731047141696\dots$, i.e. $\beta_2^{(2)} := 1/\sqrt{q_2^*} = 1.046282352687417855\dots$; neither satisfies an integer polynomial of degree ≤ 8 with height 10^{10} , and no bilinear relation ties q_1^* to q_2^* (PSLQ, 100 digits). The family interpolates between the open head ($j = 1$, the wild β_2) and the classical tail (the π^2 -law): the roots become “provably transcendental in the limit” while the first, the one of combinatorial interest, remains untouched; the confluent boundary of the step-ratio mechanism made visible inside a single object. (That the higher roots are genuine singularities of U itself is the non-vanishing $\Pi_q(q_m) \neq 0$, which is Hypothesis 52 and is open; what Corollary 82 with Theorem 50 delivers at every travel pole is the weaker $B_U(q_m) \neq 0$.)

Three refinements (all computations at 115 digits, 30 cosine and 10 sine members). First, the sine side has its own family, $H(q_j^s, 2q(1-q)) = 0$ with $1 - q_j^s \sim 2/(j\pi)^2$ (law ratios 0.913, ..., 0.9991 for $j \leq 10$), and the two families *interlace*: $q_j^* < q_j^s < q_{j+1}^*$ (verified $j \leq 9$); its head is $q_1^s = 0.815032640290 \dots$. Second, the accumulation law has an exact-looking rational second order, identical for both families:

$$1 - q_j^* = \frac{2}{\mu_j} \left(1 - \frac{8}{9\mu_j} + \frac{B}{\mu_j^2} + O(\mu_j^{-3}) \right), \quad \mu_j = \left(\frac{(2j-1)\pi}{2} \right)^2 \text{ resp. } (j\pi)^2,$$

with $A = -8/9$ matched to 12 digits (fit residual $3 \cdot 10^{-18}$ out of sample) and B derived below. The constant A is derived, and the derivation separates two regimes that must not be conflated. Writing $\varepsilon = 1 - q$, the logarithm of the ratio of the k -th term of $G(1 - \varepsilon, \varepsilon^2 y)$ to $(-1)^k y^k / (2k)!$ equals $\frac{k}{2}\varepsilon + c_2(k)\varepsilon^2 + O(\varepsilon^3)$ with

$$c_2(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{23k}{72};$$

to first order G is $\cos \sqrt{y} e^{\varepsilon/2}$, so at *fixed* j the zero branch obeys $z_j = \varepsilon^2 \mu_j (1 - \frac{1}{2}\varepsilon + O(\varepsilon^2))$: the fixed- j confluent correction is $-\frac{1}{2}$ (measured -0.5004 at $\varepsilon = 0.002$, converging linearly). The crossing, however, sits in the *uniform* regime $\mu\varepsilon \approx 2$, where the k^3 -term acts as the operator $-\frac{\varepsilon^2}{9} D^3$, $D = (u/2) d/du$ on $\cos u$, $u = \sqrt{X}$; at the zeros $D^3 \cos u = \frac{u^3}{8} \sin u + O(u^2)$, while even powers of D are cos-weighted and drop out (parity protects the order). With $m = \varepsilon\mu$ the crossing equation $2(1 - \varepsilon)e^{\varepsilon/2} = \varepsilon\mu(1 - \varepsilon^2\mu/36)$ becomes $m = 2 - \varepsilon + \varepsilon m^2/36 + O(\varepsilon^2)$, whence $m = 2 - \frac{8}{9}\varepsilon$:

$$A = -1 + \frac{1}{9} = -\frac{8}{9},$$

identically for the sine family (its ε^2 -coefficient carries the same leading $-k^3/9$). The two regimes carry different constants ($-\frac{1}{2}$ versus the crossing's $-\frac{8}{9}$). One more order of the same calculus derives B exactly. The ingredients: the exact prefactor $2(1 - \varepsilon)e^{\varepsilon/2} = 2 - \varepsilon - \frac{3}{4}\varepsilon^2 - \dots$; the cubic coefficient $c_3(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{17k}{72}$ (the m^3 -terms cancel in its defining sum, so c_3 shares the leading $-k^3/9$); the quartic $c_4(k) = \frac{k^3}{450} + O(k^4)$ together with $\frac{1}{2}c_2(D)^2 = \frac{D^6}{162} + \frac{D^5}{108} + \dots$; and one subtlety: at order $\varepsilon^6 u^9$ the cube $\frac{1}{6}(c_2(D)\varepsilon^2)^3$ contributes $+\frac{u^9}{2239488} \sin u$ through the leading term of D^9 , cancelling the $\delta^3/6$ of the cosine expansion *identically*; truncating the exponential before the cube gives a spurious m^5 -term. With that cancellation the crossing equation closes as $m = 2 - \frac{8}{9}\varepsilon - \frac{1891}{8100}\varepsilon^2 + O(\varepsilon^3)$, and inversion gives

$$B = 2m_2 + \frac{64}{81} = \frac{1309}{4050} = 0.3232098765432 \dots,$$

confirmed by a five-order free fit to the computed roots: $B = 0.3232098765429 \dots$ agrees to 13 digits (with $A = -8/9$ simultaneously reproduced to 17), and the (A, B) -constrained fit has out-of-sample residuals below $4 \cdot 10^{-16}$. The accumulation law of the β_2 family therefore begins

$$1 - q_j^* = \frac{2}{\mu_j} \left(1 - \frac{8}{9\mu_j} + \frac{1309}{4050\mu_j^2} + O(\mu_j^{-3}) \right),$$

with both corrections rational and derived. Three closing observations. (a) *Universality*: the sine family obeys the *same* law; a constrained fit to 22 sine members gives $B_{\text{sin}} \cdot 4050 = 1309.000000$ to the fit precision $2 \cdot 10^{-8}$; so through second order the two interlaced families share one accumulation series, with only μ_j distinguishing them. (b) Every ingredient of the crossing calculus is rational (the $c_n(k)$ are rational polynomials; $D^n \cos u$ has odd u -powers on sin and even on cos, an invariant of $D = (u/2)d/du$ preserved by the one-line induction; the δ -elimination therefore stays in odd

powers and the m -equation in rational coefficients), so we conjecture the expansion is rational to all orders. The third constant resists identification: $C = -0.4745357072870\dots$ is stable to 10^{-14} across model orders and members through $j = 55$, excludes every denominator up to $\sim 10^6$ (PSLQ), and the fitted higher coefficients grow ($F \approx 4.3$), marking the series as asymptotic-divergent; which caps the extractable precision and leaves C 's (conjecturally rational, large-denominator) value to the third-order calculus. (c) At $j = 1$ the same series becomes a rational-coefficient expansion in $4/\pi^2$ evaluated at the wild head: divergent, hence a resummation question. If a Borel-type summation of the rational series could be shown to reproduce q^* , the arithmetic of π^2 would enter the problem for the first time through the family; we record this as a direction and have tested its carrying capacity twice. A Borel–Padé resummation of the seven known coefficients reproduces the four accessible samples (both family heads and both second members) to residuals $7.9 \cdot 10^{-3}$, $1.7 \cdot 10^{-6}$, $4.6 \cdot 10^{-9}$, $7.4 \cdot 10^{-11}$ at $\mu = 2.47, 9.87, 22.2, 39.5$. An attempted transseries extraction from these residuals then *refuted its own premise*: the log-slopes between consecutive samples are 1.14, 0.48, 0.24, matching the truncation signature $7/\mu = 1.17, 0.47, 0.24$ exactly (three points), not an exponential $e^{-\beta\mu}$; the residuals are the x^7 -tail of the divergent series, not a nonperturbative signal. At the head this is the standard resurgence degeneracy: seven coefficients place the evaluation at optimal truncation, where truncation floor and first transseries term coincide in scale ($e^{-t_0\mu_1} \approx 0.01$, with $t_0 \approx 0.75 \pm 1.70i$ the Padé–Borel singularities), so the head residual $-0.008(4)$ *cannot* be attributed to a nonperturbative sector on present data. That cost has now been paid: the crossing calculus has been automated in exact rational arithmetic (weight-graded formal Newton for the zero shift; parity and residual controls exact), producing the law's coefficients through order thirty; *all rational*, e.g. $A_3 = -1695089/3572100$ (whose denominator $2^2 3^6 5^2 7^2$ explains the failed PSLQ) and $A_4 = 10199641/535815000$. The coefficients are computed by running the calculus over $\mathbb{F}_p$ and recovering each A_n by rational reconstruction from twenty 62-bit primes; the values are independently confirmed by extracting A_n from the accumulation law applied to the roots themselves, computed to 105 digits.

The Borel geometry is only partly determined by these coefficients. The signs of A_n follow a three-positive, three-negative pattern from $n \approx 12$, indicating a conjugate pair of singularities near 60° ; the pattern is not exact over the full range. This period-6 oscillation dominates the naive growth diagnostics, and must be removed before the Gevrey order can be read off. Setting

$$G_n := \frac{1}{6}(\log |A_n| - \log |A_{n-6}|),$$

which is phase-matched and therefore free of the oscillation, the model $|A_n| \sim C^m(n!)^s$ gives $G_n \sim \log C + s \log n$, so s is the slope of G_n against $\log n$. Least squares yields $s = 0.844$ on $16 \leq n \leq 30$, 0.897 on $23 \leq n \leq 30$ and 1.033 on $26 \leq n \leq 30$: the estimate rises towards 1 as the fitting window advances, and the data are consistent with Gevrey-1. (The unnormalised ratio $\log |A_n|/(n \log n)$, which equals $s + (\log C - s)/\log n$, is 0.313 at $n = 14$ and 0.411 at $n = 30$; its corrections are of order $1/\log n$ and it is therefore useless for extrapolation at these lengths.)

For the singularity modulus three independent estimates are available and agree only in order of magnitude: the phase-matched growth constant gives $|t_0| = 1/C = 3.59\text{--}3.98$ over $24 \leq n \leq 30$; a three-term recurrence fit, appropriate to a conjugate pair, gives $|t_0| = 3.285, 3.394, 3.493, 3.558$ at $n_0 = 22, 25, 28, 30$, extrapolating in $1/n_0$ to 4.30 , with $\arg t_0 = 60.8^\circ, 61.7^\circ, 62.4^\circ, 62.3^\circ$; and the Padé–Borel approximants give $|t| = 2.61, 2.56, 2.96, 3.09, 3.33$ for $[6/6]$ through $[14/14]$. Thus $|t_0|$ lies between 2.9 and 4.3 and is not pinned. Gevrey-1 growth, and with it the $n!$ -Borel framework, is supported but not established; the location of the singularities is not established; and the Borel representation below remains a conjecture whose analytic premise is open. One structural constraint is available: by the weight lemma, only Bernoulli numbers of index $\lesssim n$ reach A_n (a B_j occurring in c_m sits at k -degree $\deg c_m - j$ and reaches A_n only if $j \leq 2n - m + 1$), which excludes the

Gevrey-2 growth that B_{2n} would otherwise force, but does not pin the class. Where testable the resummation succeeds: at the sine head the Padé–Borel ladder converges $1.7 \cdot 10^{-6} \rightarrow 3.7 \cdot 10^{-11}$ (any transseries prefactor $\lesssim 10^{-6}$ unless phase-suppressed), while at the cosine head the plain ladder reaches its approximation wall at $\sim 10^{-3}$ with residuals oscillating about zero. We therefore pose the *Borel representation conjecture*: $1 - q_j^* = (2/\mu_j) \mathcal{S}(1/\mu_j)$ for both families and all j , where $\mathcal{S}$ is the Borel–Laplace sum of the explicit rational series $\sum A_n x^n$; in particular

$$q^* = 1 - \frac{8}{\pi^2} \mathcal{S}\left(\frac{4}{\pi^2}\right),$$

the first exact (conjectural) bridge between q^* and π^2 -arithmetic. Numerically the head representation is confirmed to $\sim 10^{-4}$ by conformal-map Borel resummation (the residual crosses zero smoothly with no plateau above the estimator scatter), and the *same* resummation machinery reproduces the three lower- x heads to 10^{-13} – 10^{-26} ; the coefficients are now computed through A_{22} , still rational (e.g. A_{21}, A_{22} have denominators $\sim 10^{93}$), consistent with the rationality of the expansion at every order.

Two honest boundaries frame the bridge. *Analytically*, its proof is *not* a cite-the-theorems exercise: it reduces to a single hard lemma; a q -uniform, turning-point (Olver/Airy-grade) factorial remainder bound $|A_n| \leq C\sigma^n n!$ for the Hahn–Exton crossing as $q \rightarrow 1$; which has no citable form in the literature and is plausibly the program’s recurring degenerate-saddle obstruction in Borel-summability costume. *Arithmetically*, the Borel transform $b(t) = \sum (A_n/n!)t^n$ carries a sharp, exceptionless prime law: for every odd prime $p \leq 43$, $p \mid \text{den}(b_n)$ exactly when $2n + 1 \geq p$, with p -adic valuation exactly 2 at the onset $n = (p - 1)/2$, and thereafter non-vanishing with valuation growing linearly *on average* (least-squares slopes 2.38, 1.17, 0.73, ... for $p = 3, 5, 7$; the valuation is non-monotone locally, e.g. v_5 dips $6 \rightarrow 5$ and v_{17} dips $2 \rightarrow 1$). The clean-before-onset half is a theorem for all p . (i) The polynomials $c_n(k)$ built from the calculus are p -integral for $n \leq p - 2$ (von Staudt–Clausen [19] at the Faulhaber layer). (ii) A_n depends only on the calculus data through order $2n$; this is now proved, in two steps. First, by series reversion A_n is a function of $m_1, \dots, m_n$ alone (the order- n inverse coefficient uses only the first n forward coefficients; verified by perturbing $m_{>n}$). Second, m_i depends only on $c_1, \dots, c_{2i}$: in the turning-point grading, where $k \sim \varepsilon^{-1/2}$ assigns $\varepsilon^a u^b$ the weight $2a - b$ (physical order = weight/2) and every operation is additive on weights, the polynomial c_m enters at minimum weight $2m - \deg_k c_m$, attained by its top-degree term. Here $\deg_k c_m = m + [m \text{ even}]$: the degree- m coefficient of the $\varepsilon^m \log$ -ratio $\ell_m(i)$ equals $(-1)^m [x^m] \log((e^x - 1)/x)$, which vanishes for every odd $m \geq 3$ because $\log((e^x - 1)/x) = \frac{x}{2} + \log(2 \sinh(x/2)/x)$ and the last term is *even* in x ; the Faulhaber sum then raises the k -degree by one. Hence c_m has minimum weight $m - [m \text{ even}]$ and first reaches physical order i only when $m \leq 2i$, so m_i (and therefore A_n) sees no c_m with $m > 2n$. (iii) The assembly introduces no prime p at output order $n < p$: the Bernoulli denominators of the Faulhaber sums acquire p only when $(p - 1) \mid 2m$, i.e. $2m \geq p - 1$, and all factorial and division indices stay below p . For $n < (p - 1)/2$ one has $2n \leq p - 3$, so every ingredient feeding A_n is p -integral and $p \nmid \text{den}(A_n)$.

The matching lower direction; that p *does* appear at $n = (p - 1)/2$, with valuation exactly 2; reduces to a unit. At $n = (p - 1)/2$ the sole prime- p source is c_{p-1} (all lower c are p -integral), and it enters A_n *linearly*, since c_{p-1}^2 has weight $2(p - 2) > 2n$. Only the top three coefficients of c_{p-1} (degrees $\geq p - 2$) survive the weight cutoff; of the two that carry p^2 ; the k^1 and k^p monomials (von Staudt–Clausen at the Faulhaber layer, $\deg_k c_{p-1} = p$); only k^p reaches A_n , as k^1 sits at weight $2p - 3 > 2n + 1$. The coefficient of that top monomial in $A_{(p-1)/2}$ is a *unit*, $\partial A_{(p-1)/2} / \partial c_{p-1}[k^p] = \pm 1$, and this now has a leading-symbol proof. The monomial perturbs the confluent expansion of G by $\varepsilon^{p-1} k^p$; the turning-point map (the D^p -action, $D = (u/2) d/du$) sends k^p to $a_p u^p \sin u + b_p u^{p-1} \cos u$ with $a_p = (-1)^{n-1} 2^{-p}$ the leading sine coefficient and $b_p = (-1)^n n(2n + 1) 2^{-p}$ the leading cosine

coefficient ($p = 2n + 1$; both closed forms follow from the recursion $(A, B) \mapsto (\frac{u}{2}(A' - B), \frac{u}{2}(B' + A))$). At a simple zero u_* of $\cos$ the cosine term vanishes and every sine power below u^p contributes only at higher order in ε , so the zero shift is $\delta = \varepsilon^{p-1} a_p u_*^p$ (the factor $\sin u_* = \pm 1$ cancels against $G' = -\sin$). With $m = \varepsilon u^2$ and the leading balance $u_*^2 = 2/\varepsilon$ (from $\varepsilon u^2 = 2(1 - \varepsilon)e^{\varepsilon/2} \rightarrow 2$),

$$\delta m = 2\varepsilon u_* \delta = 2a_p \varepsilon^p u_*^{p+1} = a_p 2^{n+2} \varepsilon^n, \quad \text{so} \quad \frac{\partial m_n}{\partial c_{p-1}[k^p]} = a_p 2^{n+2} = \pm 2^{1-n},$$

and with the reversion Jacobian $\partial A_n / \partial m_n = 2^{n-1}$ (from $\varepsilon \sim 2/\mu$) the product is ± 1 . Only the top sine power u^p reaches order ε^n , so the magnitude is exact; the sign is $(-1)^n$ (verified for all $n \leq 10$, independently of p). Since $c_{p-1}[k^p]$ has p -adic valuation -2 while the other reaching monomials (k^{p-1}, k^{p-2}) have valuation ≥ -1 against p -integral multipliers, $v_p(\text{den } A_{(p-1)/2}) = 2$ exactly. Thus the onset law; $p \mid \text{den}(A_n) \iff 2n + 1 \geq p$, with valuation 2 at $n = (p - 1)/2$; is *proved for all odd* p . The average-linear (Legendre/factorial) growth; not the logarithmic growth an $\text{lcm}(1, \dots, 2n)$ or ζ -type denominator would give; makes $\text{den}(A_n)$ super-geometric ($\log \text{den}(A_n)/n$ rises $5.4 \rightarrow 10.2$ over $n \leq 20$, unbounded). Consequently $b(t)$ is *not* a G -function (super-geometric denominators alone suffice, independent of holonomy), so $\sum A_n x^n$ is not of André arithmetic-Gevrey type [2] and lies outside the G -function, E -function, and E -operator (Fischler–Rivoal [15]) classes entirely. The closest existing frame, the E -operator theory, has exactly one blocking gap; the hypothesis that b is a G -function; which the prime law refutes rather than supplies. The missing tool is therefore genuinely new, not an incremental hypothesis-removal: a value theory for Borel–Laplace sums whose Borel transform is non-holonomic with factorial denominators, evaluated at a point $(4/\pi^2)$ transcendental over $\mathbb{Q}$. This constant is a canonical first target for such a theory. Third, the family sum converges to

$$T := \sum_{j \geq 1} (1 - q_j^*) = 0.7357490877157094 \dots (\pm 3 \cdot 10^{-17}, \text{ tail summed via the fitted law and Hurwitz zeta}),$$

for which no relation with $\{1, \pi^{-2}, \pi^{-4}, \log 2, \gamma, \pi^2/6\}$ exists at height 10^4 within the stated uncertainty (and $T \neq 2/e$): the classical tail is π^2 -rational piece by piece, but the wild head shifts the sum off every tested constant; a sum rule, if one exists, must come from global structure not yet visible.

The failure pattern above is not an artifact of particular constructions; it is forced by exact arithmetic facts, which we record as a theorem-level ledger.

Proposition 85 (Arithmetic no-go ledger at $\rho = \frac{1}{2}$). *Let $q = s/t \in (0, 1)$ with $\gcd(s, t) = 1$, and let $g_k = (-1)^k q^{k(k-1)} / (q; q)_{2k}$ be the Taylor coefficients of $G(q, \cdot)$. Then:*

- (i) (exact denominators) *in lowest terms $g_k = (-1)^k s^{k^2-k} t^{k^2+2k} / \prod_{i=1}^{2k} (t^i - s^i)$; in particular $\text{den}(g_k) = \prod_{i=1}^{2k} (t^i - s^i) = t^{k(2k+1)(1+o(1))}$ exactly, while $|g_k| = (s/t)^{k^2(1+o(1))}$, so the arithmetic ratio is $\rho = \log(t/s)/(2 \log t) \leq \frac{1}{2}$, with equality iff $s = 1$.*
- (ii) (ν -ladder arithmetic) *the q -Lommel polynomials of Koelink–Swarttouw [22] satisfy $\deg_q p_m = m^2$ exactly, with leading q -coefficient $(-1)^m$; hence at $q = s/t$ their denominators are the pure power t^{m^2} , carrying no cyclotomic structure.*
- (iii) (ν -direction regularity) *the ladder recurrence $u_{m+1} = ((1 - q^{2m+1})/x_0 + x_0)u_m - u_{m-1}$ has coefficients converging exponentially to $x_0 + 1/x_0$ with distinct characteristic roots $x_0^{\pm 1}$; by Poincaré–Perron [37, 36] every nonzero solution satisfies $|u_m|^{1/m} \rightarrow x_0^{\pm 1}$. No theta-type (q^{cm^2} -decaying) solution exists in the ν -direction, for either kind. Consequently the cleared ν -ladder forms are $\gg t^{m^2} x_0^{-m} \rightarrow \infty$ for every rational q , including $s = 1$.*

- (iv) (cyclotomic floor) any linear form built over the coefficients $g_0, \dots, g_M$ has decay at best $q^{M^2(1+o(1))}$, while its cyclotomic support is $\{\Phi_d(t, s) : d \leq 2M\}$; even if every Φ_d appeared at most once; the idealized best case of any lcm/acceleration argument; the denominator exceeds $t^{\sum_{d \leq 2M} \varphi(d)} = t^{(12/\pi^2 + o(1))M^2}$, and $12/\pi^2 = 1.2159 \dots > 1$. Hence no cyclotomic-compression (Rhin–Viola-type [40]) refinement of the Padé, continued-fraction, or ladder families can produce integer forms tending to 0. Moreover the true solved Padé denominators contain growing non-cyclotomic irreducible factors (at $n = 3$, one of degree 26 with coefficients up to 4; no integer d has $\varphi(d) = 26$), which no lattice-of-cyclotomes argument addresses.

Proof. (i) $q^{k(k-1)} = s^{k(k-1)} / t^{k(k-1)}$ and $1/(q; q)_{2k} = t^{k(2k+1)} / \prod_{i \leq 2k} (t^i - s^i)$; the numerator $s^{k^2-k} t^{k^2+2k}$ is coprime to each factor since $\gcd(st, t^i - s^i) = 1$. (ii) Induction: $p_{m+1} = (X+1-q^{2m+1})p_m - Xp_{m-1}$, and the top q -degree term of p_{m+1} is $-q^{2m+1} \cdot (\text{top of } p_m)$, of degree $m^2 + 2m + 1 = (m+1)^2$, which cannot cancel against Xp_{m-1} (degree $(m-1)^2$). (iii) The perturbation $(1-q^{2m+1})/x_0 - 1/x_0$ decays geometrically, so the Poincaré–Perron theorem [37, 36] applies with the limit equation $u_{m+1} = (x_0 + 1/x_0)u_m - u_{m-1}$, whose roots $x_0 \neq 1/x_0$ are simple. (iv) A form using $g_0, \dots, g_M$ vanishes to order at most M at $z = 0$, so its value at a fixed z_0 inside the disc is $\gg |g_{M+1}z_0^{M+1}| = q^{M^2(1+o(1))}$ in the truncation-based families; its denominators involve $(t^i - s^i)$ for $i \leq 2M$, i.e. $\Phi_d(t, s)$ for $d \leq 2M$, and $\sum_{d \leq N} \varphi(d) = \frac{3}{\pi^2} N^2 + O(N \log N)$ gives the floor with $N = 2M$. The degree-26 factor is exhibited by the exact $n = 3$ computation (`code/zeta_probe/beta2_qniven_ladder.py`). $\square$

Remark 86 (scope). Proposition 85 closes every construction whose coefficients live in the natural $\mathbb{Z}[q^{\pm 1}, z]$ -lattices of the q -difference module; Taylor Padé and Hermite–Padé, the argument-direction continued fraction, the ν -ladders of both kinds; together with all cyclotomic-compression refinements of them. It does *not* exclude a fundamentally different family of forms; producing one is exactly what a $\rho = \frac{1}{2}$ criterion would mean. The ledger thereby locates β_2 ’s arithmetic at a precise frontier of q -Diophantine approximation.

The remaining freedom can be narrowed further: the finite places of $\mathbb{Q}$ are provably useless here. The series $S(q) = \sum_k g_k z^{*k}$, $z^* = 2q(1-q)$, has rational terms, so besides its real sum it possesses s -adic and t -adic *avatars*; the same series summed in $\mathbb{Q}_s$ and $\mathbb{Q}_t$, where it also converges (the valuations of the terms tend to $+\infty$).

Lemma 87 (adelic units). *Let $q = s/t \in (0, 1)$, $\gcd(s, t) = 1$, and $z^* = 2q(1-q)$. For every $k \geq 1$ and every prime p : if $p \mid t$ with $e = v_p(t)$, then*

$$v_p(g_k z^{*k}) = k^2 e + k[p=2] \geq k^2,$$

(exact, verified symbolically), and if $p \mid s$ then $v_p(g_k z^{*k}) \geq k^2$. Consequently the series converges in $\mathbb{Q}_p$ for every $p \mid st$, and the finite-place avatars satisfy $S_p \equiv 1 \pmod{p^{v_p(st)}}$; in particular $S_t \equiv 1 \pmod{t}$ and $S_s \equiv 1 \pmod{s}$: they are units, hence nonzero; regardless of whether the real avatar vanishes.

Proof. Each factor $t^i - s^i$ of $\text{den}(g_k)$ is a p -unit for every $p \mid st$ (it is $\equiv -s^i$ or $\equiv t^i \pmod{p}$), so by Proposition 85(i), $v_p(g_k) = (k^2 + 2k)e$ for $p \mid t$ and $v_p(g_k) = (k^2 - k)v_p(s)$ for $p \mid s$. From $z^* = 2s(t-s)/t^2$: for $p \mid t$, $v_p(t-s) = 0$ and $v_p(z^*) = [p=2] - 2e$, giving $v_p(g_k z^{*k}) = (k^2 + 2k)e + k([p=2] - 2e) = k^2 e + k[p=2]$; for $p \mid s$, $v_p(z^*) = v_p(2s) \geq v_p(s) \geq 1$, giving $v_p(g_k z^{*k}) \geq (k^2 - k) + k = k^2$. All are ≥ 1 for $k \geq 1$, so each avatar is 1 plus terms of positive valuation. $\square$

Remark 88 (exact adelic balance, and the shape of any future criterion). Lemma 87 has two consequences. First, it extends the no-go ledger from the module’s coefficient lattices to the *values*:

no divisibility or smallness can be sourced from the finite places, since the finite-place avatars are units; a hypothetical rationality of β_2 would be a maximally place-asymmetric event, the real avatar vanishing while every finite avatar equals a unit $\equiv 1$. Second, it makes the $\rho = \frac{1}{2}$ balance exact in the sense of the product formula: the term $g_k z^{*k}$ carries archimedean decay $(s/t)^{k^2(1+o(1))}$, s -adic and t -adic numerator content $s^{k^2} t^{k^2}$, and denominator $t^{2k^2(1+o(1))}$; the books balance with *zero margin* at the k^2 scale, so any successful construction must be lossless at all three places simultaneously and win at the geometric scale. Every mechanism surveyed above loses a factor t^{ck^2} somewhere; a $\rho = \frac{1}{2}$ criterion is therefore forced to be a purely *archimedean cancellation* phenomenon. This suggests the precise principle that would settle β_2 , which we record as a conjecture: for the irregular rank-two q -difference module with polynomial companion matrix and $\det A = q$, the vanishing locus is *adelically rigid*: the entire function $G_v(q, \cdot)$ cannot vanish at a rational point z_0 at which all finite-place avatars are units (as at $z_0 = z^*$, by Lemma 87) at one place v of $\mathbb{Q}$ while the avatars at all other convergent places are units. The finite-place half of this rigidity is exactly Lemma 87; its archimedean half implies $\beta_2 \notin \mathbb{Q}$. This is a q -analogue, for irregular q -difference modules, of the place-uniformity enjoyed by E - and G -function values, where the exceptional sets of Siegel–Shidlovskii–André theory are independent of the place.

Further analysis of the archimedean obstruction yields several exact structural facts and a faithful reduction. All are verified numerically to 40–60 digits (`code/zeta_probe/beta2_*.py`).

Proposition 89 (structure of the zero curve). *Write $Q = q^2$ and let $0 < z_1(q) < z_2(q) < \dots$ be the positive zeros of $G(q, \cdot)$.*

- (i) (base change) $G(q, z) = \frac{(q^2; q^2)_\infty}{\sqrt{q} (q; q^2)_\infty} z^{1/4} J_{-1/2}^{(3)}(\sqrt{z}/q; q^2)$, where $J_\nu^{(3)}$ is the Hahn–Exton (third Jackson) q -Bessel function; thus G is, up to an explicit prefactor, the Hahn–Exton q -cosine, and its zeros are those of a classical, well-studied object.
- (ii) (zero lattice) $z_k(q) Q^{k-1} \rightarrow 1$ as $k \rightarrow \infty$, with theta-suppressed dressing $\delta_k := z_{k+1} Q^k - 1 = O(c^k Q^{k^2})$ (numerically $\delta_0 = -0.5051$, $\delta_1 = -1.17 \cdot 10^{-2}$, $\delta_2 = -1.03 \cdot 10^{-5}, \dots$); $z^* = z_1(q^*)$ is the maximally dressed member, displaced from the bare lattice point $Q^0 = 1$ by an $O(1)$ correction; it is not a torsion/lattice point of the connection curve.
- (iii) (power sums are rational) the symmetric functions $\sigma_p(q) := \sum_{n \geq 1} z_n(q)^{-p}$ lie in $\mathbb{Q}(q)$ for every $p \geq 1$ (Newton’s identities on the $\mathbb{Q}(q)$ -coefficients g_k); e.g. $\sigma_1 = 1/((1-q)(1-q^2))$. Consequently $z^*(q) = \lim_{p \rightarrow \infty} \sigma_p(q)/\sigma_{p+1}(q)$ with geometric rate $(z_1/z_2)^p$, giving an all-orders $\mathbb{Q}(q)$ -approximation of the zero curve.

Two of the no-go’s ingredients also sharpen. First, the *cause* of $\rho = \frac{1}{2}$ is exactly the double Pochhammer: $\rho = \log(t/s)/(2 \log t)$, and the factor 2 in the denominator is the $2k$ in $(q; q)_{2k}$; a single Pochhammer would give $\rho > 1$ (whence Liouville) when $t > s^2$. The intractability is intrinsic to the base- q^2 Hahn–Exton structure, not to the choice of construction. Second, the finite-place primes are *disjoint* from st : $\prod_{i \leq 2N} (t^i - s^i)$ is coprime to st (only the prime 2, via lifting-the-exponent when s, t are both odd, contributes an $O(N)$ term), so the arithmetic of every partial-sum denominator is supported off $\{s, t\}$; orthogonal to the unit residues of Lemma 87. The Hankel/determinantal escape is likewise closed exactly: $\det(g_{i+j})_{0 \leq i, j \leq n}$ has denominator of *cubic* archimedean size ($\log |\text{den}| \sim cn^3$), one power worse than the quadratic remainder.

Finally, the Nesterenko route is blocked more precisely than by non-holonomicity alone. Nesterenko’s method needs a finitely generated δ -closed ($\delta = q d/dq$) $\mathbb{C}(q)$ -algebra containing S , i.e. that S be *differentially algebraic*; non-holonomicity does not suffice (the quasimodular E_2 is non-holonomic yet

δ -algebraic). One computes $\delta S = \sum_k a_k(k^2 - \frac{kq}{1-q} + L_k)$ with $L_k = \sum_{i=1}^{2k} iq^i/(1-q^i) \rightarrow (1-E_2)/24$: S carries *every* finite Eisenstein truncation L_k , not merely the limit, which is precisely why it escapes the modular differential ring. Whether S is differentially algebraic is itself open, but the exclusion is now proof-grade over a wide window: S satisfies no algebraic differential equation in the δ -jets at any of the profiles (order, degree, $\deg_q$) = (2, 12, 3), (3, 8, 3), (4, 5, 6), (5, 4, 8), (6, 3, 15), (8, 2, 35), (12, 1, 120); exact linear algebra on 2000 coefficients modulo a prime, every matrix of full column rank; since a primitive integer relation survives reduction, full rank modulo one prime excludes a relation over $\mathbb{Q}$ at that profile outright. (The pipeline is validated on E_2 , for which it recovers the q -Chazy equation $\delta^3 E_2 = E_2 \delta^2 E_2 - \frac{3}{2}(\delta E_2)^2$ at its exact profile.) The same exclusion holds verbatim for the zero branch $z_1(q)$. The Nesterenko route is blocked *modulo the differential transcendence of S* , which is itself open; and, with no functional equation in q available for S (no shift, q -difference, or Mahler equation exists), no current method can even attack it.

Remark 90 (reduction to a theta non-resonance). All routes converge on the same faithful reformulation: β_2 is irrational iff $G(q^*, z^*) \neq 0$ is forced by the arithmetic, and (Ramis–Sauloy–Zhang [39]) this is a *theta non-resonance* statement; the connection ratio $R_q(z) = [P_{21} y_{\text{irr}}]/[P_{11} y_{\text{reg}}]$, an explicit θ_{q^2} -quotient, must satisfy $R_q(2q(1-q)) \neq -1$ at every algebraic q . The evidence points toward truth: in the confluent limit $q \rightarrow 1$ this is exactly Lindemann’s theorem that $\cos$ has no algebraic zero. But the reduction is not yet shown *easier*: $R_q(z) = -1$ is a functional-equation-consistent value that a nonconstant theta-quotient must attain somewhere, and on the diagonal $z = 2q(1-q)$ the theta-arguments appear algebraically dependent, so no current independence theorem (Bertrand; Nesterenko [30]; Schneider–Lang [24]) forbids it. The archimedean obstruction is thus the exact mirror of the proven finite-place balance: units ($\equiv 1$) at every $p \mid st$, and a putative -1 at ∞ , with neither forbidden. The required connection-entry computation is carried out below.

Proposition 91 (the connection entries). *Write $Q = q^2$, $\theta_b(x) = \sum_{n \in \mathbb{Z}} b^{n(n-1)/2} x^n$. The equation for G has, at $z = \infty$, the formal solution basis $f_1(z) = \theta_Q(-z) \hat{g}(z)$ and $f_2(z) = h(z)/\theta_Q(-z/q)$, where $h(z) = \sum_k c_k z^{-k}$ is convergent (entire in $1/z$, $c_k = O(Q^{k^2/2})$) while $\hat{g}(z) = \sum_k b_k z^{-k}$ is q -Gevrey divergent, with the closed form*

$$b_k = \frac{q^{k-k(k-1)/2}}{(q; q)_k}$$

(equivalent to the recursion $b_k = [-(q+1)Q b_{k-1} - Q^{3-k} b_{k-2}]/(q(Q^k - 1))$; verified symbolically for $k \leq 8$). By Euler’s theorem $\sum_j x^j/(q; q)_j = 1/(x; q)_\infty$, the q -Borel transform is therefore explicit:

$$\hat{B}(\xi) = \sum_{k \geq 0} b_k q^{k(k+1)/2} \xi^{-k} = \sum_{k \geq 0} \frac{(q^2/\xi)^k}{(q; q)_k} = \frac{1}{(q^2/\xi; q)_\infty},$$

meromorphic on $\mathbb{C}^*$ with poles exactly on $\xi \in q^{2+\mathbb{N}_0}$, and satisfying $\hat{B}(\xi) + \frac{q+1}{\xi} \hat{B}(q\xi) + (\xi^{-2} - 1) \hat{B}(q^2\xi) = 0$ identically (direct computation from the product form; in particular $\hat{B}(-1) = (q+1) \hat{B}(-q)$). The theta-kernel q -Laplace

$$g^{[\lambda]}(z) = \frac{\sum_{m \in \mathbb{Z}} q^{m(m-1)/2} (\lambda/z)^m \hat{B}(\lambda q^m)}{\theta_q(\lambda/z)}, \quad f_1^{[\lambda]} = \theta_Q(-z) g^{[\lambda]}(z),$$

is then singular only for $\lambda q^{\mathbb{Z}}$ meeting the pole set, i.e. for the single class $[\lambda] \in q^{\mathbb{Z}}$ of $E_q = \mathbb{C}^*/q^{\mathbb{Z}}$ (the positive-real lattice class, near which the zeros of G lie); every other class is regular. For $\lambda \notin q^{\mathbb{Z}}$ the bilateral sum converges absolutely on compacts of $\mathbb{C}^* \setminus (-\lambda q^{\mathbb{Z}})$ and $L f_1^{[\lambda]} = 0$ exactly: the

theta shifts $\theta_q(q^{-2}x) = q^{-3}x^2\theta_q(x)$ and $\theta_q(q^{-4}x) = q^{-10}x^4\theta_q(x)$ turn $f_1^{[\lambda]}(Qz)$ and $f_1^{[\lambda]}(Q^2z)$ into the same kernel sum with $\widehat{B}(\lambda q^m)$ replaced by $\widehat{B}(\lambda q^{m+2})$, $\widehat{B}(\lambda q^{m+4})$; collecting the coefficient of $q^{m(m-1)/2}(\lambda/z)^m/\theta_q(\lambda/z)$ in $Lf_1^{[\lambda]}/\theta_Q(-z)$ then yields $q[\widehat{B}(\xi) + \frac{q+1}{\xi}\widehat{B}(q\xi) + (\xi^{-2} - 1)\widehat{B}(q^2\xi)]$ at $\xi = \lambda q^m$, identically zero. (Each step was verified numerically to 40 digits; the kernel normalization was validated on convergent input to 34 digits, and $Lf_1^{[\lambda]} \approx 10^{-61}$ numerically.) Writing $G = C_1 f_1^{[\lambda]} + C_2^{[\lambda]} f_2$, numerically to 59–60 digits:

- (i) since $\det A = q$, every Casoratian $\text{Cas}(u, v)(z) := u(z)v(Qz) - u(Qz)v(z)$ of solutions satisfies $\text{Cas}(Qz) = q \text{Cas}(z)$ (immediate from $u(Q^2z) = a u(Qz) - q u(z)$), so $C_1 = \text{Cas}(G, f_2) / \text{Cas}(f_1^{[\lambda]}, f_2)$ and $C_2^{[\lambda]} = \text{Cas}(f_1^{[\lambda]}, G) / \text{Cas}(f_1^{[\lambda]}, f_2)$ are Q -elliptic; proven, and verified to 10^{-60} ;
- (ii) C_1 is a constant, independent of both z and the resummation direction λ : this is proven in Theorem 92 below (the limit argument applies along every orbit class $\neq [1]_Q$, and ellipticity extends the value); the slope-1 component of G is Stokes-invariant, and all q -Stokes ambiguity is carried by $C_2^{[\lambda]}$, nonconstant elliptic with λ^2 in its zero divisor;
- (iii) (dressing-product formula, verified to 59 digits)

$$C_1 = \frac{1}{(Q; Q)_\infty \prod_{k \geq 1} z_k Q^{k-1}} :$$

the Stokes-invariant connection constant is the regularized product of the zero dressings;

- (iv) at $q = q^*$ the entire apparatus evaluates: $C_1(q^*) = 2.69898382616391375\dots$, and the zero condition $G(q^*, z^*) = 0$ takes the pinned form

$$C_2^{[\lambda]}(z^*) = -C_1 \frac{f_1^{[\lambda]}(z^*)}{f_2(z^*)} = 0.011592994120624266\dots \quad (\lambda = \tfrac{3}{2}),$$

verified to 30 digits. This is the explicit instance of the theta non-resonance: every object in it is now constructed.

The constant C_1 is in fact *modular*, and this upgrades the dressing-product formula into an exact evaluation of the zero product.

Theorem 92 (the connection constant, and the zero product of the Hahn–Exton q -cosine). *For $0 < q < 1$,*

$$C_1 = \frac{1}{(q; q)_\infty}, \quad \text{and consequently} \quad \prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty,$$

where $z_1 < z_2 < \dots$ are the positive zeros of $G(q, \cdot)$. Likewise, for the second solution's underlying entire function H with zeros $w_1 < w_2 < \dots$, $\prod_{k \geq 1} w_k(q) q Q^{k-1} = (q^3; q^2)_\infty$. (Verified at three bases including q^* , to 41, 23, 26 digits for C_1 and 58 digits for the product.)

Proof. (1) *An exact identity.* From $(q; q)_{2k} = (q; q)_\infty / (q^{2k+1}; q)_\infty$ and Euler's expansion $(x; q)_\infty = \sum_{j \geq 0} (-1)^j q^{j(j-1)/2} x^j / (q; q)_j$ at $x = q^{2k+1}$, together with $k(k-1) + 2kj = (k+j)(k+j-1) - j(j-1)$ and the substitution $m = k+j$, absolute convergence of the double sum permits the rearrangement

$$(q; q)_\infty G(z) = \sum_{m \geq 0} (-1)^m q^{m(m-1)/2} z^m P_m(1/z), \quad P_m(w) := \sum_{j=0}^m c_j w^j, \quad c_j = \frac{q^{j-j(j-1)/2}}{(q; q)_j}$$

(the c_j coincide with the b_j of Proposition 91; identity checked symbolically through z^8 and numerically to 41 digits).

(2) *The constant.* Fix $u < 0$ and put $z = uQ^{-N}$. The moduli $q^{m(m-1)}|z|^m$ peak at $m_0 = N + O(1)$ and decay like $q^{(m-m_0)^2 + O(1)}$ away from it. For $1 \leq j \leq m \leq 2N$, $2Nj - j(j-1)/2 \geq Nj$, so $|c_j z^{-j}| \leq Cq^{Nj}|u|^{-j}$ and $|P_m(1/z) - 1| \leq C'q^N$ uniformly; the terms with $m > 2N$ contribute a relative $O(q^{N^2})$. The negative- m part of $\theta_Q(-z)$ is $O(|z|^{-1})$, negligible against the peak. Hence $(q; q)_\infty G(z)/\theta_Q(-z) \rightarrow 1$ along $z = uQ^{-N}$, for each fixed $u < 0$.

(3) *Geometric zero dressing.* On the circle $|z - Q^{1-k}| = \varepsilon q^k Q^{1-k}$ the identity of step (1) and the bounds of step (2) give $|(q; q)_\infty G - \theta_Q(-z)| \leq Cq^k |\theta_Q(-z)|_{\text{peak}}$, while $\theta_Q(-z)$ has a simple zero at Q^{1-k} with derivative bounded below there; Rouché gives $z_k = Q^{1-k}(1 + O(q^k))$, i.e. $\log(z_k Q^{k-1}) = O(q^k)$, absolutely summable.

(4) *The product.* $G(z) = \prod_k (1 - z/z_k)$ (order 0, genus 0, $G(0) = 1$) and $\theta_Q(-z) = (Q; Q)_\infty (z; Q)_\infty (Q/z; Q)_\infty$ (Jacobi). Along $z = -R$, $R \rightarrow \infty$,

$$\frac{G(-R)}{\theta_Q(R)} = \frac{1}{(Q; Q)_\infty (-Q/R; Q)_\infty} \prod_{k \geq 1} \frac{1 + R/z_k}{1 + RQ^{k-1}},$$

and for $x, y > 0$ the function $R \mapsto \log \frac{1+Rx}{1+Ry}$ is monotone from 0 to $\log(x/y)$, so each factor is dominated by $|\log(z_k Q^{k-1})|$, summable by step (3). Dominated convergence gives $G(-R)/\theta_Q(R) \rightarrow 1/[(Q; Q)_\infty \prod_k z_k Q^{k-1}]$ as $R \rightarrow \infty$; evaluating along the subsequence $R = |u|Q^{-N}$ by step (2), the limit equals $1/(q; q)_\infty$, whence $\prod_k z_k Q^{k-1} = (q; q)_\infty / (Q; Q)_\infty = (q; q^2)_\infty$ by Euler's $(q; q)_\infty = (q; q^2)_\infty (q^2; q^2)_\infty$. Finally, in the basis of Proposition 91: along the negative axis (a regular class, $[-1] \notin q^{\mathbb{Z}}$) one has $g^{[\lambda]}(z) \rightarrow 1$ and $f_2(z)/\theta_Q(-z) = O(1/(\theta_Q(R)\theta_Q(R/q))) \rightarrow 0$, so the limit just computed is the entry C_1 ; hence $C_1 = 1/(q; q)_\infty$. The H -statement is identical with comparison function $\theta_Q(-qz)$ and ratio $(1-q)/(q; q)_\infty$. Finally, steps (1)–(2) apply verbatim along $z = uQ^{-N}$ for any $u \in \mathbb{C}^*$ whose orbit avoids the zero class $[1]_Q$ of $\theta_Q(\cdot)$, using the standard theta lower bound $|\theta_Q(-z)| \geq c([u]) \cdot (\text{peak term})$ off that class; along each such orbit $C_1(z)$ is constant (ellipticity) and $f_2/f_1 \rightarrow 0$, $g^{[\lambda]} \rightarrow 1$, so $C_1([u]) = 1/(q; q)_\infty$ on every class $\neq [1]_Q$, and by meromorphy $C_1 \equiv 1/(q; q)_\infty$: for every regular λ . In particular C_1 is z - and λ -independent. $\square$

Remark 93 (Diophantine consequences at z^*). At $q = q^*$ the identity splits off the candidate rational zero:

$$\prod_{k \geq 2} z_k(q^*) Q^{k-1} = \frac{(q; q^2)_\infty}{z^*} = 0.988299884628746 \dots \quad (\text{verified to 30 digits}).$$

Were β_2 rational, the *modular* value $(q; q^2)_\infty$ (transcendental at algebraic q by Nesterenko's theorem applied to the Euler products) would equal the rational z^* times the tail dressing product $\prod_{k \geq 2} z_k Q^{k-1}$. Thus the entire modular sector of the problem is now explicit; the Stokes-invariant connection constant is $1/(q; q)_\infty$, the zero product is $(q; q^2)_\infty$; and *all* remaining unknown arithmetic is provably confined to the tail dressing product, equivalently to the q -Stokes cocycle $C_2^{[\lambda]}$: the non-modularity of the β_2 problem *is* its q -Stokes data. A transcendence statement for that tail (an algebraic independence of the higher-zero dressings from the modular values, or a rigidity theorem for the q -Stokes cocycle) would settle β_2 ; no current theorem provides it.

The cocycle itself is now explicit. Writing $g = g^{[\lambda]}$ for the resummed series and h for f_2 's convergent one,

$$C_2^{[\lambda]}(z) = \theta_Q(-z/q) \frac{g(z) G(Qz) + z^{-1} g(Qz) G(z)}{-(z/q) g(z) h(Qz) + z^{-1} g(Qz) h(z)}$$

This is an identity: it is the ratio of the Casoratian evaluations $\text{Cas}(f_1, G) = \theta_Q(-z)[g(z)G(Qz) + z^{-1}g(Qz)G(z)]$ and $\text{Cas}(f_1, f_2) = \frac{\theta_Q(-z)}{\theta_Q(-z/q)}[-(z/q)g(z)h(Qz) + z^{-1}g(Qz)h(z)]$, both instances of $\theta_Q(Qw) = w^{-1}\theta_Q(w)$ (and verified numerically to 10^{-61}); with divisor now fully described: an *explicit* zero lattice $z \in q^{-1}Q\mathbb{Z}$ from the theta factor (the computed zero at $z = 1/q$, exact to 10^{-63}), a direction-dependent zero at $z = \lambda^2$ arising from the exact *resummation resonance* $g^{[\lambda]}(\lambda^2)G(Q\lambda^2) + \lambda^{-2}g^{[\lambda]}(Q\lambda^2)G(\lambda^2) = 0$ (verified to 10^{-62} at $\lambda \in \{\frac{3}{2}, 2, \frac{13}{5}\}$; a proof for general regular λ is open), and poles at the zeros of the h -bracket. Substituting into the pinned zero condition at z^* and cancelling the common theta factor reduces the entire β_2 question to *one* relation between the two modular values $\theta_Q(-z^*)$, $(q; q)_\infty$ and five values of the non-modular pair $(g^{[\lambda]}, h)$ on the orbit of z^* . The remaining input is a transcendence/independence statement for those five values; supplying it is one of three research programs (a non-modular extension of Nesterenko's method via the deformation of this connection, a $\rho < \frac{1}{2}$ -regime irrationality criterion, or adelic rigidity of the vanishing locus); none is currently available.

The first of those programs now has its foundation: the deformation of the module in q closes finitely.

Proposition 94 (the deformation module). *Let $L = q - a\sigma + \sigma^2$ with $a = q + 1 - qz$, $\sigma f(z) = f(Qz)$, and let $\theta = z\partial_z$, $\delta = q\partial_q$ (acting on all q -dependence, including the base $Q = q^2$, so that $\delta\sigma = \sigma\delta + 2\sigma\theta$). Then*

$$L(\theta G) = -qz\sigma G, \quad L(\delta G) = -qG + q(1 + 3z)\sigma G + 4q\theta G - 2a\sigma\theta G$$

(two commutator computations; verified to 51 digits). Consequently the $\mathbb{Q}(q, z)\langle\sigma\rangle$ -module generated by $\{G, \theta G, \delta G\}$ has finite rank ≤ 6 , with explicit σ -companion relations $\sigma^2 G = a\sigma G - qG$, $\sigma^2 \theta G = a\sigma \theta G - q\theta G - qz\sigma G$, and $\sigma^2 \delta G = a\sigma \delta G - q\delta G - qG + q(1 + 3z)\sigma G + 4q\theta G - 2a\sigma\theta G$.

This finiteness is of the first jet only: the module is not closed under a second application of θ or δ . Both $\theta^2 G$ and $\delta^2 G$ escape $\text{span}_{\mathbb{Q}(q, z)}\{G, \sigma G, \theta G, \sigma\theta G, \delta G, \sigma\delta G\}$ (verified by exact modular linear algebra at $q = \frac{1}{3}$, z -series to 70 terms, coefficient degrees ≤ 6 : the augmented rank strictly exceeds the plain rank). The two derivations are moreover linked by the exact identity

$$\delta G = (\theta^2 - \theta)G + C, \quad C = \sum_{k \geq 0} (-1)^k \left(\sum_{i=1}^{2k} \frac{iq^i}{1-q^i} \right) \frac{q^{k(k-1)} z^k}{(q; q)_{2k}}$$

($= z^2 G_{zz} + C$; immediate from g_k 's logarithmic q -derivative), whose weight C carries the truncated Eisenstein sums $\sum_{i \leq 2k} iq^i/(1-q^i)$ term by term; the same off-modular data that obstructs the differential closure. In particular, differentiating the pinned relation of Proposition 91 in q does not close: the second q -derivative leaves the module, consistent with Proposition 102. The value of the rank-6 structure is at the function level; it is the module whose difference Galois group is computed in Theorem 99; not as a Nesterenko precondition in the descent variable, which Proposition 102 shows does not exist.

Corollary 95 (equation of the zero curve). *Implicit differentiation of $G(q, z_1(q)) = 0$ gives*

$$z_1'(q) = - \frac{z_1(\delta G)(q, z_1)}{q(\theta G)(q, z_1)},$$

with $\theta G, \delta G$ the first-jet generators of Proposition 94 (formula checked against a finite-difference derivative to 20 digits at $q = 0.3$). The right-hand side is not a closed vector field; iterating d/dq leaves the rank-6 module (Proposition 94); so this is one exact relation, not an autonomous system. The number q^* is the crossing of the zero curve with the diagonal: $z_1(q^*) = 2q^*(1 - q^*)$ (residual $< 10^{-27}$).

Lemma 96 (irreducibility of the σ -module). *For $0 < q < 1$ the rank-2 σ -module of G (in the sense of [38]) is irreducible over $\mathbb{C}(z)$: the Riccati equation*

$$u(z)u(Qz) - a(z)u(z) + q = 0, \quad a = q + 1 - qz,$$

has no solution $u \in \mathbb{C}(z)$.

Proof. Suppose $u = P/R$ with $P, R \in \mathbb{C}[z]$ coprime, of degrees p, r , leading coefficients α, β . Clearing denominators,

$$F(z) := P(Qz)P(z) - a(z)P(z)R(Qz) + qR(z)R(Qz) \equiv 0.$$

At $z = 0$: $F(0) = P(0)^2 - (q+1)P(0)R(0) + qR(0)^2 = 0$; if $R(0) = 0$ then $P(0) = 0$, contradicting coprimality, and symmetrically; so $P(0)R(0) \neq 0$ and $u(0) \in \{1, q\}$. *Top degree*: the three terms have degrees $2p, p+r+1, 2r$. If $p = r$, the top degree $2p+1$ is attained by the middle term alone, with coefficient $q\alpha\beta Q^r \neq 0$; if $|p-r| \geq 2$, the top degree is attained by the first or third term alone, with coefficient $\alpha^2 Q^p$ or $q\beta^2 Q^r$. Both are impossible, so $|p-r| = 1$. *Root pairing*: let $\zeta \neq 0$ be a root of R of multiplicity m , so $P(\zeta) \neq 0$. At $z = \zeta/Q$ the three terms of F have orders $k, k+m+[a(\zeta/Q)=0], m_0+m$, where $k = \text{ord}_{\zeta/Q} P$ and $m_0 = \text{ord}_{\zeta/Q} R$; since $F \equiv 0$ the minimum order cannot be attained by a single term, forcing $k \geq m_0 + m \geq m$. Symmetrically, a root $\xi \neq 0$ of P of multiplicity k forces $\text{ord}_{Q\xi} R \geq k$ (coprimality gives $\text{ord}_{Q\xi} P = 0$, so the first term has order k at ξ , the third $\text{ord}_{Q\xi} R$, and the middle at least $k + \text{ord}_{Q\xi} R$; again no single minimum). The two maps $\zeta \mapsto \zeta/Q, \xi \mapsto Q\xi$ are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities, $p \leq r \leq p$, so $p = r$; contradicting $|p-r| = 1$. $\square$

Proposition 97 (the symmetric square). *Products $w = y\tilde{y}$ of solutions of L satisfy the third-order equation*

$$w(Q^3 z) = \frac{A_1(A_0 A_1 - q)}{A_0} w(Q^2 z) + q(q - A_0 A_1) w(Qz) + \frac{q^3 A_1}{A_0} w(z),$$

with $A_0 = a(z), A_1 = a(Qz)$, obtained by eliminating the mixed product $m = y\tilde{y}(Q\cdot) + \tilde{y}y(Q\cdot)$ through $m(Qz) = 2A_0 w(Qz) - qm(z)$. Its characteristic roots at $z = 0$ are $\{1, q, q^2\}$; verified on the solutions $G^2, z^{1/2}GH, zH^2$ to 41–43 digits.

Lemma 98 (divisor orientation). *Let $r \in \mathbb{C}(z)^\times$ satisfy the rational-quotient relation of Proposition 97, $r_0 r_1 r_2 = c_2 r_0 r_1 + c_1 r_0 + c_0$ with $r_k = r(Q^k z)$, $c_2 = A_1(A_0 A_1 - q)/A_0$, $c_1 = q(q - A_0 A_1)$, $c_0 = q^3 A_1/A_0$. On every Q -orbit in $\mathbb{C}^*$ disjoint from the singular orbit $z_a Q^\mathbb{Z}$ ($z_a = (q+1)/q$, the zero of a), the maximal-modulus zero-or-pole of r cannot be a pole, and the minimal-modulus zero-or-pole cannot be a zero.*

Proof. Off $z_a Q^\mathbb{Z}$ the c_i are finite and $c_0 \neq 0$. If the maximal-modulus zero-or-pole ζ on such an orbit were a pole, evaluate at $z = Q^{-2}\zeta$: there r_0, r_1 are finite and nonzero (all higher-modulus orbit points carry no zero or pole), r_2 has a pole, so the left side has a pole while the right side is finite. If the minimal-modulus zero-or-pole ζ' were a zero, rewrite the relation for $s = 1/r$ as $c_0 s_0 s_1 s_2 + c_1 s_1 s_2 + c_2 s_2 = 1$ and evaluate at $z = \zeta'$: s_0 has a pole, s_1, s_2 are finite and nonzero, $c_0(\zeta') \neq 0$, so the left side has a pole while the right side is 1. $\square$

Theorem 99 (the difference Galois group contains SL_2). *For every $0 < q < 1$, the difference Galois group over $\mathbb{C}(z)$ of the module of G contains SL_2 .*

Proof. For $N \geq 1$ suppose $\text{Sym}^N M$ has a rank-1 submodule, i.e. there is a nonzero solution w of the N -th symmetric-power equation with $r := w(Q\cdot)/w \in \mathbb{C}(z)$. The products $f_1^{[\lambda]N-k} f_2^k$ ($0 \leq k \leq N$) form a solution basis (their Casoratian is a power of $\text{Cas}(f_1, f_2) \neq 0$; checked numerically for $N = 2$),

so $w = \sum_k c_k f_1^{N-k} f_2^k$ with constants c_k , and the case split is λ -robust since Stokes jumps preserve $k_0 := \min\{k : c_k \neq 0\}$.

Case $k_0 < N$. Along a ray in a regular class, f_2/f_1 decays super-polynomially, so the rational function r has the Poincaré asymptotic expansion of the quotient of the dominant component:

$$r \sim (-1/z)^{N-k_0} (-z/q)^{k_0} \left(\frac{\hat{g}(Qz)}{\hat{g}(z)} \right)^{N-k_0} \left(\frac{h(Qz)}{h(z)} \right)^{k_0}.$$

The order- j coefficient d_j of $\hat{g}(Qz)/\hat{g}(z)$ satisfies $d_j \sim -b_j$ with $b_j = q^{(3j-j^2)/2}/(q; q)_j$ (verified: $d_j/(-b_j) = 0.99986, 0.99998, 0.999998$ at $j = 10, 12, 14$, $q = \frac{1}{3}$, converging to 1). Since this factor enters r to the power $N - k_0 \geq 1$ against rational, hence geometrically bounded, prefactors, the coefficients of r inherit the growth $b_j \sim q^{-j^2/2}$, which is super-geometric, whereas a rational function's expansion at ∞ has geometrically bounded coefficients. Since matching asymptotic expansions are unique, no rational r exists.

Case $k_0 = N$. Then $r = (-z/q)^N \rho^N$ with $\rho := h(Q\cdot)/h$ meromorphic on $\mathbb{C}^*$; $\rho^N \in \mathbb{C}(z)$ and single-valuedness force $\rho \in \mathbb{C}(z)$ (an N -th root of a rational function is single-valued on $\mathbb{C}^*$ only if every divisor multiplicity, including at 0 and ∞ , is divisible by N). Then $f_2(Qz)/f_2(z) = -(z/q)\rho(z)$ is rational, giving a rational solution of the rank-2 Riccati equation; contradicting Lemma 96.

Hence no $\text{Sym}^N M$ has a rank-1 submodule. $N = 1$ is irreducibility; $N = 2$ excludes the imprimitive (dihedral) case; and a projectively finite irreducible subgroup (A_4, S_4, A_5) fixes a line in some Sym^N with $N \leq 12$ (the classical binary polyhedral invariants), which is likewise excluded. By the classification of algebraic subgroups of GL_2 , the group contains SL_2 . $\square$

Proposition 100 (no $\sigma\delta$ -integrability). *For every $0 < q < 1$ there is no $B \in \text{gl}_2(\mathbb{C}(z))$ and no $c \in \mathbb{C}(z)$ with $\theta(A) = \sigma(B)A - AB + cA$, where $A = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$ and $\theta = z\partial_z$.*

Proof. Writing $B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, the entries give $\gamma = -q\beta_1$, $\delta = \alpha_1 + a\beta_1 - c$, and eliminating α (the central terms c cancel identically; verified symbolically) leaves the necessary condition

$$qa\beta_3 = a_1(aa_1 - q)\beta_2 - a(aa_1 - q)\beta_1 + qa_1\beta + qz(a_1 + q^2a),$$

$\beta_k = \beta(Q^k z)$, $a = a(z)$, $a_1 = a(Qz)$. A polynomial part of degree $m \geq 0$ contributes top coefficient $\propto \beta_m(1 - q^{2+2m}) \neq 0$ at degree $m + 3$, above the inhomogeneity: so β has no polynomial part. If β has a pole off the orbit $z_a Q^{\mathbb{Z}}$, the maximal-modulus pole appears in exactly one term (the slot $z = \zeta Q^{-3}$, coefficient $qa \neq 0$ there), and the minimal-modulus pole likewise (slot $z = \zeta'$, coefficient qa_1): both impossible; so the poles lie in the window $\{z_a Q^{-1}, \dots, z_a Q^3\}$ ($a(z_a Q^k) = (q+1)(1 - Q^k)$ vanishes only at $k = 0$). Sweeping the slots $z = z_a Q^{-4}, \dots, z_a Q^{-1}$ kills the orders at $z_a Q^{-1}, \dots, z_a Q^2$ one by one (each pole appears in a single term with nonvanishing coefficient), and the order at $z_a Q^3$ is killed at the slot $z = z_a Q$; coefficient $a_1(aa_1 - q)$ with $(aa_1 - q)(z_a Q) = (q+1)^2(1 - q^2)(1 - q^4) - q$; or, where that vanishes (one root $q \approx 0.78027$), at the slot $z = z_a Q^2$; coefficient $a(aa_1 - q)$ with $(aa_1 - q)(z_a Q^2) = (q+1)^2(1 - q^4)(1 - q^6) - q$ (one root $q \approx 0.86753$); the two polynomials are coprime, so for every $q \in (0, 1)$ at least one slot applies. Hence $\beta \equiv 0$, contradicting $z(a_1 + q^2a) = z[(q+1)(1 + q^2) - 2q^3z] \neq 0$. (Cross-check: at $q = \frac{1}{3}$ the exact linear system for poles of order ≤ 3 in the window has $\text{rank}(A) = 15$, $\text{rank}[A|b] = 16$.) $\square$

Corollary 101 (differential transcendence of G). *For every $0 < q < 1$ the function $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$: the Hahn–Exton q -cosine is differentially transcendental.*

Proof, via parametrized difference-Galois theory. We apply the parametrized (σ, θ) -difference Galois theory with derivation $\theta = z\partial_z$, over the difference-Galois formalism of [38] and in the form of

[12, 3]. The determinant equation $\sigma(y) = \det(A)y = qy$ has the solution $y = z^{1/2}$ (indeed $\sigma(z^{1/2}) = (Qz)^{1/2} = qz^{1/2}$, as $Q = q^2$), which is algebraic over $\mathbb{C}(z)$; hence the parametrized difference Galois group of $\sigma(y) = \det(A)y$ lies in $\mathbb{C}^\times$. This is exactly the hypothesis of Dreyfus–Hardouin–Roques [12, Prop. 2.9], which for a rank-2 system whose difference Galois group contains SL_2 (here Theorem 99) gives the dichotomy: the parametrized group Gal^θ either is conjugate into $\mathrm{GL}_2(\mathbb{C})$ (all solutions differentially algebraic) or contains $\mathrm{SL}_2(\tilde{\mathbb{C}})$ (all nonzero solutions differentially transcendental), and the *first* alternative holds if and only if there exists $B \in \mathrm{gl}_2(\mathbb{C}(z))$ with $\theta(A) = \sigma(B)A - AB$ (their integrability condition (7); $\theta(\det A) = 0$ since $\det A = q$ is constant in z , so no trace term appears). By Proposition 100 no such B exists; indeed none exists even allowing a scalar twist cA . Therefore the second alternative holds: $\mathrm{Gal}^\theta \supseteq \mathrm{SL}_2(\tilde{\mathbb{C}})$, and every nonzero solution, in particular $G(q, \cdot)$, is differentially transcendental over $\mathbb{C}(z)$. $\square$

The transcendence of the *number* q^* is a value-level statement, and the natural route (Nesterenko’s multiplicity method fed by Corollary 101) meets a precise structural obstruction, which we isolate.

Proposition 102 (where the differential closure fails in the descent variable). *Write $\delta = q\partial_q$, $\theta = z\partial_z$, and let $u_j(q) := G(q, Q^j z^*(q))$ denote the orbit values along the diagonal $z^*(q) = 2q(1 - q)$.*

- (i) (orbit collapse) *The σ -orbit is not an obstruction to finiteness: the z -recurrence evaluated on the diagonal has coefficients $a(Q^j z^*) = q + 1 - qQ^j z^* \in \mathbb{Q}(q)$, so $u_{j+2} = a(Q^j z^*) u_{j+1} - q u_j$ and every orbit value lies in $\mathrm{span}_{\mathbb{Q}(q)}\{u_0, u_1\}$ (verified for $j \leq 3$ to 32 digits).*
- (ii) (the E_2 -coupled jet structure) *The q -derivative of G obeys the exact identity*

$$\delta G = (\theta^2 - \theta)G + \varphi(q)G - T, \quad \varphi(q) = \sum_{i \geq 1} \frac{iq^i}{1 - q^i}, \quad T = \sum_{k \geq 0} (-1)^k \left(\sum_{i > 2k} \frac{iq^i}{1 - q^i} \right) \frac{q^{k(k-1)} z^k}{(q; q)_{2k}}$$

(verified to 41 digits), where φ is the quasimodular Eisenstein weight ($24\varphi = 1 - E_2$ up to normalization) and the correction T has coefficients smaller than G ’s by the factor $\sum_{i > 2k} iq^i / (1 - q^i) = O(kq^{2k})$. Expanding these weights as Lambert series and interchanging the (absolutely convergent) sums exhibits T as an Eisenstein-weighted trace of the module over the forward σ -orbit:

$$T(q, z) = \sum_{d \geq 1} \frac{2q^d}{1 - q^d} (\theta G)(q, q^{2d}z) + \sum_{d \geq 1} \frac{q^d}{(1 - q^d)^2} G(q, q^{2d}z)$$

(verified to 41 digits): the entire non-modular content of δG is an orbit convolution of $(G, \theta G)$ against E_2 -type weights. On the diagonal the orbit values collapse by (i) into four generators u_0, u_1, θ -analogues v_0, v_1 (the v ’s obey the inhomogeneous transported recurrence $v_{j+2} = a_j v_{j+1} - qv_j - qQ^j z^* u_{j+1}$), but the collapse coefficients accumulate into transfer-product series $A, B, C, D \in \mathbb{Z}[[q]]$ (integrality is immediate: transfer data and Lambert weights lie in $\mathbb{Z}[[q]]$), which are not of any classical type: they satisfy no relation over $\mathrm{span}\{1, \varphi, \sum \sigma_3(n)q^n\}$ with rational-function coefficients of degree ≤ 8 , and no linear q -ODE of order ≤ 3 and degree ≤ 6 (exact linear algebra on 45 coefficients). The identity relocates, and does not remove, the obstruction. Since δ and θ commute, each application of δ raises the z -jet order by two: $\delta^n S$ involves $\theta^{2n} G|_{z=z^*}$ with unit leading coefficient.

- (iii) (the actual obstruction) *The z -jets do not close: $\theta^2 G$ and $\delta^2 G$ escape the rank-6 module of Proposition 94, and by Corollary 101 the tower $\{\theta^m G\}_m$ satisfies no algebraic relation over $\mathbb{C}(z)$. A finitely generated δ -closed $\mathbb{C}(q)$ -algebra containing S would force an algebraic relation*

over $\mathbb{C}(q)$ among the diagonal restrictions $\{\theta^m G(q, z^*(q))\}_{m \leq M}$ for some finite M ; a statement equivalent to S being differentially algebraic in q , for which there is no supporting structure and proof-grade negative evidence across a wide window: no algebraic differential equation at profiles up to $(2, 12, 3)$, $(3, 8, 3)$, $(4, 5, 6)$, $(5, 4, 8)$, $(6, 3, 15)$, $(8, 2, 35)$, $(12, 1, 120)$ in (order, degree, $\deg_q$), by full column rank on 2000 coefficients modulo a prime (`beta2_diffalg_search.py`, `beta2_ade_sweep.py`; E_2/q -Chazy control recovered).

Thus the Nesterenko route closes if and only if S is differentially algebraic in q ; the obstruction is carried by the z -jet tower; whose function-level independence is exactly Corollary 101; and not by the σ -orbit, which collapses to rank two.

Remark 103 (the value-level descent as a joint multiplicity estimate). Nesterenko's method derives transcendence of a value from three ingredients: a finite ring closed under the relevant operator, a *multiplicity estimate* bounding $\text{ord } P(\text{generators}) \leq c(\deg P)^\kappa$, and routine analytic size/Siegel-lemma constructions. Theorem 99 and Proposition 100 furnish precisely the nondegeneracy a multiplicity estimate consumes (no invariant subvariety; no first integral). But Proposition 102 shows the first ingredient fails in the descent variable q : the finiteness of the β_2 module is intrinsically *joint* in (σ_z, δ_q) , and σ_z is a non-infinitesimal q -lattice shift, outside the Pfaffian/differential multiplicity theories of Nesterenko and Binyamini. Thus the transcendence of $q^* = \beta_2^{-2}$ reduces to a *joint q -difference–differential multiplicity estimate* for the rank- ≤ 6 module at (q^*, z^*) : a bound $\text{ord}_{(q^*, z^*)} P(G, \theta G, \delta G) \leq c N^\kappa$ ($N = \deg P$) in the local ring, uniform along the σ -orbit. Such an estimate for finite (σ, δ) -modules is not available in the literature; it is the sole non-routine input remaining, and the precise open problem on which the (ir)rationality of β_2 now turns.

Remark 104 (the obstruction inside q -Siegel–Shidlovskii theory). The same wall appears, quantified, in the value-transcendence theory built on Siegel's method. The approach of Amou–Matala-aho–Väänänen [1] (and the Matala-aho-school Baker-type refinements) proves non-vanishing of values of Poincaré-type q -difference solutions at algebraic points under two hypotheses: the system is Poincaré-type at 0 (met here: $A(0)$ is invertible, $\det A = q$), and a *Siegel margin* $\gamma < \Gamma$ holds, where γ is the ratio of the logarithmic height of the cleared denominators to that of the analytic decay. For the Hahn–Exton class that ratio is exactly $\gamma = \rho = \log(t/s)/(2 \log t) \leq \frac{1}{2}$ of Proposition 85: the double Pochhammer $(q; q)_{2k}$ is the factor 2; so the margin *fails*, and the theory does not apply. Thus Conjecture Z (transcendence of the Hahn–Exton zeros at algebraic base, which by $z^* = z_1(q^*)$ implies β_2 transcendental) reduces to extending q -Siegel–Shidlovskii past the margin for a class that is additionally irregular at ∞ . The connection theory of this paper supplies the irregular- ∞ data explicitly; the margin itself is the value-theoretic face of the multiplicity estimate of Remark 103, and is equal in strength to β_2 -transcendence, not weaker. A detailed program is recorded in the repository (`paper/journal/qSS_program.md`).

One further mechanism deserves record, because it isolates which half of the Siegel scheme is actually blocked. Under the rationality hypothesis all tail power sums $\tau_p = \sum_{k \geq 2} z_k^{-p} = \sigma_p - z^{*-p}$ are rational, and they are the moments of a positive measure with infinite support (unit atoms at $1/z_k$); hence every Hankel determinant $\det[\tau_{i+j+2}]_{i,j \leq n}$ is *strictly positive for free*; no auxiliary nonvanishing argument is needed; and, being a positive rational, is bounded below by the reciprocal of its denominator ($\leq t^{8n^3/3+O(n^2)}$). Analytically these determinants decay like $Q^{n^3/3(1+o(1))}$ (atoms at $\approx Q^{k-1}$; verified: strictly positive with super-quadratic exponent growth at q^*). The comparison $2 \log(t/s) > 8 \log t$ fails for every rational q : the positivity route conserves the ledger at a factor 4. The instructive point is that *nonvanishing was never the bottleneck* of any construction in this ledger; positivity supplies it gratis here; the entire difficulty of the problem resides in the size-versus-height comparison. Nor is there an Apéry-type denominator miracle hiding in that comparison: the

measured reduced denominators of the partial sums $S_N(s/t)$ equal 96–100% of the naive clearing $\prod_{i \leq 2N} (t^i - s^i)$ in logarithmic size, with the ratio tending to 1 as N grows (exact rational arithmetic, $N \leq 12$, at $q = \frac{1}{2}$ and $\frac{2}{5}$); the cancellation is an $O(N)$ boundary effect in an $O(N^2)$ exponent. The arithmetic of the ledger is exactly its worst-case costing.

Remark 105 (backward-orbit normal form of the zero condition). The zero condition admits a compression into a single explicit rational sequence. If $G(q, z_1) = 0$, the three-term recurrence along the backward σ -orbit, $u_{-m} = (a(Q^{-m}z_1)u_{-m+1} - u_{-m+2})/q$ with $u_n = G(q, Q^n z_1)$, loses one solution constant: $u_{-m} = u_1 r_m(q)$ with $r_0 = 0$, $r_1 = -1/q$, and $r_m \in \mathbb{Q}(q, z_1)$ computable by the same recurrence. The asymptotics of Theorem 92 then give the exact evaluation

$$\lim_{m \rightarrow \infty} r_m(q) Q^{m(m+1)/2} (-z_1)^{-m} = \frac{\theta_Q(-z_1)}{(q; q)_\infty G(q, Qz_1)},$$

with geometric rate Q^m and q -Gevrey correction series $\sum_j b_j z_1^{-j} Q^{mj}$ (verified to 17 digits at q^* by $m = 24$). Under the hypothesis $q^* = s/t$ the normalized terms are rationals of height $(st)^{m^2 + O(m)}$ approximating the limit to $O(Q^m)$; accelerated forms that cancel J correction terms gain decay Q^{mJ} but pay the quadratic heights of the b_j , and the arithmetic ratio stays $\leq \frac{1}{2}$; the balance of Proposition 85 once more. The normal form is recorded as the sharpest single-sequence statement of what any future method must exploit: the entire rationality hypothesis is equivalent to the arithmetic of this one explicit sequence. The θ -jet admits the same normal form one level up: along the backward orbit,

$$\frac{(\theta G)(q, Q^{-m}z_1)}{G(q, Q^{-m}z_1)} - m \longrightarrow -\frac{z_1 \theta'_Q(-z_1)}{\theta_Q(-z_1)} \quad (\text{rate } Q^m; \text{ verified to 10 digits at } q = 0.3),$$

an exact theta log-derivative: each backward step increments $\theta \log \theta_Q(-z)$ by exactly one (quasi-periodicity), and the $\hat{g}$ -correction vanishes in the limit.

The *forward* orbit admits the complementary normal form, in the language of rational dynamics. The Riccati relation gives the explicit continued fraction $q/(a(z) - q/(a(Qz) - \dots))$, which by Pincherle's theorem [25] converges to the minimal-solution ratio $q H(Qz)/H(z)$; the zero condition $G(q, z_1) = 0$ forces the *dominant* ratios $w_n := G(q, Q^{n+1}z_1)/G(q, Q^n z_1)$ to obey $w_1 = a(z_1)$ exactly and thereafter the rational map

$$w_{n+1} = a(Q^n z_1) - \frac{q}{w_n},$$

so that under the hypothesis $q, z_1 \in \mathbb{Q}$ the entire forward orbit is an explicit $\mathbb{Q}$ -orbit converging to the attracting fixed point 1 with $w_n - 1 = \Theta(Q^n)$ (verified to 25 digits at q^*). Its heights compound quadratically ($t^{O(n^2)}$), so the Liouville comparison Q^n vs. t^{-n^2} is again consistent: both normal forms; backward theta-growth and forward rational dynamics; conserve the balance of Proposition 85, and together they exhaust the one-orbit reformulations of the zero condition.

Combining the two orbits with the q -Wronskian makes the relation-count of the rationality hypothesis explicit. Under $G(q, z_1) = 0$ the unknown transcendental data reduces to three numbers ($g_1 := G(q, Qz_1)$, $H(q, z_1)$, and the backward limit ρ_∞), bound by exactly two relations: $g_1 H(q, z_1) = 1 - q$ (the q -Wronskian at the zero) and $\rho_\infty g_1 = \theta_Q(-z_1)/(q; q)_\infty$ (the backward normal form). Eliminating g_1 ,

$$\frac{\rho_\infty}{H(q, z_1)} = \frac{\theta_Q(-z_1)}{(1 - q)(q; q)_\infty}$$

(verified to 32 digits at q^*), with modular right-hand side. Three unknowns, two relations: the rationality hypothesis is consistent at this level, and a proof requires precisely one further independent relation; the multiplicity statement of Remark 103, in yet another equivalent form.

Remark 106 (truncation roots, and the exact deficit). A scheme not covered by the lattice ledger prices the wall in its simplest form. Clearing the truncation of S gives integer polynomials $P_N = (q; q)_{2N} \sum_{k \leq N} (-2)^k q^{k^2} (1 - q)^k / (q; q)_{2k} \in \mathbb{Z}[q]$ of degree $2N^2 + N$, whose heights are only $e^{O(N)}$ (the pentagonal sparsity of q -Pochhammer coefficients; measured $\log_2 H = 2.6, 4.6, \dots, 12.1$ for $N = 2, \dots, 7$), and whose root nearest q^* satisfies $|q_N - q^*| \asymp q^{*(N+1)^2}$ (verified $N \leq 7$). If $q^* = s/t$ then $|P_N(q^*)| \geq t^{-(2N^2+N)}$, while analytically $|P_N(q^*)| = e^{O(N)} q^{*(N+1)^2}$; contradiction would require $q^* < t^{-2}$, impossible for $q^* = s/t \geq 1/t$. The deficit is exactly the factor 2 in the clearing degree; the double Pochhammer once more, now in s -independent form. The same computation shows that an equivalent representation of the root condition with *single*-Pochhammer denominators (clearing degree $\sim N^2/2$) would yield $q^* \neq s/t$ for all $s < \sqrt{t}$. No such representation exists within the Rogers–Ramanujan/Bailey framework [16]: those transformations require the argument to be a q -power, and the argument here is the catalytic point $z = 2q(1 - q)$. In this precise sense the obstruction to β_2 's irrationality is the non- q -power (catalytic) argument itself.

The catalytic locus is moreover *seat-independent*. Writing $S(q) = {}_1\phi_1(0; q; Q, z^*)$ ($Q = q^2$), the parameter–argument transformation ${}_1\phi_1(0; b; Q, x) = \frac{(x; Q)_\infty}{(b; Q)_\infty} {}_1\phi_1(0; x; Q, b)$ gives the equivalent zero condition

$$T(q) := {}_1\phi_1(0; 2q(1 - q); q^2, q) = \sum_{k \geq 0} \frac{(-1)^k q^{k^2}}{(q^2; q^2)_k (2q(1 - q); q^2)_k} = 0$$

(verified to 51 digits; the prefactor cannot vanish since $z^* < 1$), in which the argument is the q -power q and the catalytic value sits in the *parameter*; equivalently, q^* is the point where the Hahn–Exton q -Bessel of q -dependent order $\nu(q) = \log_{q^2}(2q(1 - q)) - 1$ vanishes at the fixed argument $q^{-1/2}$. The clearing degree of T 's truncations is again $\sim 2N^2$: the deficit is invariant under the swap, and in either seat it is the non-special value $2q(1 - q)$ that lies outside every Rogers–Ramanujan/Bailey lattice. The well-poised generalization does not help: a WP-Bailey specialization producing a single-Pochhammer form would require $2q(1 - q)$ to be a q -power (the only special-value type such summations admit), whereas at $q = q^*$ one has $q^* < 2q^*(1 - q^*) < 1$, strictly between consecutive q -powers (the two loci where $2q(1 - q)$ is a q -power, $q = \frac{1}{2}$ and $q = \frac{2}{3}$, both miss q^*). The catalytic argument is thus q -power-free in every seat, which is the transformation-theoretic content of the $\rho = \frac{1}{2}$ wall.

Proposition 107 (effective non-rationality of q^*). *The least positive root q^* of S admits no rational representation s/t with $t \leq 10^{1044}$. Consequently, if $\beta_2 = u/v$ in lowest terms then $u > 10^{522}$.*

Proof. All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrational`) with the series tail bounded by $(q; q)_{2k} \geq (1 - 2q)/(1 - q) > 0$ for $q < \frac{1}{2}$ and term ratios $< \frac{1}{2}$ beyond the truncation. (i) $S > 0$ on $[0, 0.40]$ (400 certified chunks). (ii) $S' \leq -1$ on $[0.40, 0.46]$ (600 certified chunks; S' evaluated term-by-term with the logarithmic-derivative weights), so S has at most one zero in $[0.40, 0.46]$. (iii) with c the 2090-digit truncation of q^* , $a = c - 2 \cdot 10^{-2090}$ and $b = c + 2 \cdot 10^{-2090}$, certified evaluation gives $S(a) > 0 > S(b)$ (working precision 2130 digits, series tail enclosed in $\pm 10^{-2125}$); by (i)–(ii), $q^* \in [a, b]$. (iv) the interval $[a, b]$ contains the rational s_0/t_0 (Stern–Brocot construction) with $9.5 \cdot 10^{1044} < t_0 < 10^{1045}$, verified by exact rational comparison, and certified $S(s_0/t_0) \neq 0$. If $m/n \in [a, b]$ with $n \leq 10^{1044}$ and $m/n \neq s_0/t_0$, then $|m/n - s_0/t_0| \geq 1/(nt_0) > 10^{-2089} > b - a = 4 \cdot 10^{-2090}$, impossible; so $m/n = s_0/t_0$, whence $n = t_0 > 10^{1044}$, a contradiction. Thus no rational with denominator $\leq 10^{1044}$ lies in $[a, b]$ at all, in particular none equals q^* . For $\beta_2 = u/v$ in lowest terms, $q^* = v^2/u^2$ in lowest terms, so $u^2 > 10^{1044}$. $\square$

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